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1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

J40G

MLB-GS

REV

ECN

DESCRIPTION OF REVISION

CK APPD  
DATE

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ENGINEERING RELEASED

2011-04-07

Thu Apr 7 15:35:59 2011

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| 820-3059    | 1   | PCBF, MLB-GS, J40G               | MLB           |          |                  |
| 639-1822    |     | PCBA, MLB-GS, GDDR-HYNIX, J40G   |               |          |                  |
| 639-1823    |     | PCBA, MLB-GS, GDDR-SAMSUNG, J40G |               |          |                  |
| 085-2724    | 1   | DEV LIST, MLB-GS, J40G           | DEV1          |          | DEVELOPMENT_LIST |

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ABBREV=DRAWING

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SCH, MLB-GS, J40G

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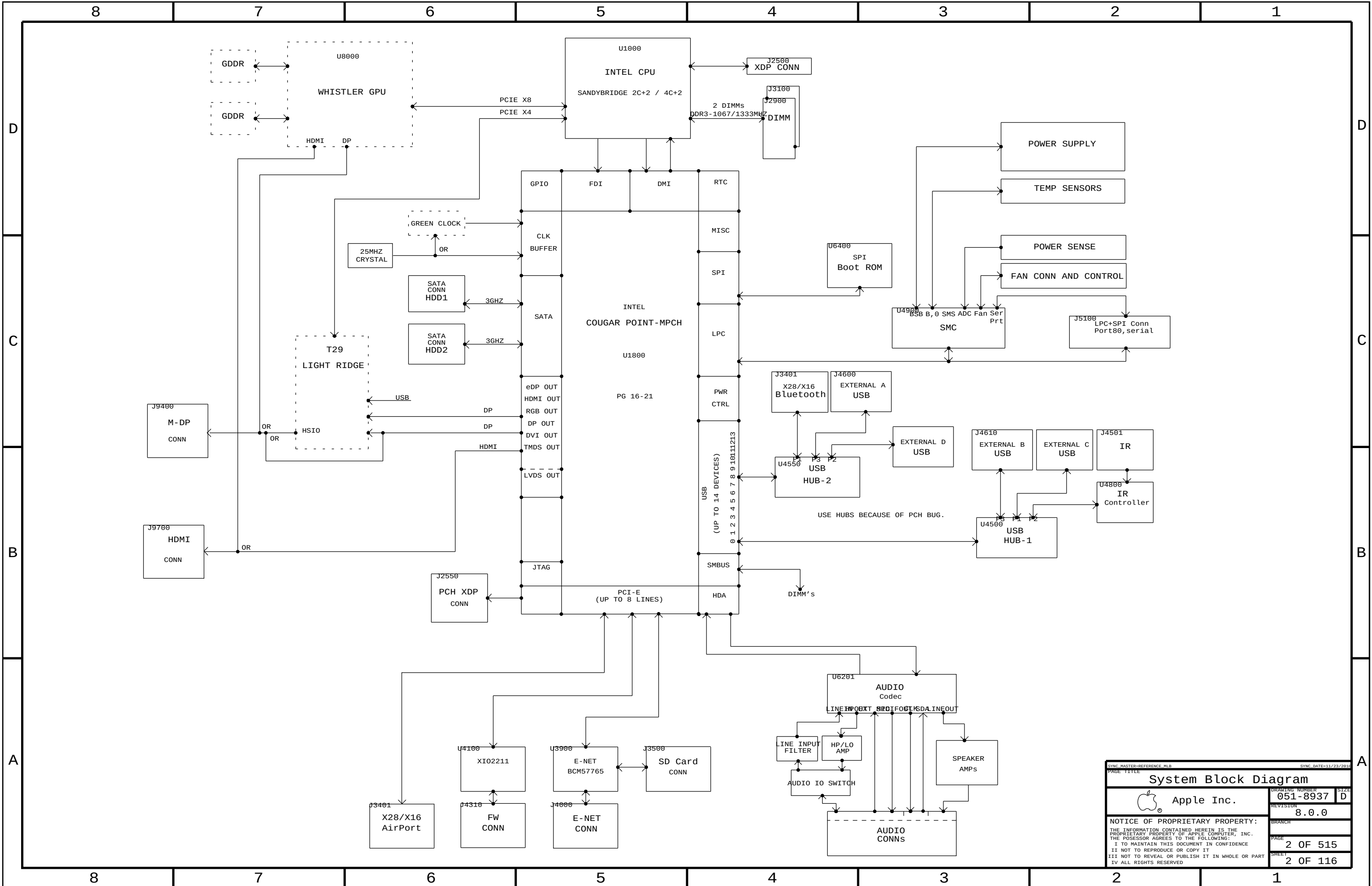
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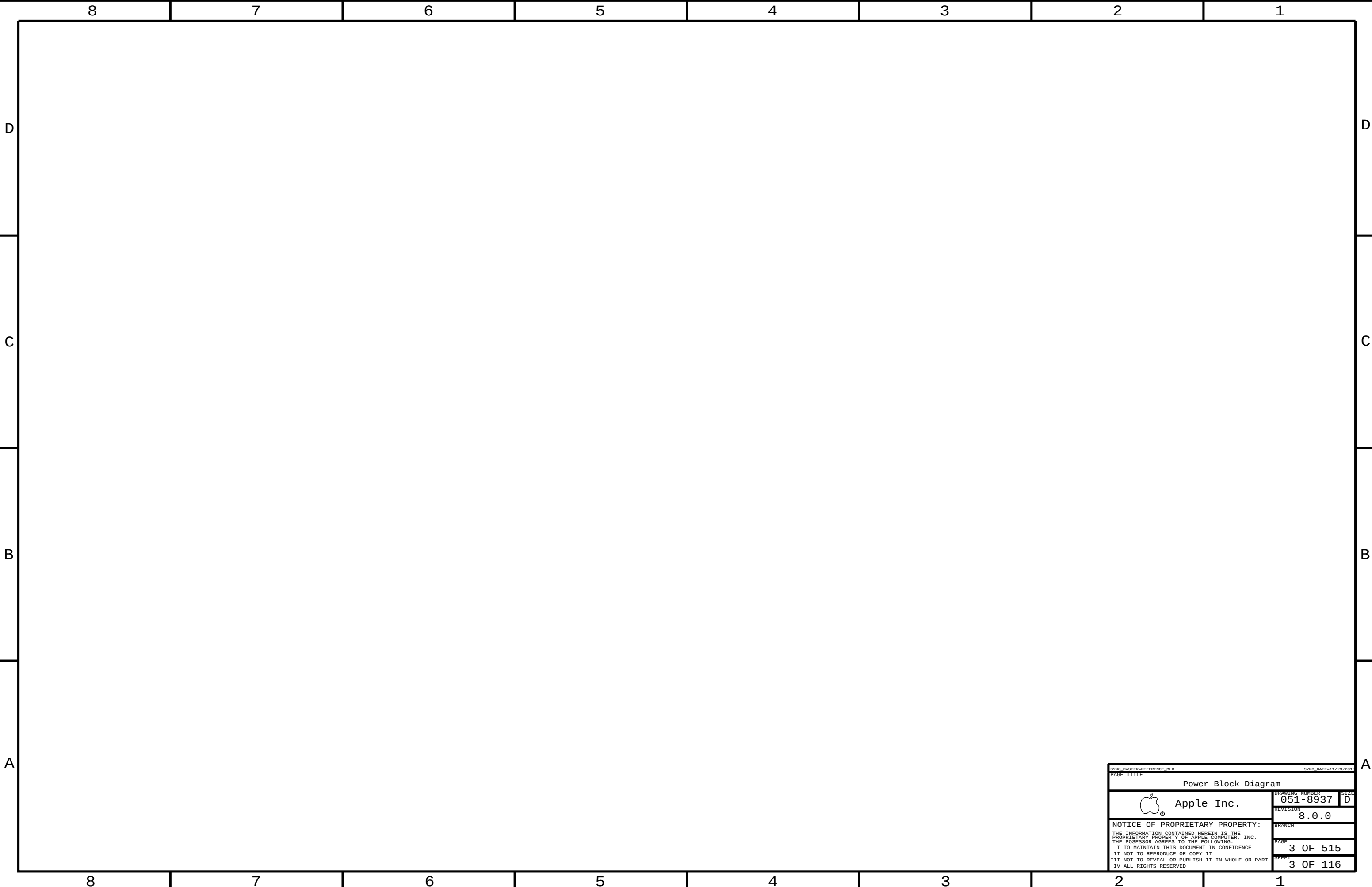
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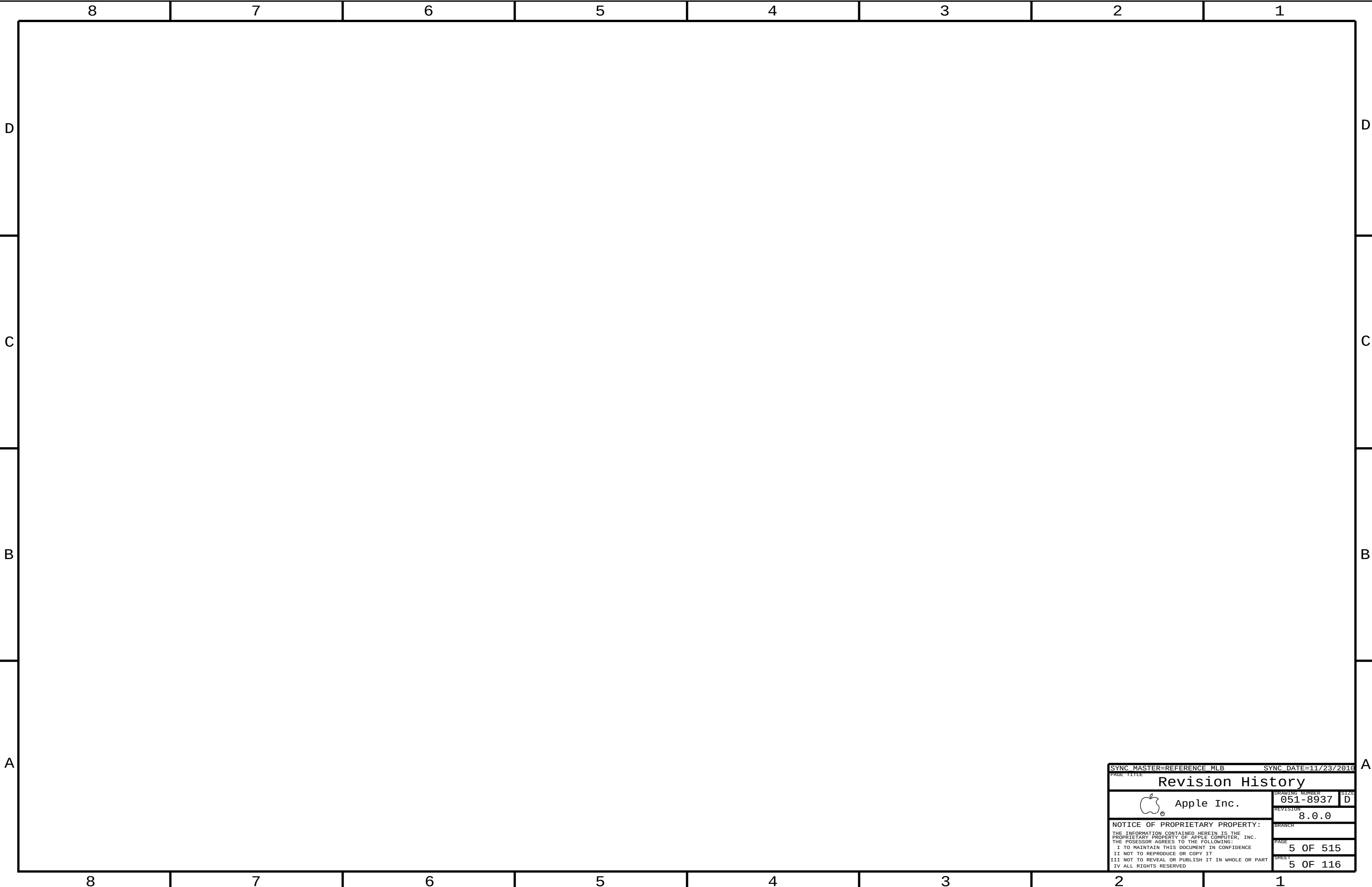
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|                                                                                                                                                                                                                                                                                                                             |                |          |
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| System Block Diagram                                                                                                                                                                                                                                                                                                        |                |          |
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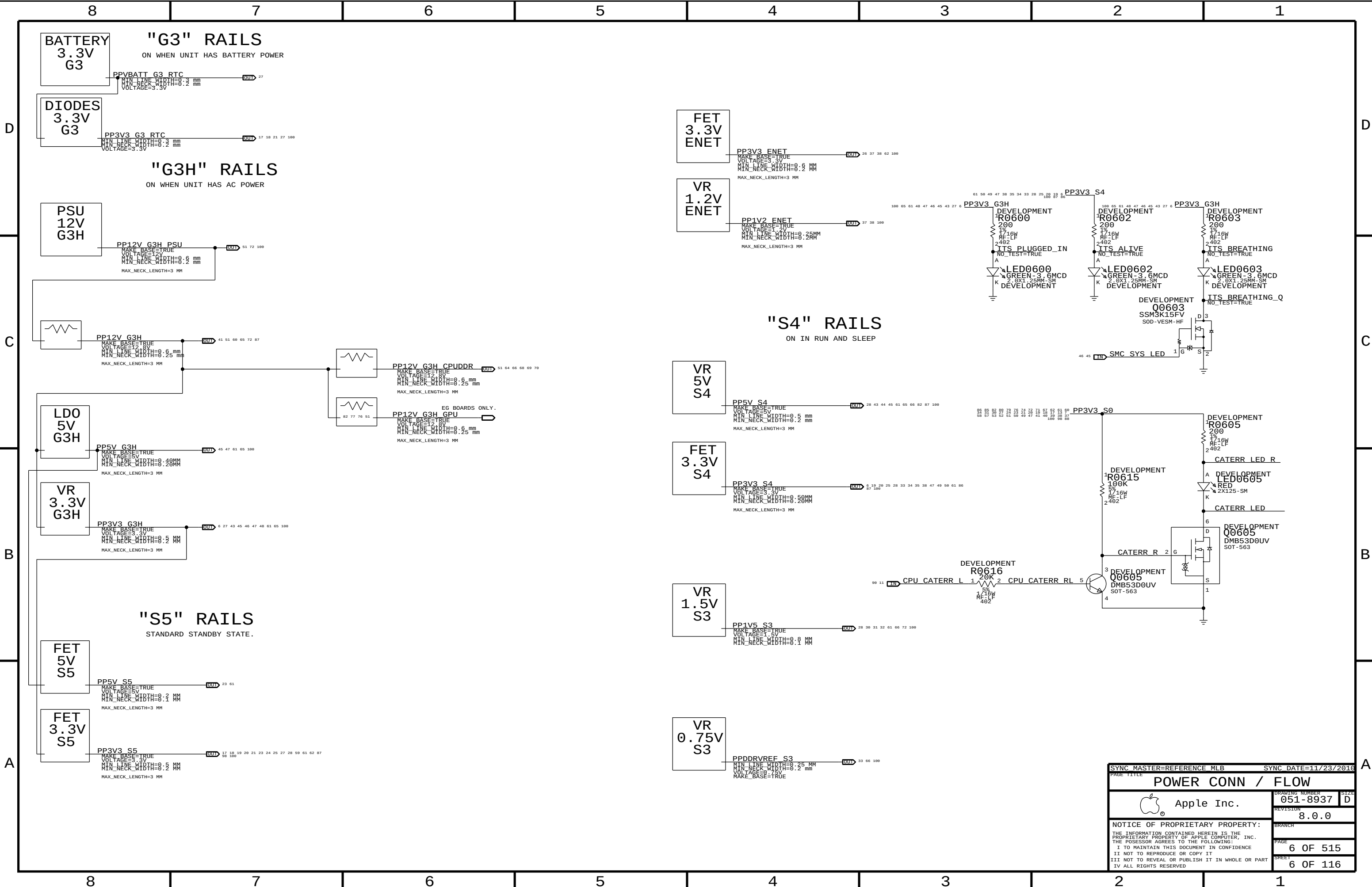
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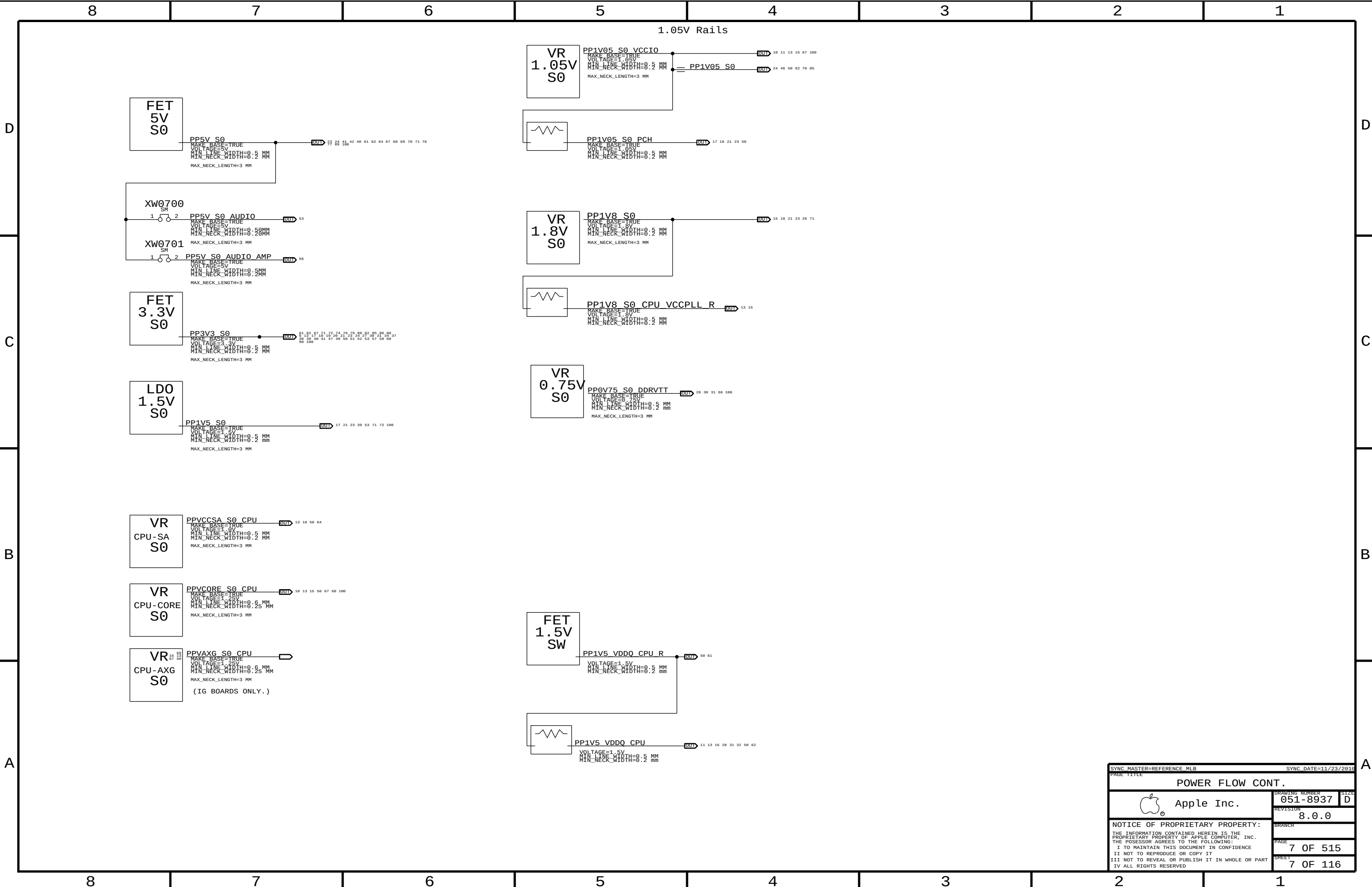
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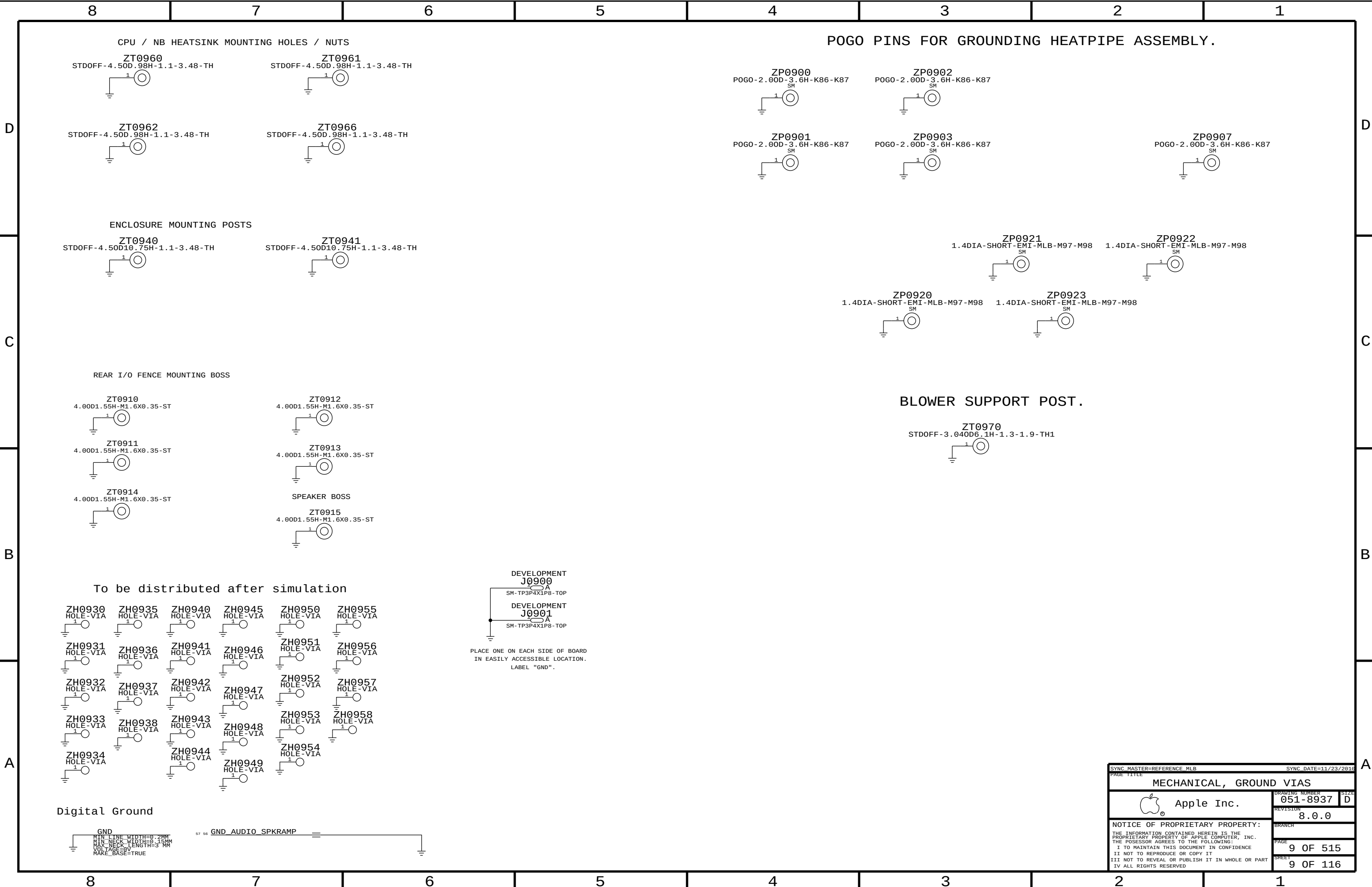
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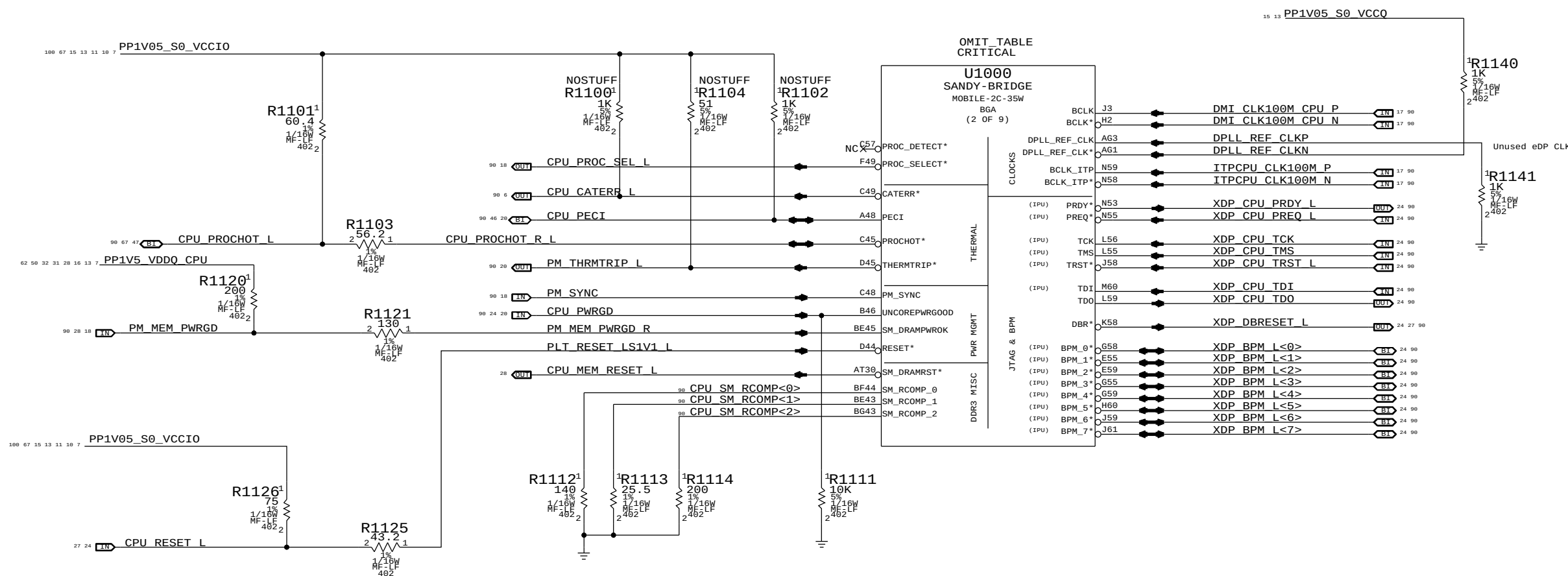


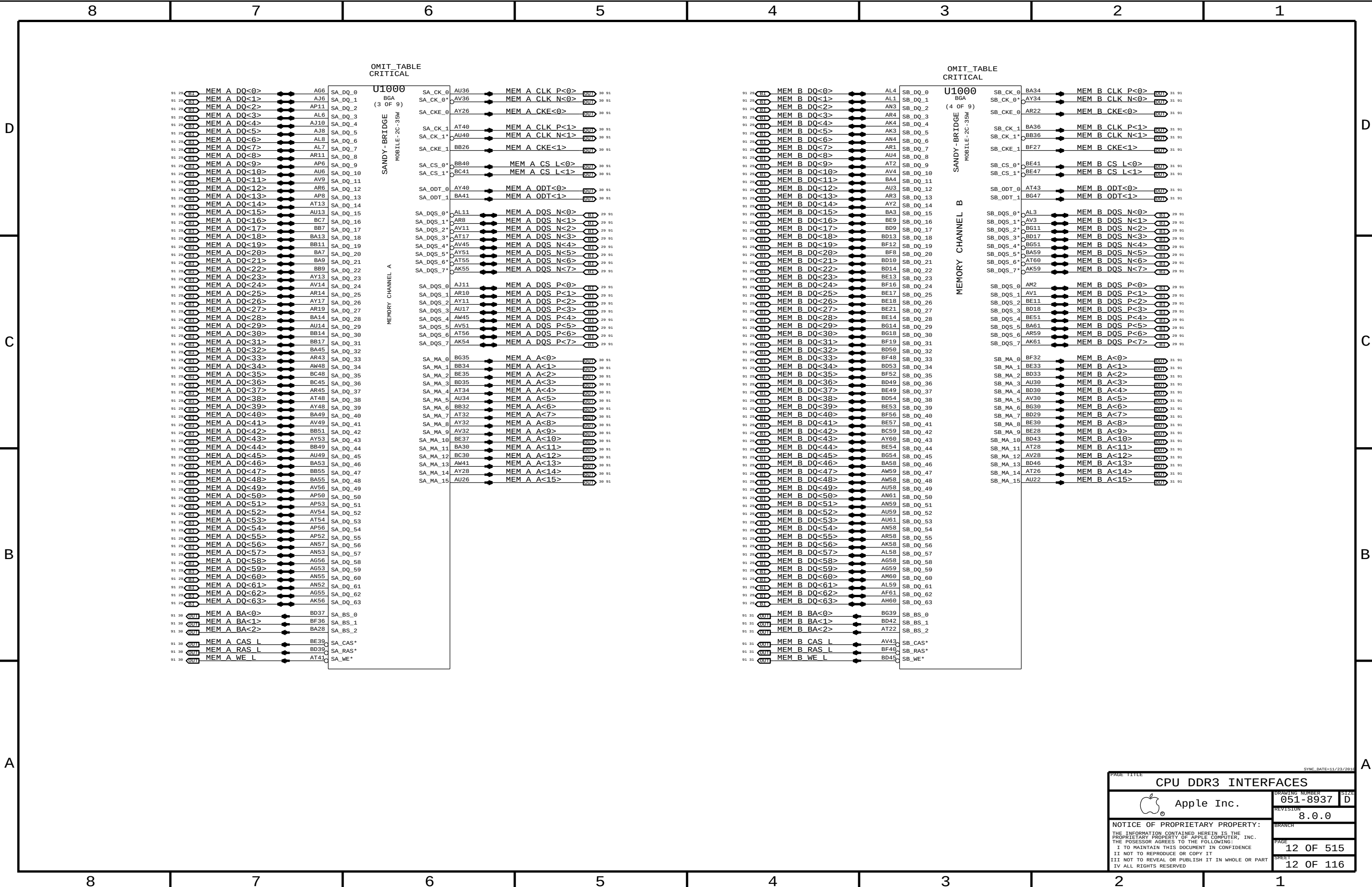


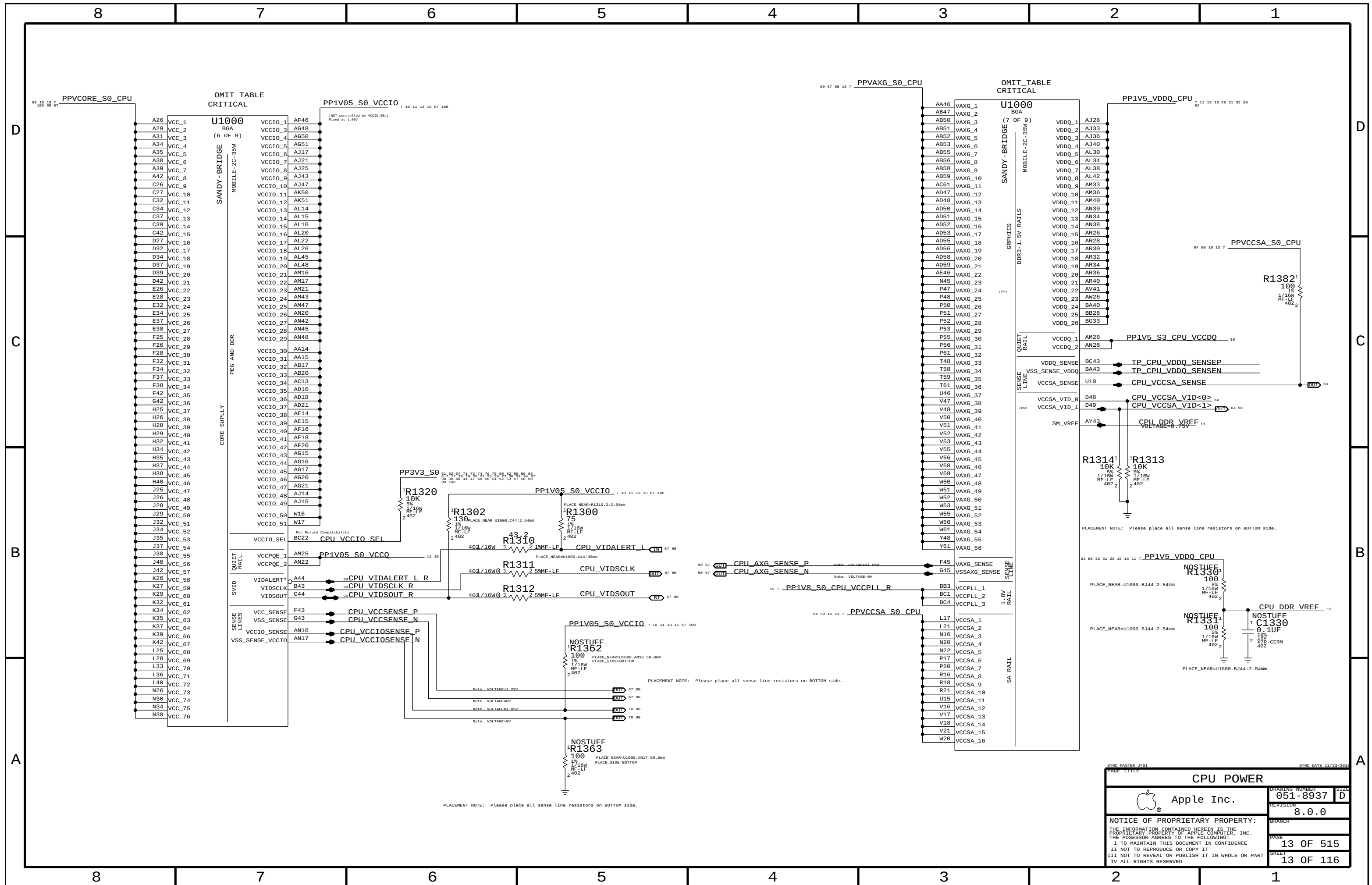


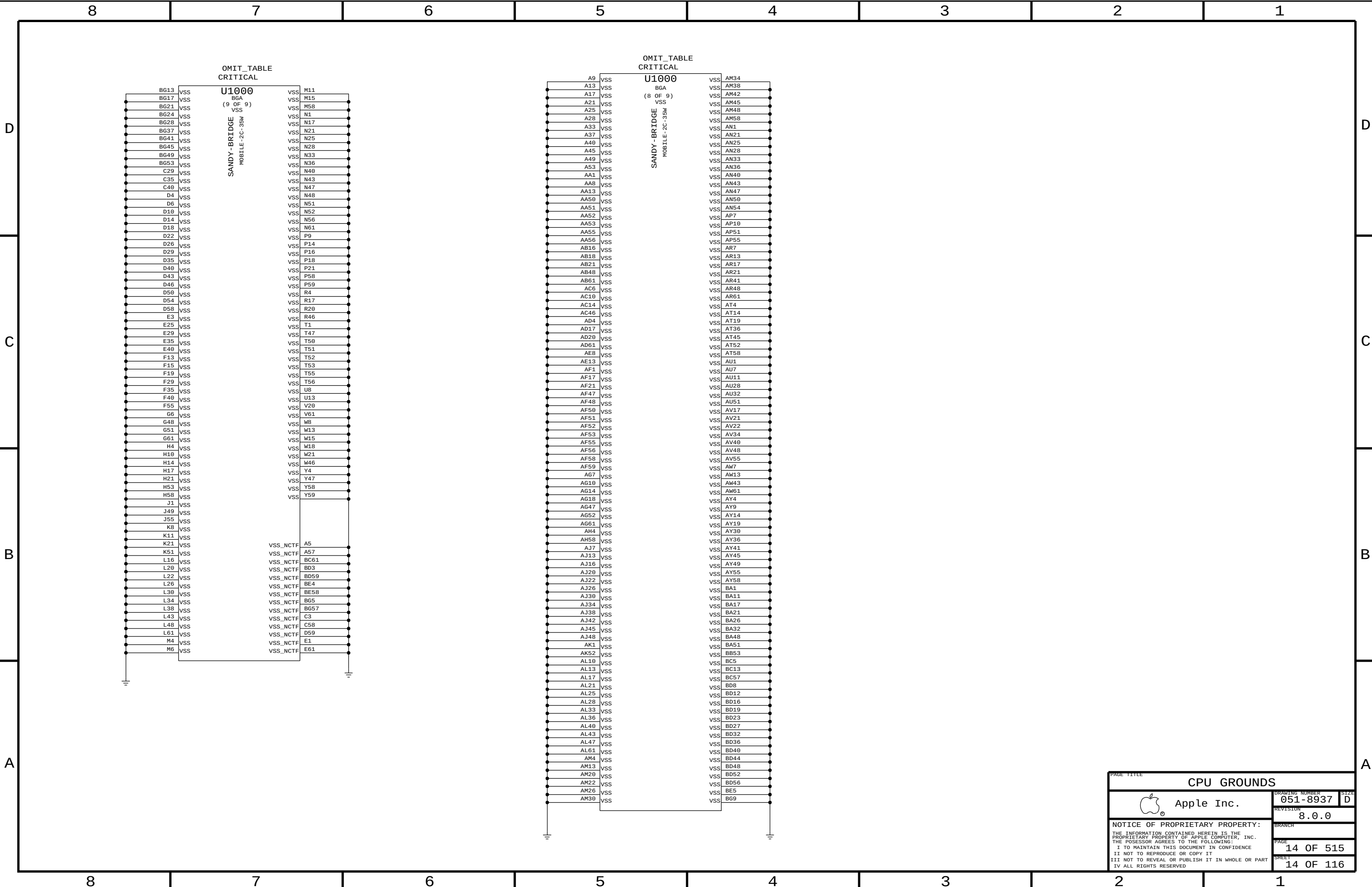
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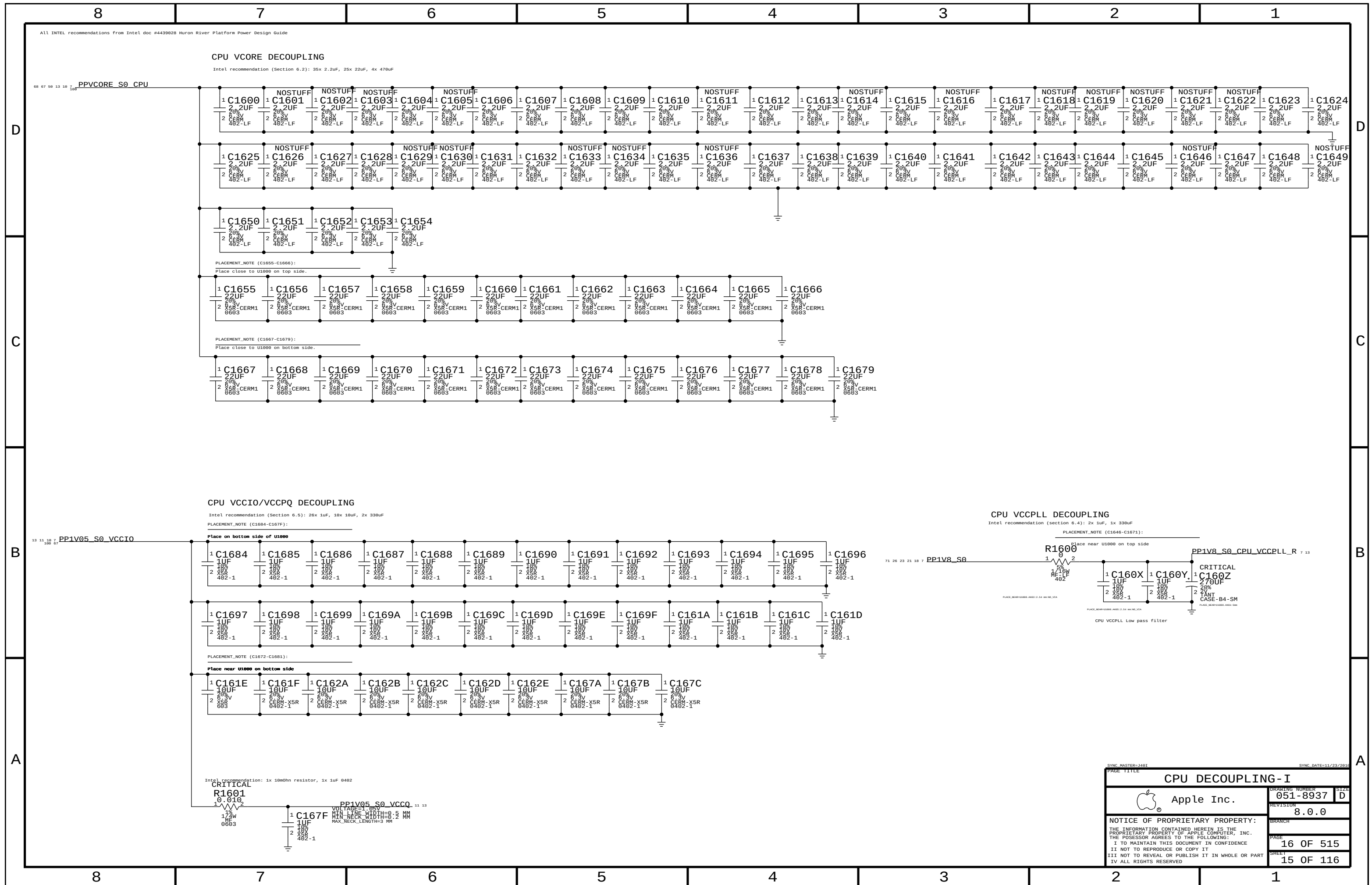


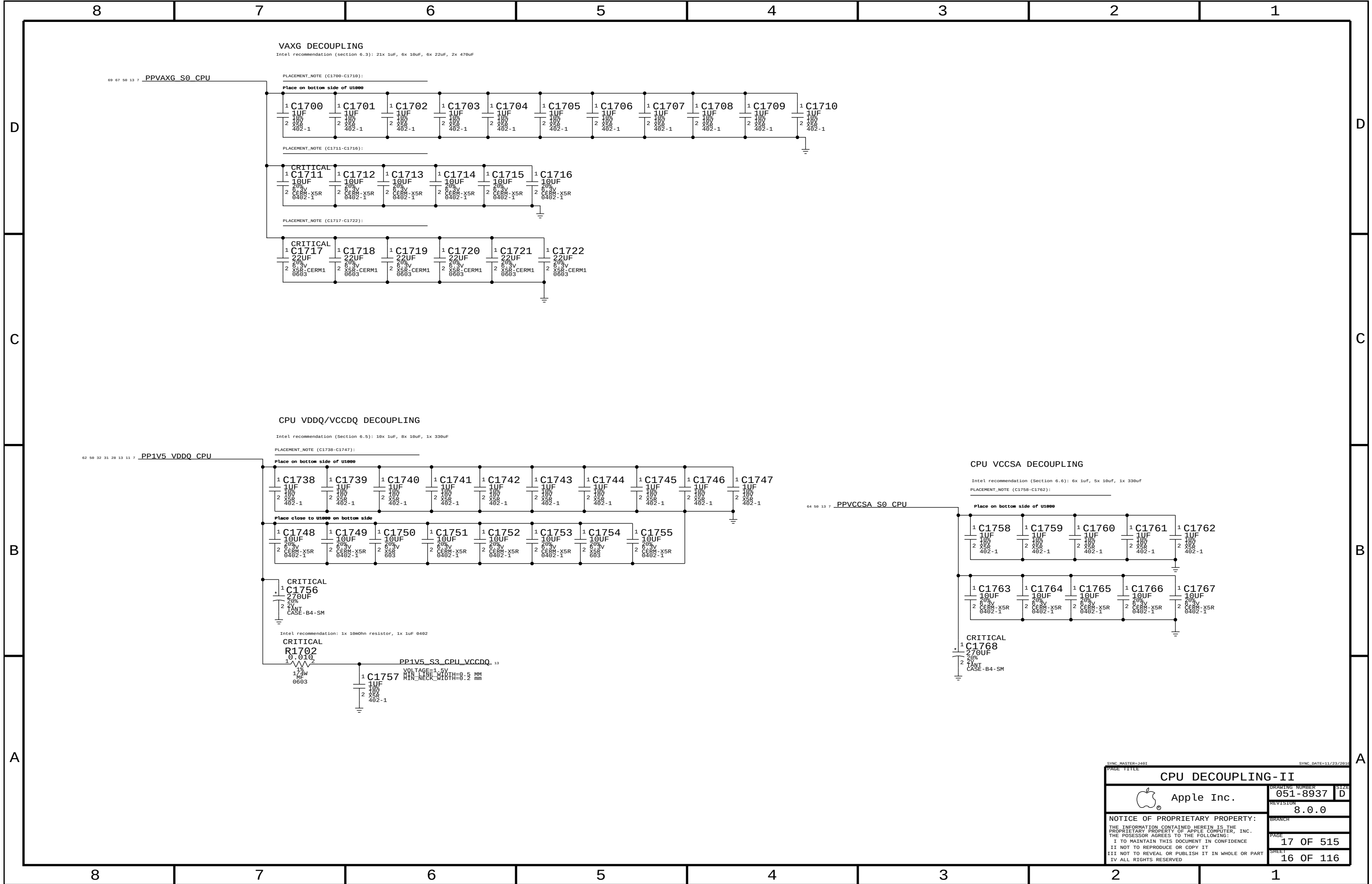






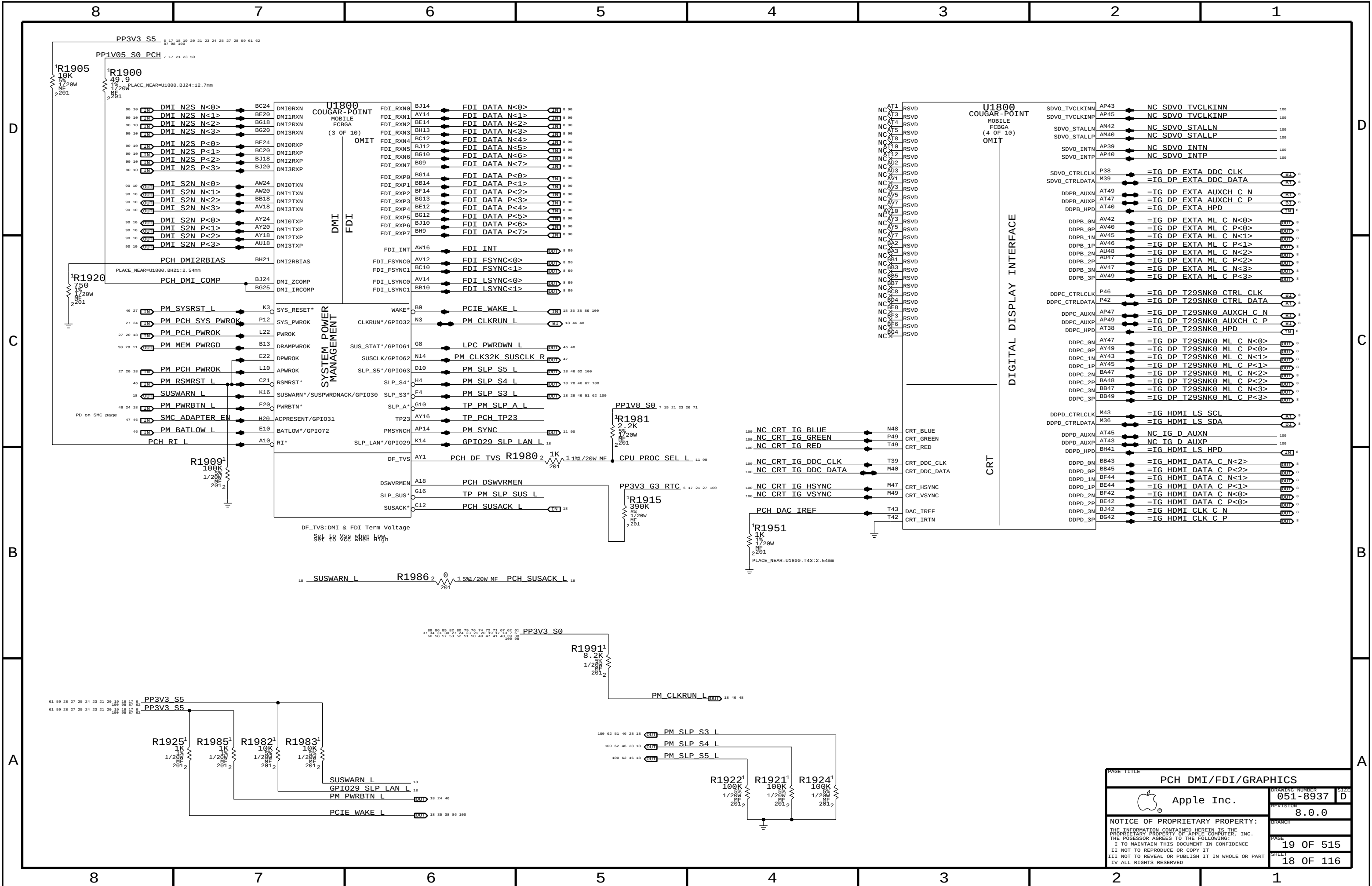


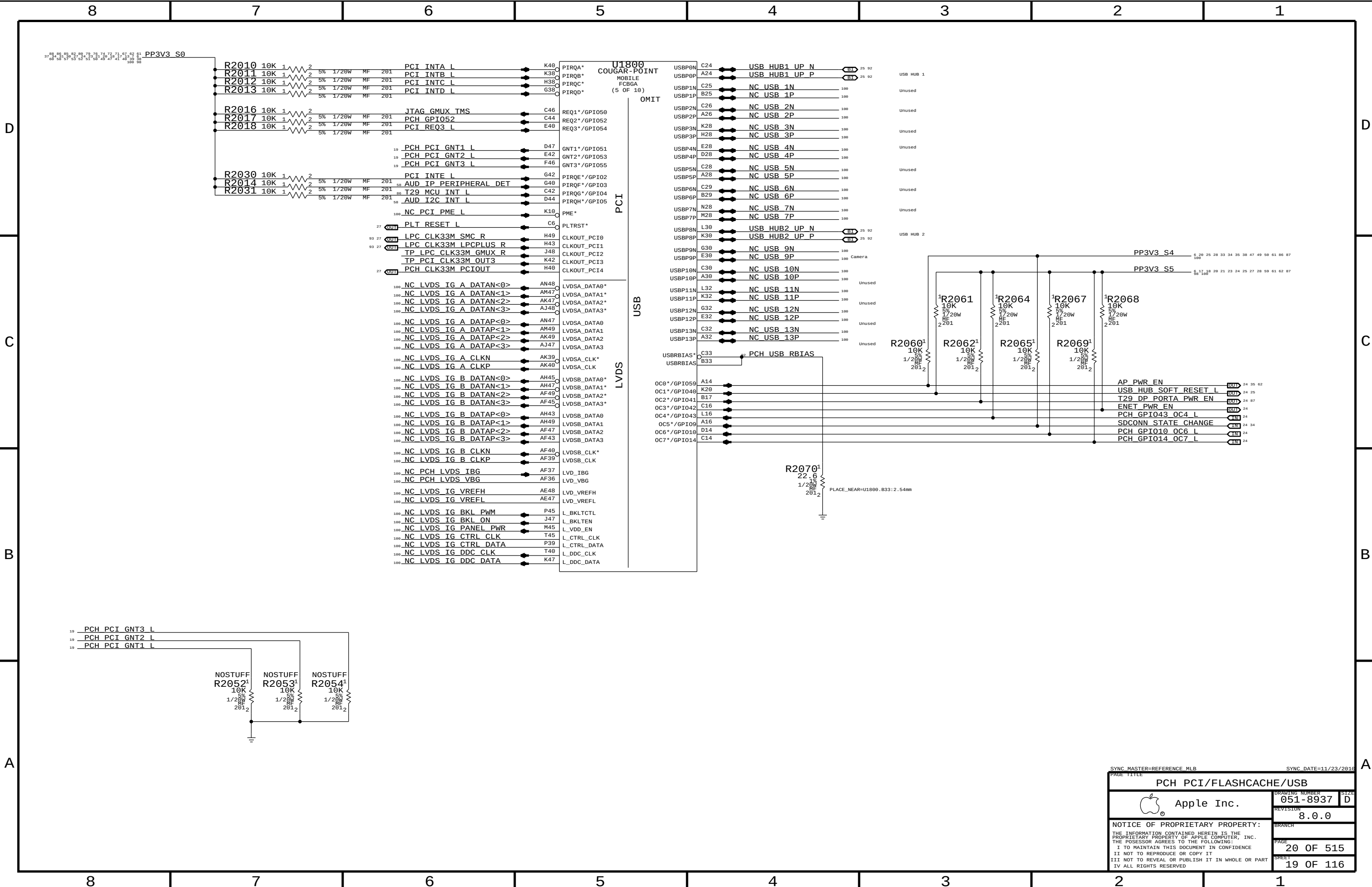


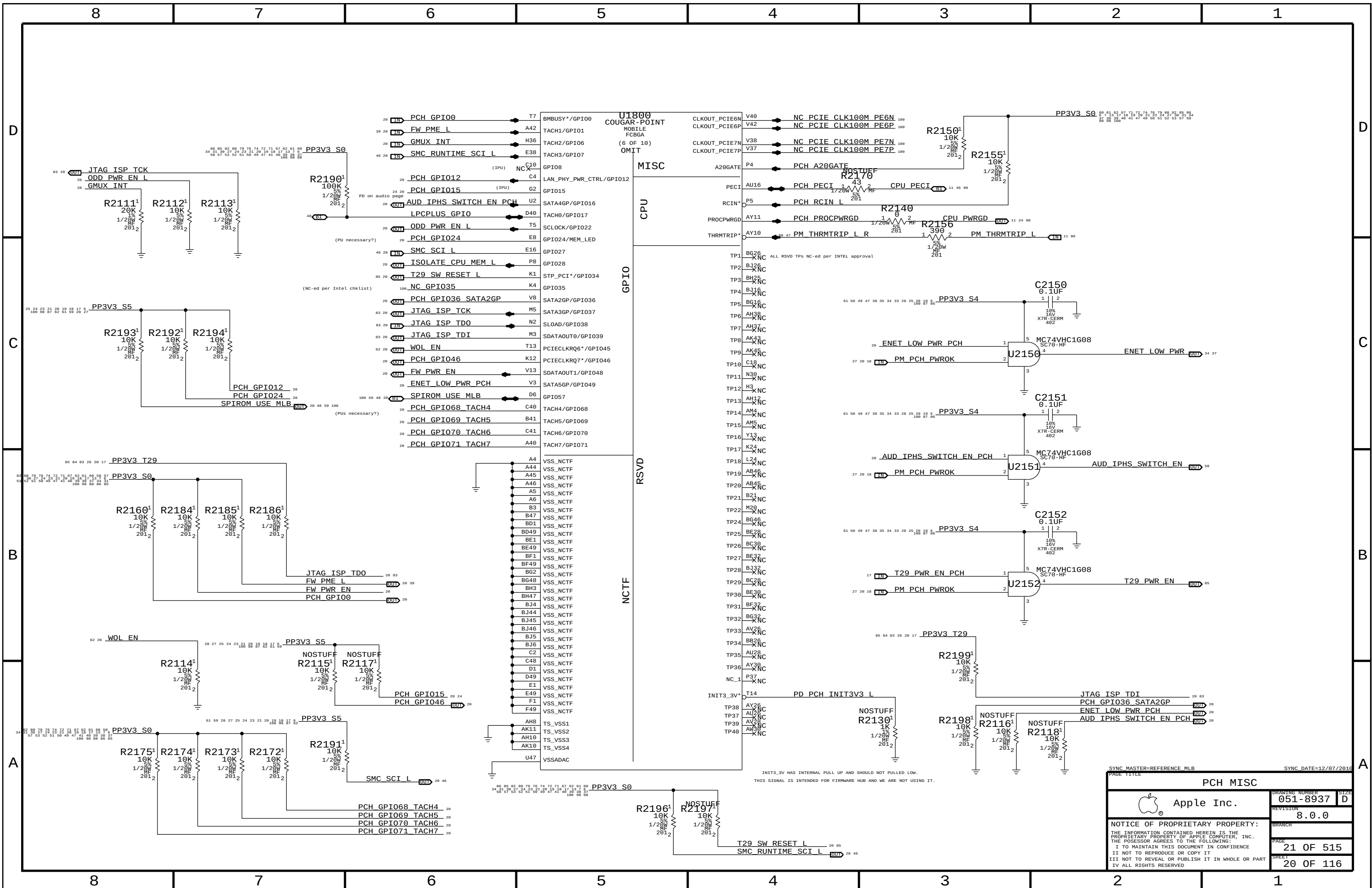




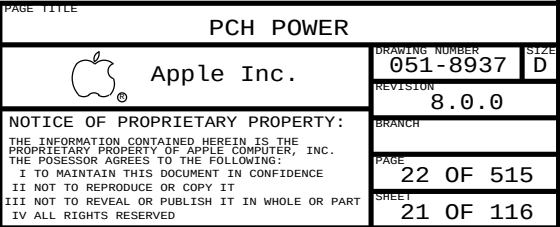


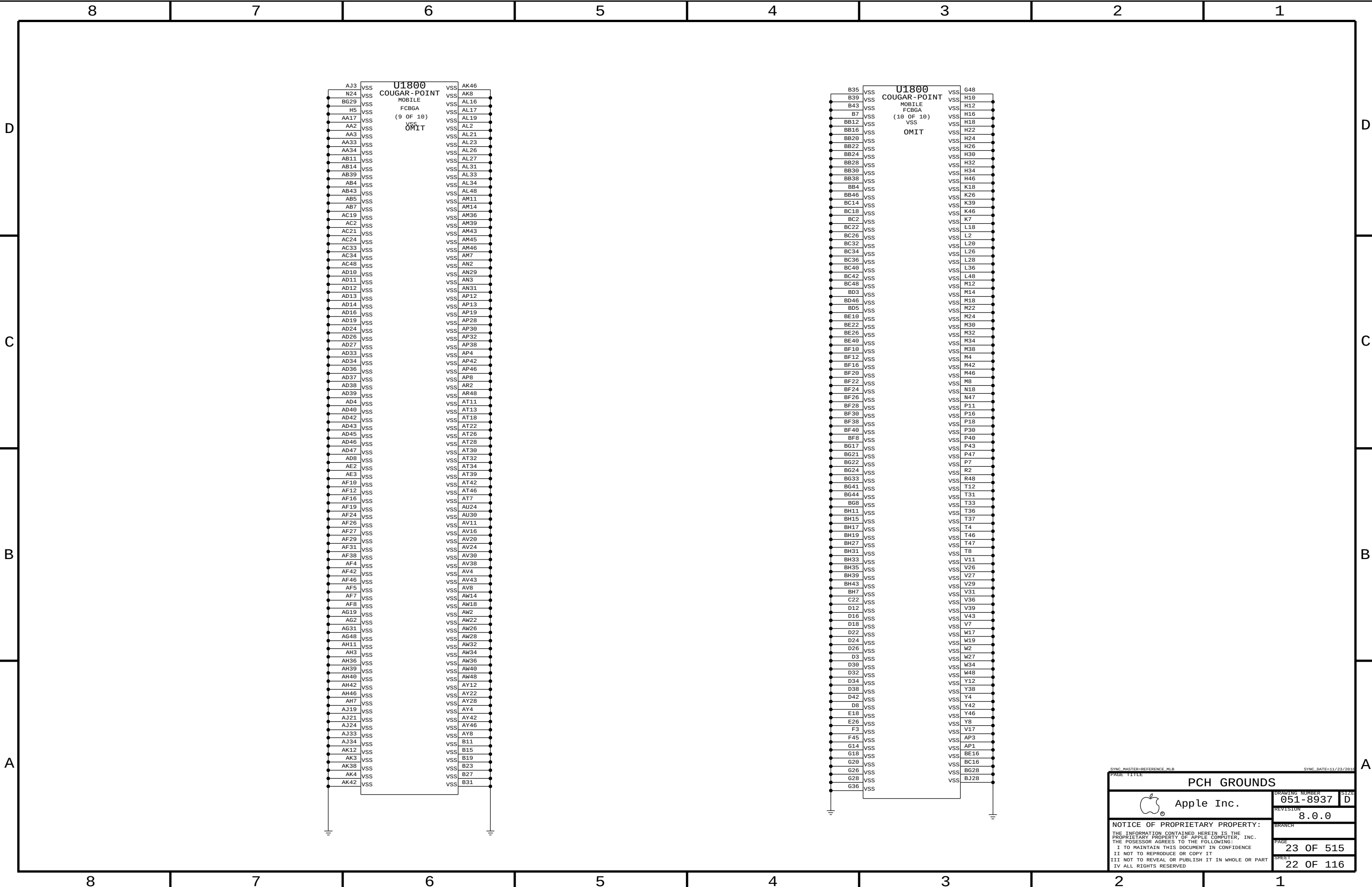






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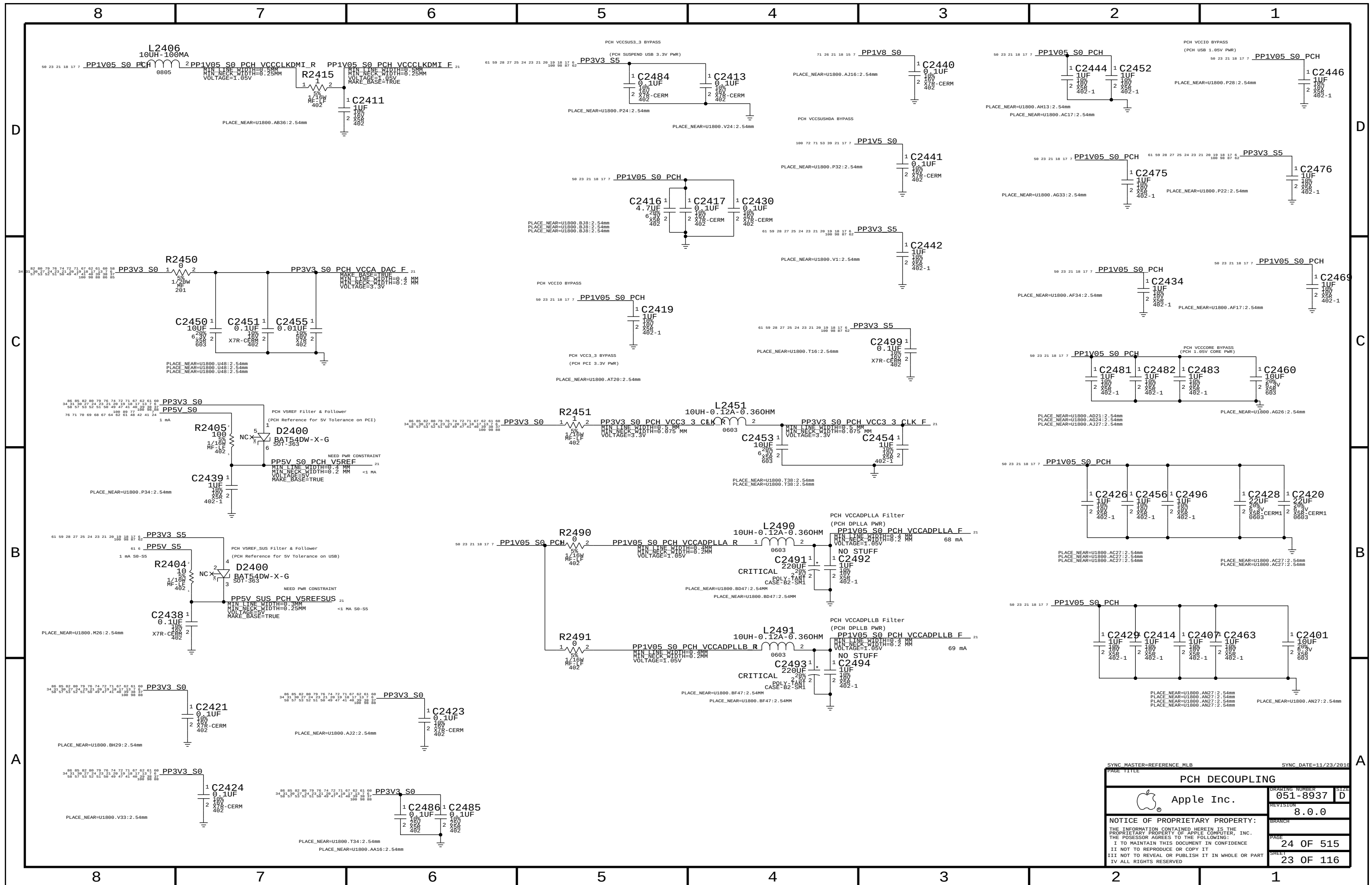


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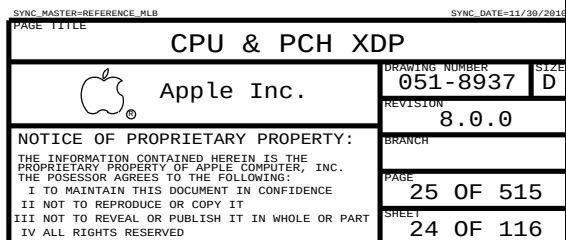
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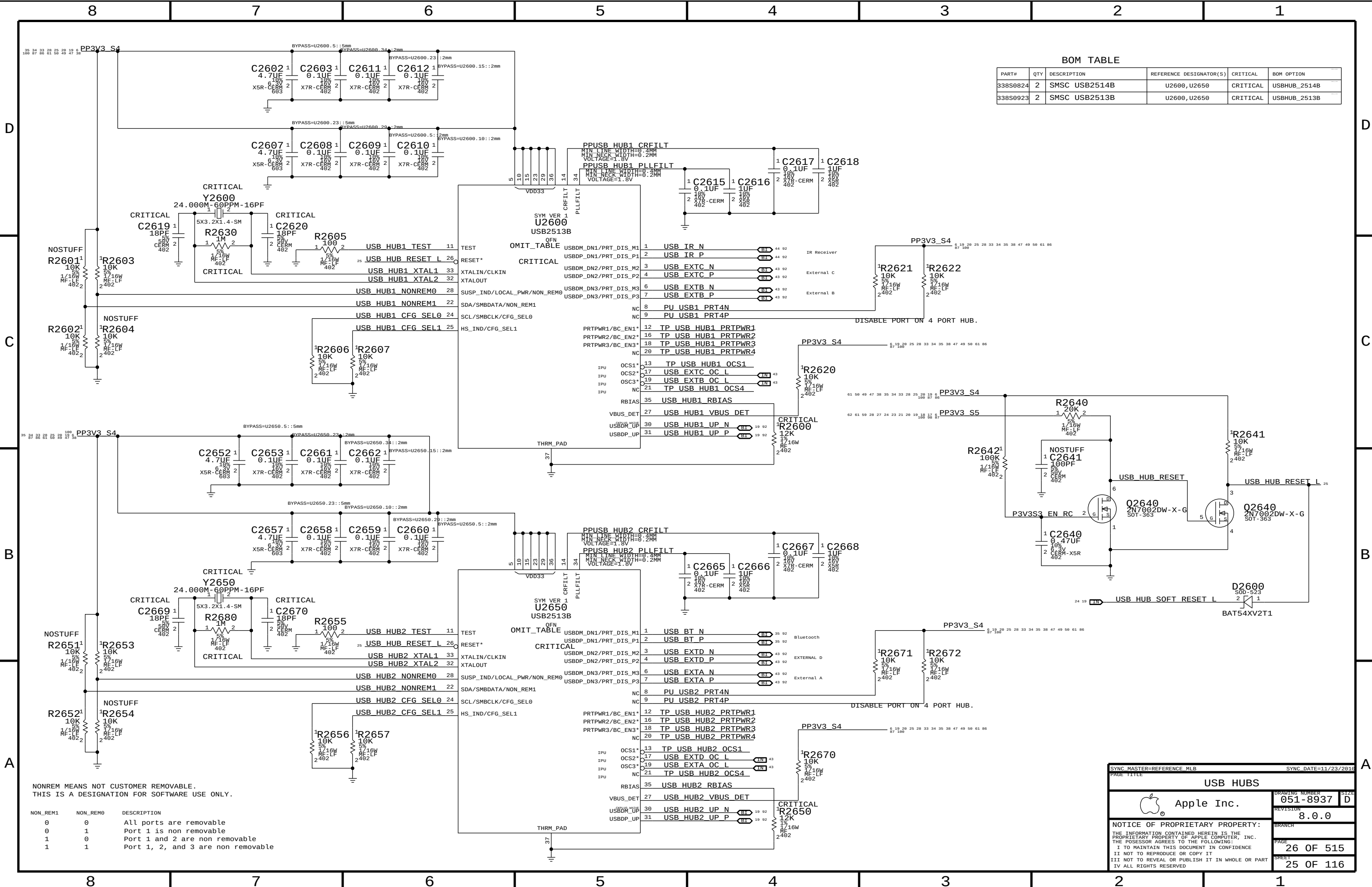




NOTE: This is not the standard XDP pinout.  
USE WITH 920-0620 ADAPTER BOARD TO SUPPORT CPU, PCH DEBUGGING.







| BOM TABLE |     |               |                         |          |              |
|-----------|-----|---------------|-------------------------|----------|--------------|
| PART#     | QTY | DESCRIPTION   | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION   |
| 338S0824  | 2   | SMSC USB2514B | U2600,U2650             | CRITICAL | USBHUB_2514B |
| 338S0923  | 2   | SMSC USB2513B | U2600,U2650             | CRITICAL | USBHUB_2513B |

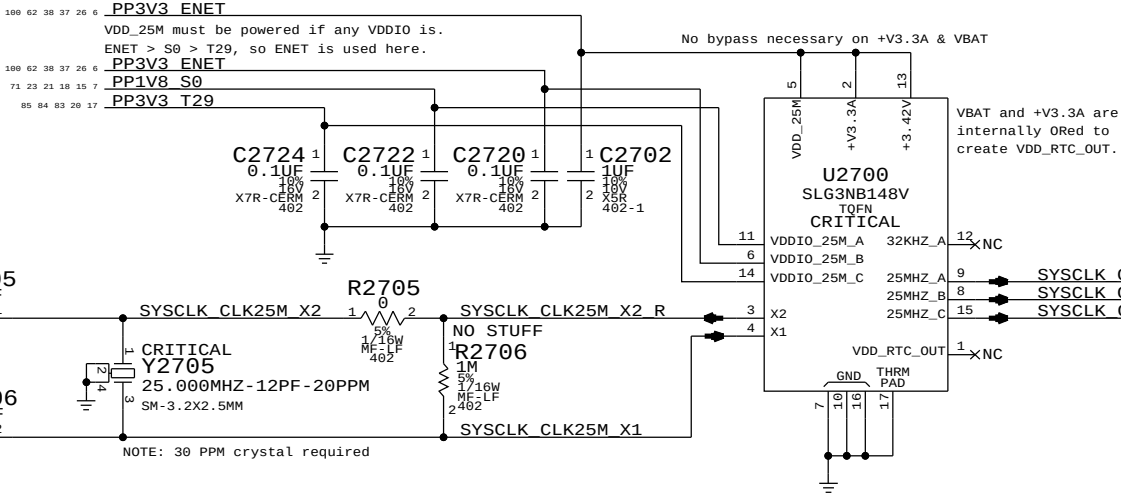
| NONREM MEANS NOT CUSTOMER REMOVABLE. THIS IS A DESIGNATION FOR SOFTWARE USE ONLY. |          |                                    |
|-----------------------------------------------------------------------------------|----------|------------------------------------|
| NON_REM1                                                                          | NON_REM0 | DESCRIPTION                        |
| 0                                                                                 | 0        | All ports are removable            |
| 0                                                                                 | 1        | Port 1 is non removable            |
| 1                                                                                 | 0        | Port 1 and 2 are non removable     |
| 1                                                                                 | 1        | Port 1, 2, and 3 are non removable |

|                                                                                                                                                                                                                                                                                                                             |  |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=REFERENCE_MLB                                                                                                                                                                                                                                                                                                   |  | SYNC DATE=11/23/2010 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  | USB HUBS             |           |
|                                                                                                                                                                                                                                                                                                                             |  | DRAWING NUMBER       | 051-8937  |
|                                                                                                                                                                                                                                                                                                                             |  | REVISION             | 8.0.0     |
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|                                                                                                                                                                                                                                                                                                                             |  | PAGE                 | 26 OF 515 |
|                                                                                                                                                                                                                                                                                                                             |  | SHEET                | 25 OF 116 |

System 25MHz Clock Generator

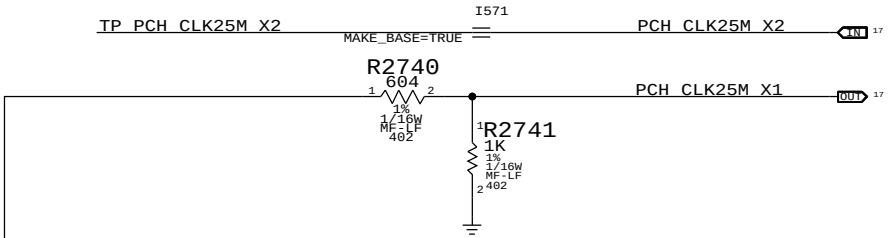
Low-Power 32kHz Source (not used)  
Normal-Power 32kHz Source (not used)  
GreenClk 25MHz Power

Ethernet XTAL Power  
SB XTAL Power  
T29 XTAL Power

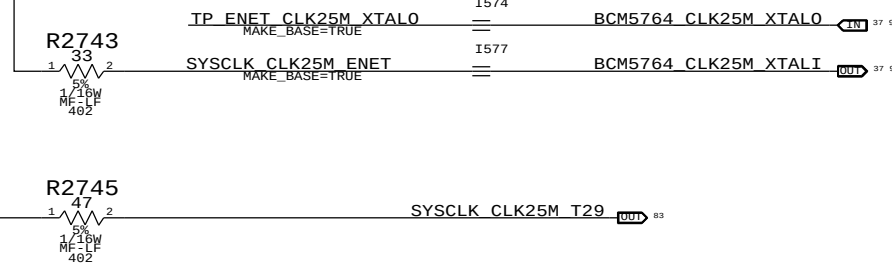


J40 NOTE: WILL NOT BE USING RTC POWER OR CLK FEATURES OF THIS CHIP DUE TO EXTRA COIN CELL LOAD  
Working with Silego on a simplified version without this support

COUGAR POINT (PCH) 25MHZ CRYSTAL INPUT

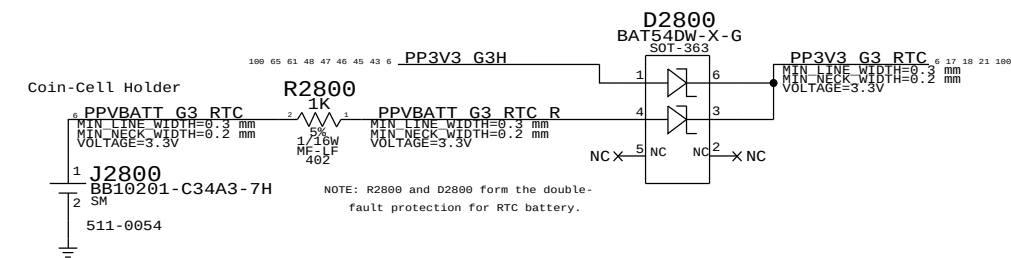


CAESAR IV (ENET) 25MHZ CRYSTAL INPUT



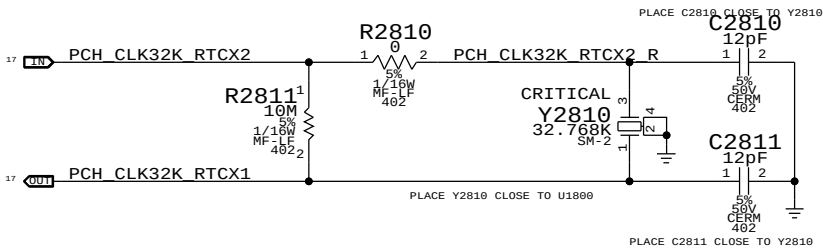
GREEN CLK

RTC Power Sources

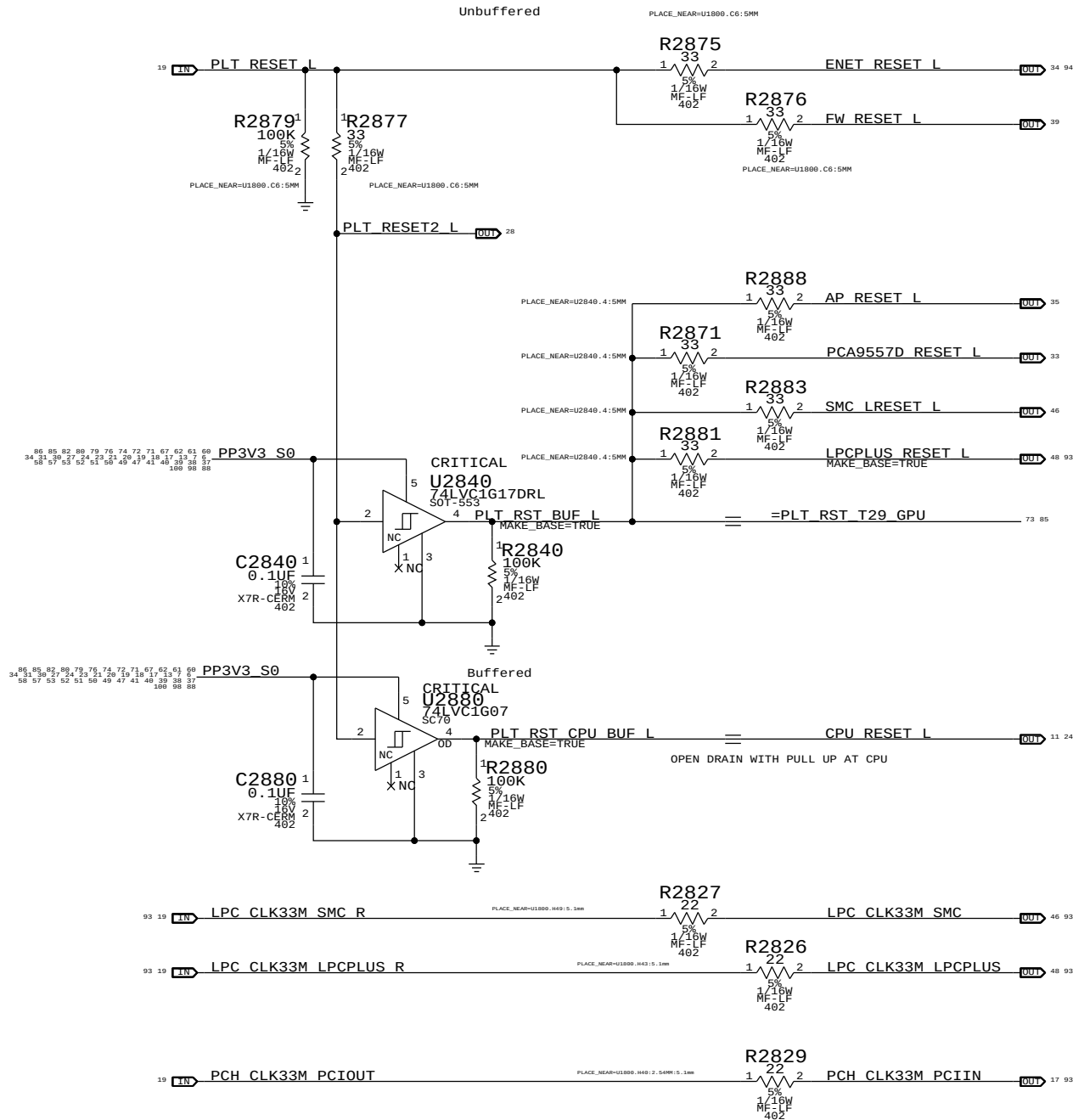


| PART NUMBER | QTY | DESCRIPTION                         | REFERENCE DES | CRITICAL | BOM OPTION |
|-------------|-----|-------------------------------------|---------------|----------|------------|
| 742-0062    | 1   | BAT, COIN, LICF, BR2032, 3V, 190MAH | BAT           | CRITICAL |            |

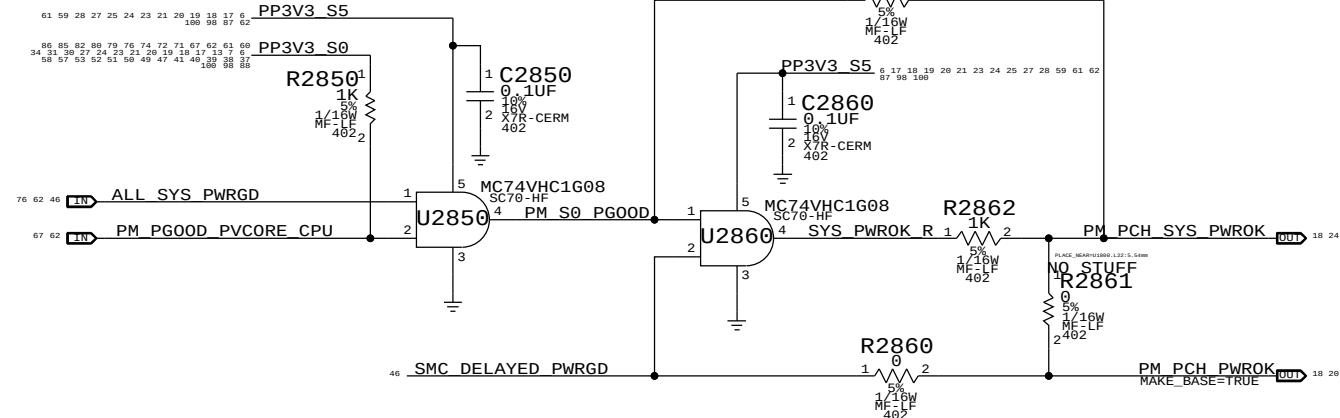
PCH RTC Crystal



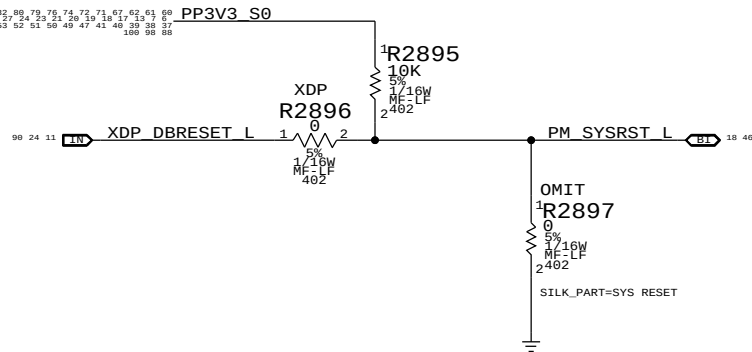
Platform Reset Connections



PCH S0 PWRGD



PCH Reset Button



|                                                                                                                             |  |             |  |
|-----------------------------------------------------------------------------------------------------------------------------|--|-------------|--|
| PAGE TITLE                                                                                                                  |  | PAGE NUMBER |  |
| RTC SUPPORT, RESETS                                                                                                         |  | 051-8937    |  |
| Apple Inc.                                                                                                                  |  | REVISION    |  |
|                                                                                                                             |  | 8.0.0       |  |
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The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM\_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE\_CPU\_MEM\_L GPIO state during S3->S0 transitions determines behavior of signals.

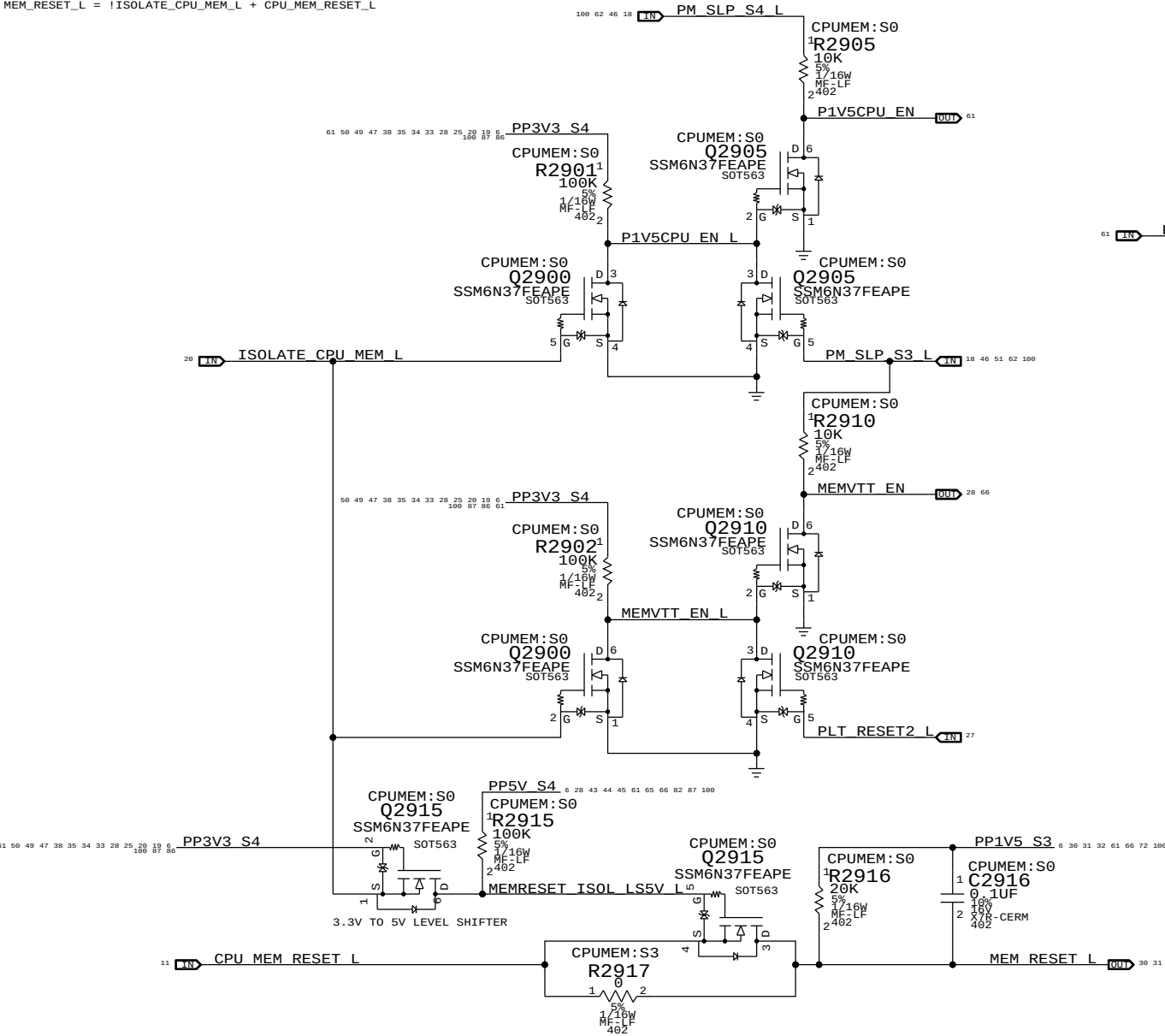
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM\_RESET\_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM\_RESET\_L isolated.

P1V5CPU\_EN = (ISOLATE\_CPU\_MEM\_L + PM\_SLP\_S3\_L) \* PM\_SLP\_S4\_L

MEMVTT\_EN = (ISOLATE\_CPU\_MEM\_L + PLT\_RST\_L) \* PM\_SLP\_S3\_L

MEM\_RESET\_L = !ISOLATE\_CPU\_MEM\_L + CPU\_MEM\_RESET\_L

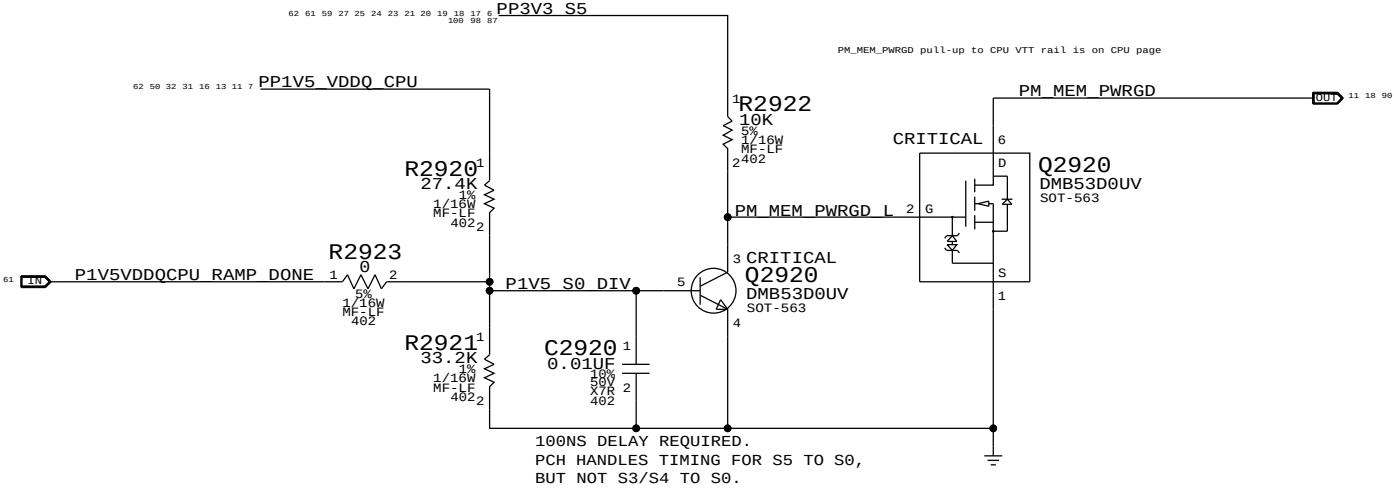


| Step  | ISOLATE_CPU_MEM_L | PLT_RESET_L | PM_SLP_S3_L | PM_SLP_S4_L | CPU_MEM_RESET_L | MEM_RESET_L     | MEMVTT_EN | P1V5CPU_EN |
|-------|-------------------|-------------|-------------|-------------|-----------------|-----------------|-----------|------------|
| S0    | 0                 | 1           | 1           | 1           | 1               | CPU_MEM_RESET_L | 1         | 1          |
|       | 1                 | 0           | 1           | 1           | 1               | 1               | 1         | 1          |
|       | 2                 | 0           | 0           | 1           | 1               | 1               | 0         | 1          |
| to S3 | 3                 | 0           | 0           | 1           | X               | 1               | 0         | 0          |
|       | 4                 | 0           | 0           | 1           | 1               | X (*)           | 1         | 1          |
|       | 5                 | 0           | 1           | 1           | 1               | 1               | 1         | 1          |
| S0    | 6                 | 0           | 1           | 1           | 1               | 1               | 1         | 1          |
|       | 7                 | 1           | 1           | 1           | 1               | CPU_MEM_RESET_L | 1         | 1          |

(\*) CPU\_MEM\_RESET\_L asserts due to loss of PM\_MEM\_PWRGD, must wait for software to clear before deasserting ISOLATE\_CPU\_MEM\_L GPIO.

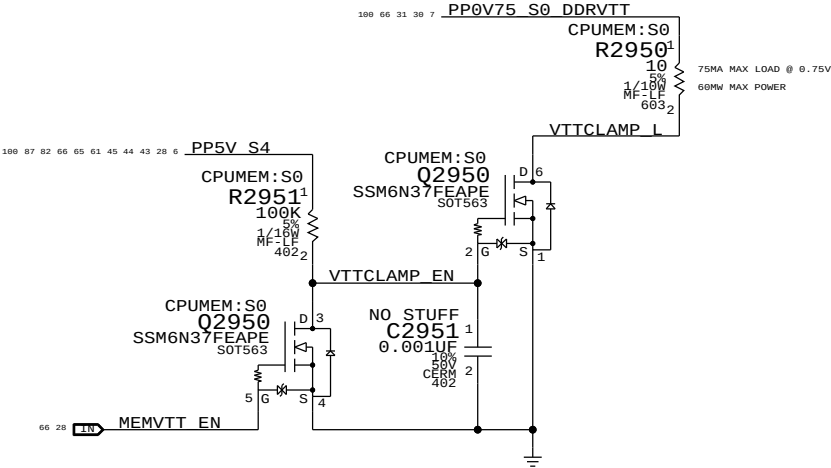
NOTE: In the event of a S3->S5 transition ISOLATE\_CPU\_MEM\_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM\_RESET\_L will not properly assert. Software must deassert ISOLATE\_CPU\_MEM\_L and then generate a valid reset cycle on CPU\_MEM\_RESET\_L.

### 1V5 S0 "PGOOD" for CPU



### MEMVTT Clamp

Ensures CKE signals are held low in S3



SYNC MASTER-REFERENCE: HLB

SYNC DATE: 11/23/2011

CPU Memory S3 Support

Apple Inc.

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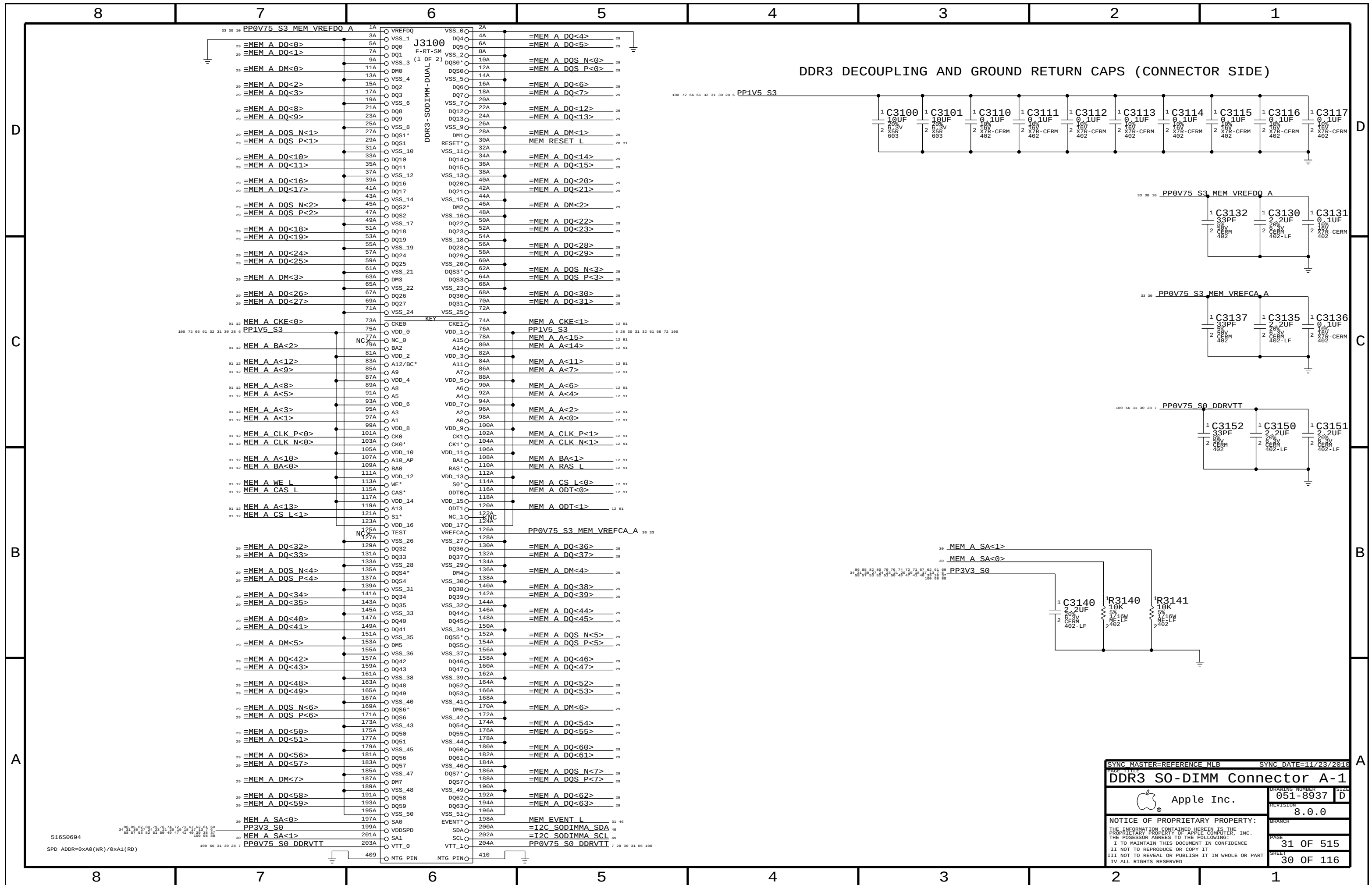
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SIZE  
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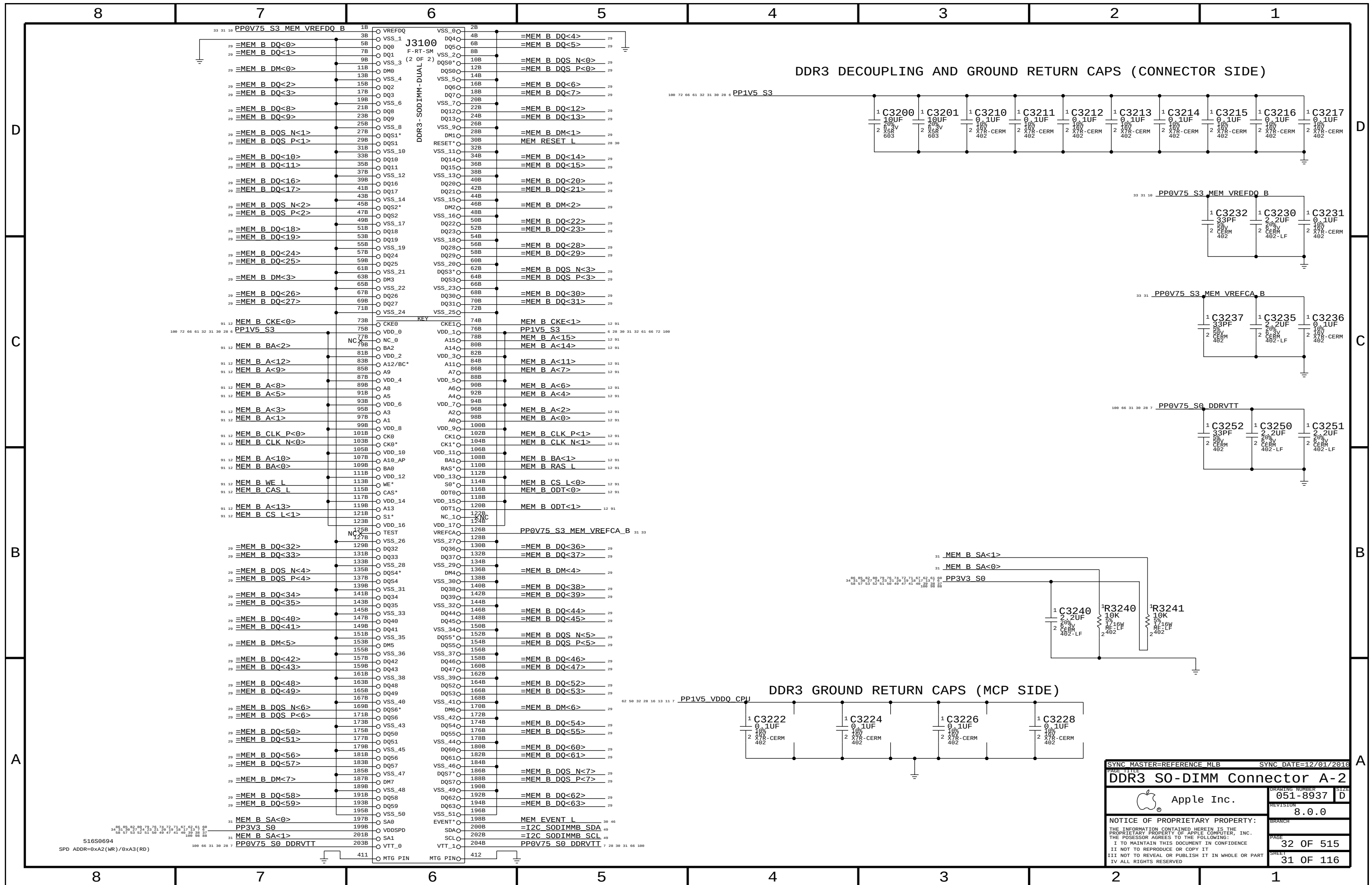


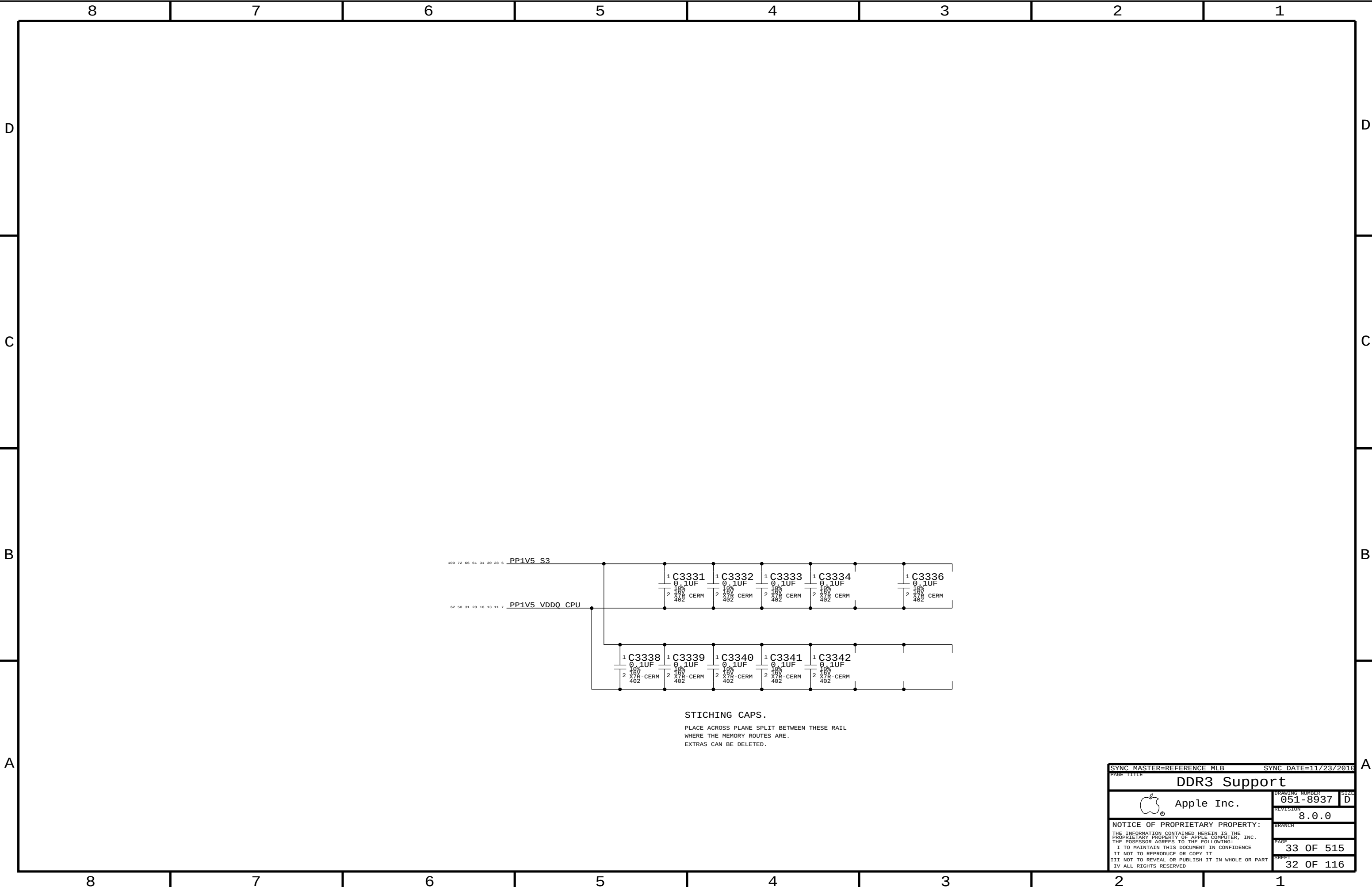
SANDYBRIDGE CPU DOES NOT USE PER PDG, THE DM BITS TO GND.

|                                                                                                                             |  |                      |          |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|----------|
| SYNC MASTER=REFERENCE_MLB                                                                                                   |  | SYNC DATE=11/23/2010 |          |
| PAGE TITLE                                                                                                                  |  |                      |          |
| S0-DIMM Pinswaps                                                                                                            |  |                      |          |
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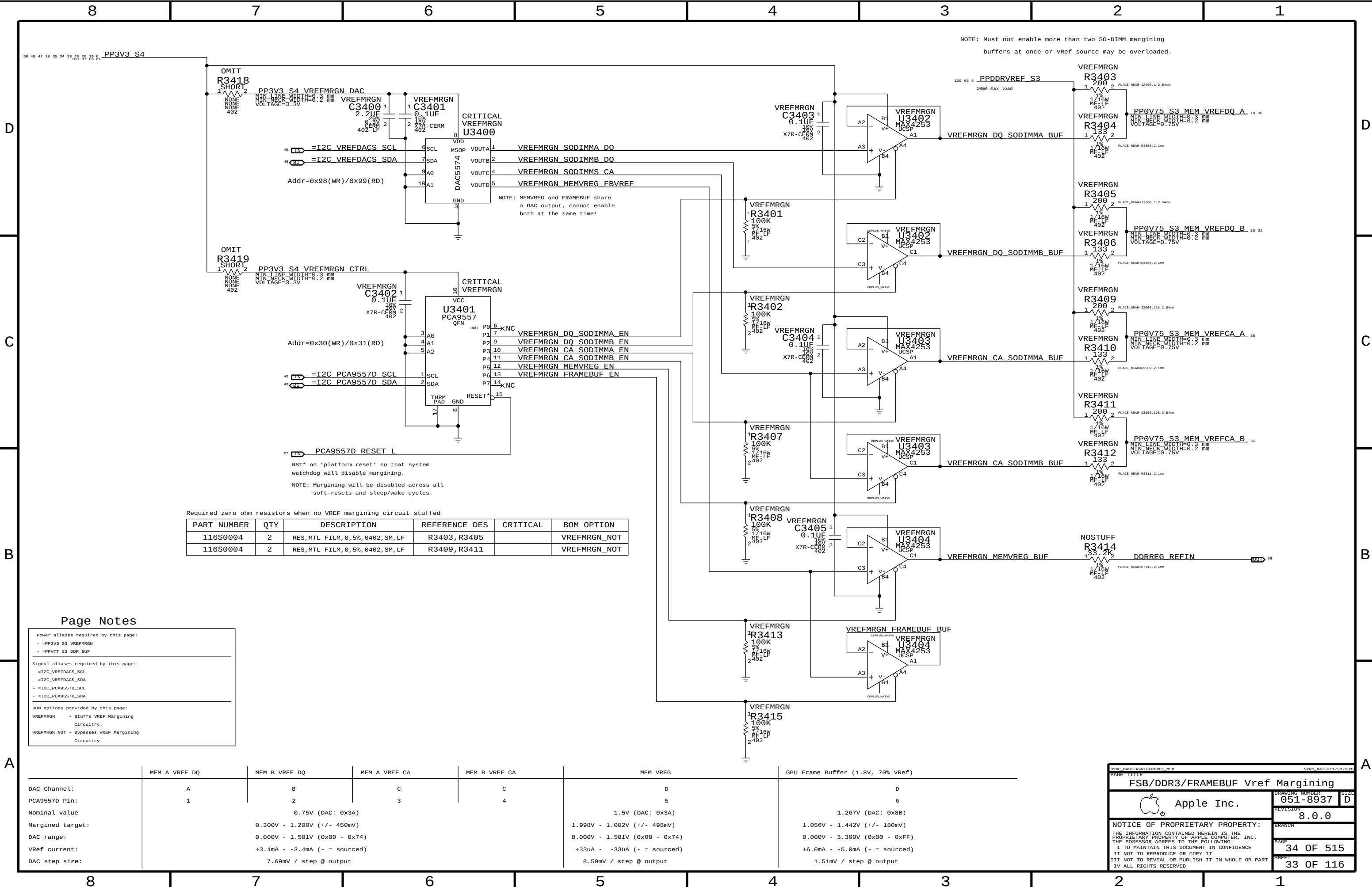






|                                                                                                                                                                                                                                                                                                                                |  |                      |           |
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| SYNC MASTER=REFERENCE MLB                                                                                                                                                                                                                                                                                                      |  | SYNC DATE=11/23/2016 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                     |  |                      |           |
| DDR3 Support                                                                                                                                                                                                                                                                                                                   |  |                      |           |
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Page Notes

Power aliases required by this page:

- =PP3V3\_S3\_VREFMRGN
- =PPVTT\_S3\_DDR\_BUF

Signal aliases required by this page:

- =I2C\_VREFDACS\_SCL
- =I2C\_VREFDACS\_SDA
- =I2C\_PCA9557D\_SCL
- =I2C\_PCA9557D\_SDA

BOM options provided by this page:

- VREFMRGN - Stuffs VREF Margining Circuitry.
- VREFMRGN\_NOT - Bypasses VREF Margining Circuitry.

|                  | MEM A VREF DQ | MEM B VREF DQ                 | MEM A VREF CA | MEM B VREF CA | MEM VREG                      | GPU Frame Buffer (1.8V, 70% Vref) |
|------------------|---------------|-------------------------------|---------------|---------------|-------------------------------|-----------------------------------|
| DAC Channel:     | A             | B                             | C             | C             | D                             | D                                 |
| PCA9557D Pin:    | 1             | 2                             | 3             | 4             | 5                             | 6                                 |
| Nominal value    |               | 0.75V (DAC: 0x3A)             |               |               | 1.5V (DAC: 0x3A)              | 1.267V (DAC: 0x8B)                |
| Margined target: |               | 0.300V - 1.200V (+/- 450mV)   |               |               | 1.998V - 1.002V (+/- 498mV)   | 1.056V - 1.442V (+/- 180mV)       |
| DAC range:       |               | 0.000V - 1.501V (0x00 - 0x74) |               |               | 0.000V - 1.501V (0x00 - 0x74) | 0.000V - 3.300V (0x00 - 0xFF)     |
| VRef current:    |               | +3.4mA - -3.4mA (- = sourced) |               |               | +33uA - -33uA (- = sourced)   | +6.0mA - -5.0mA (- = sourced)     |
| DAC step size:   |               | 7.69mV / step @ output        |               |               | 8.59mV / step @ output        | 1.51mV / step @ output            |

SYNC MASTER-REFERENCE: HLB

SYNC DATE: 11/23/2016

PAGE TITLE

FSB/DDR3/FRAMEBUF Vref Margining

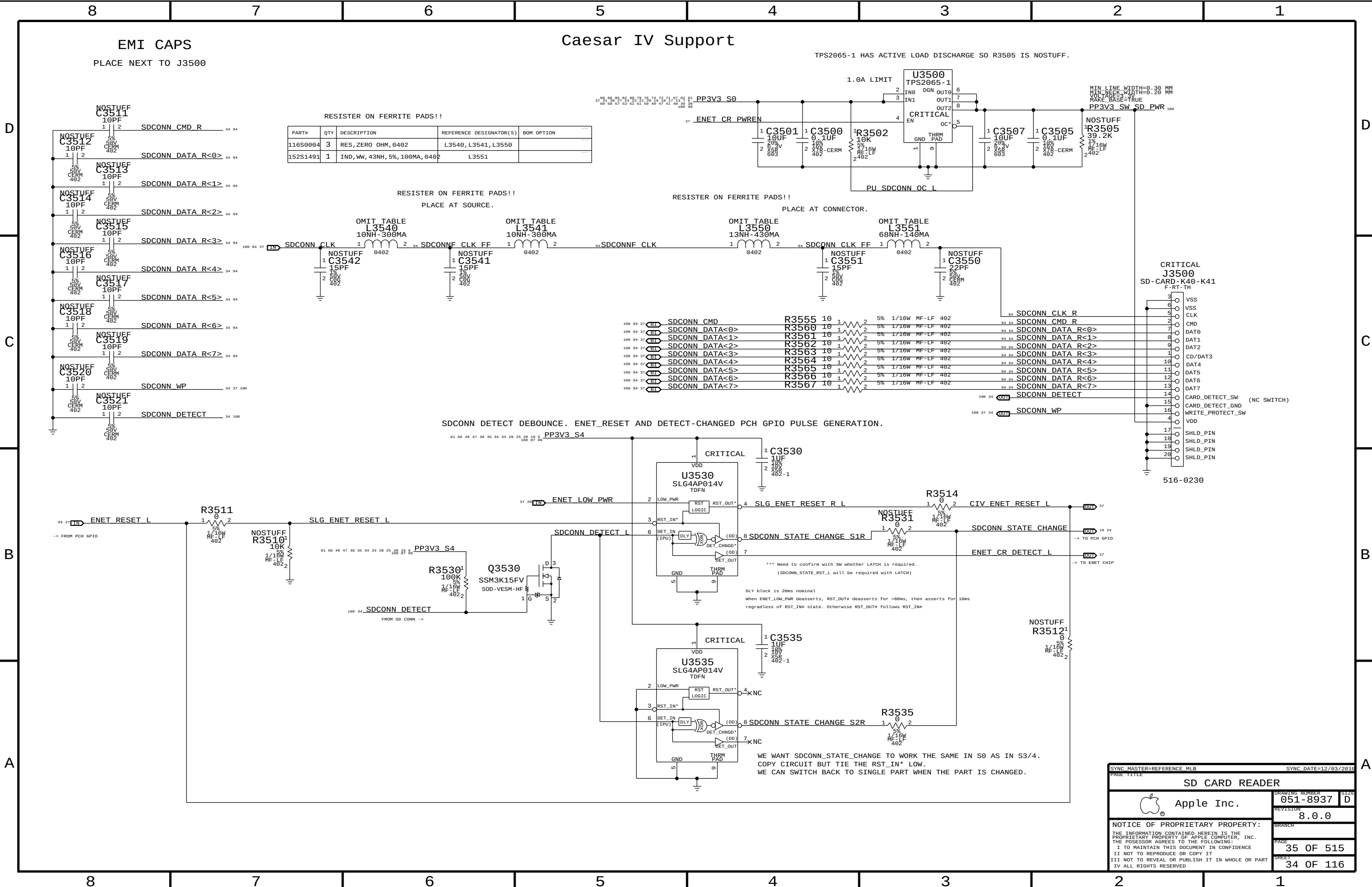
 Apple Inc.

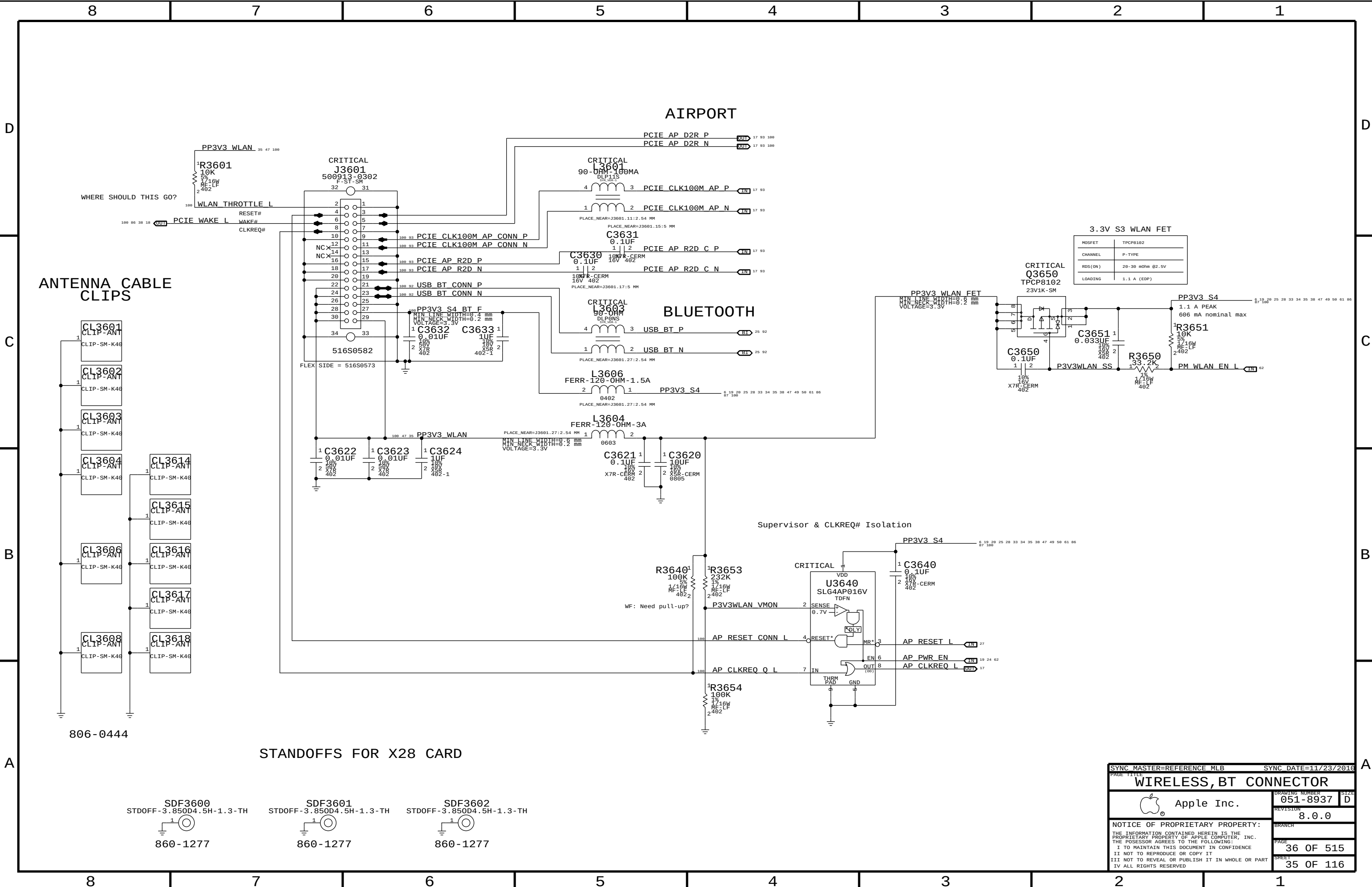
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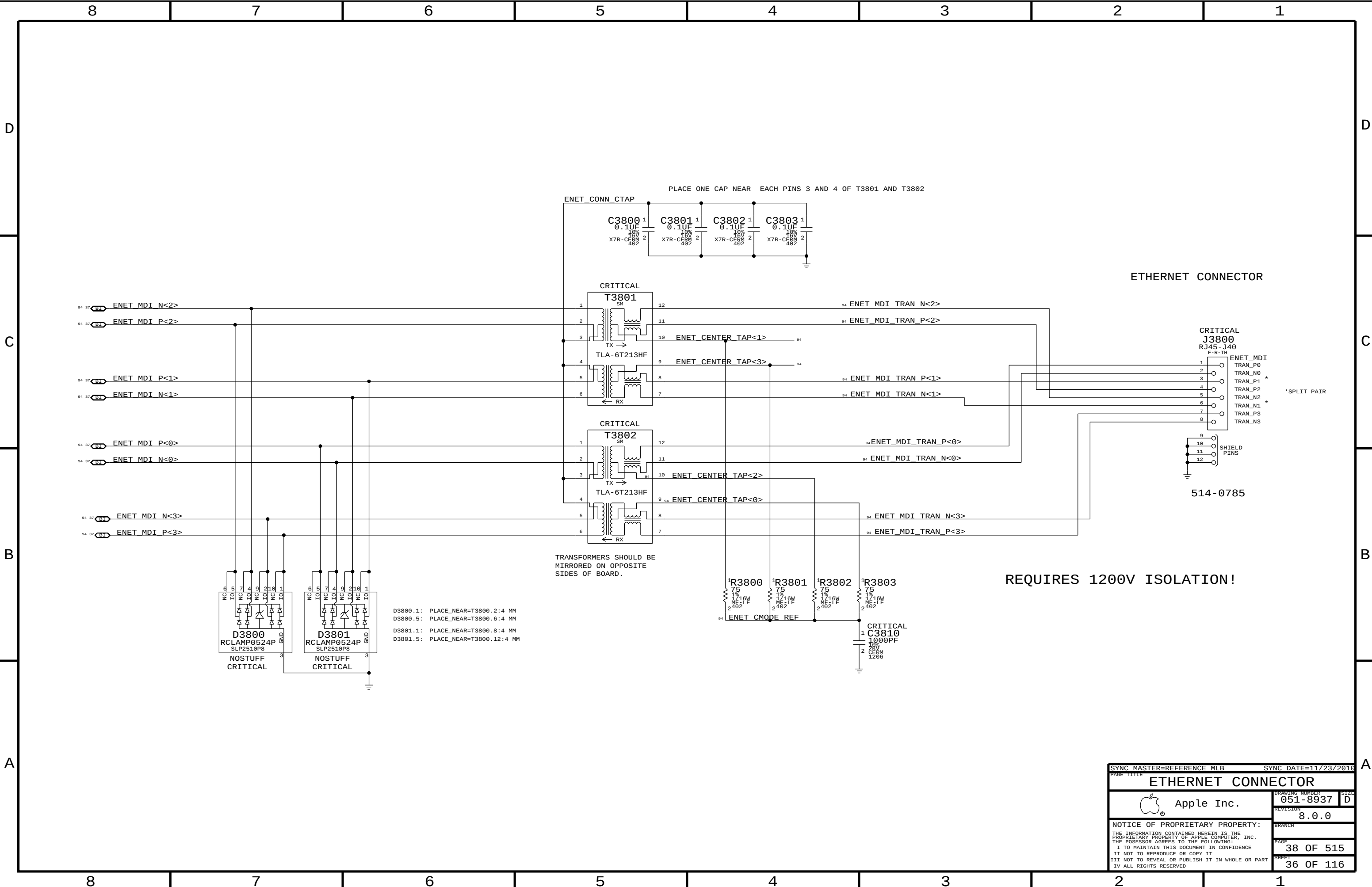
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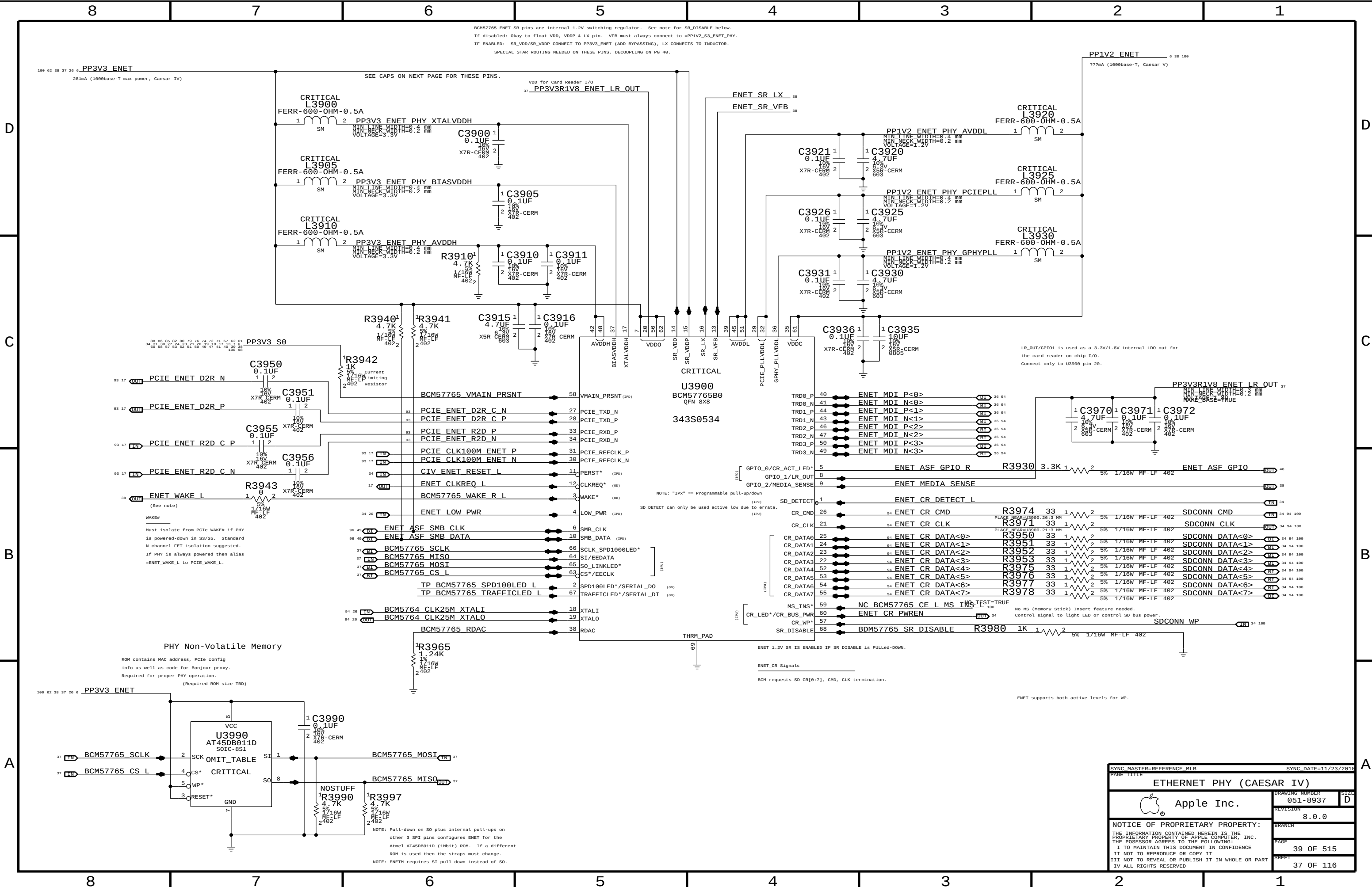
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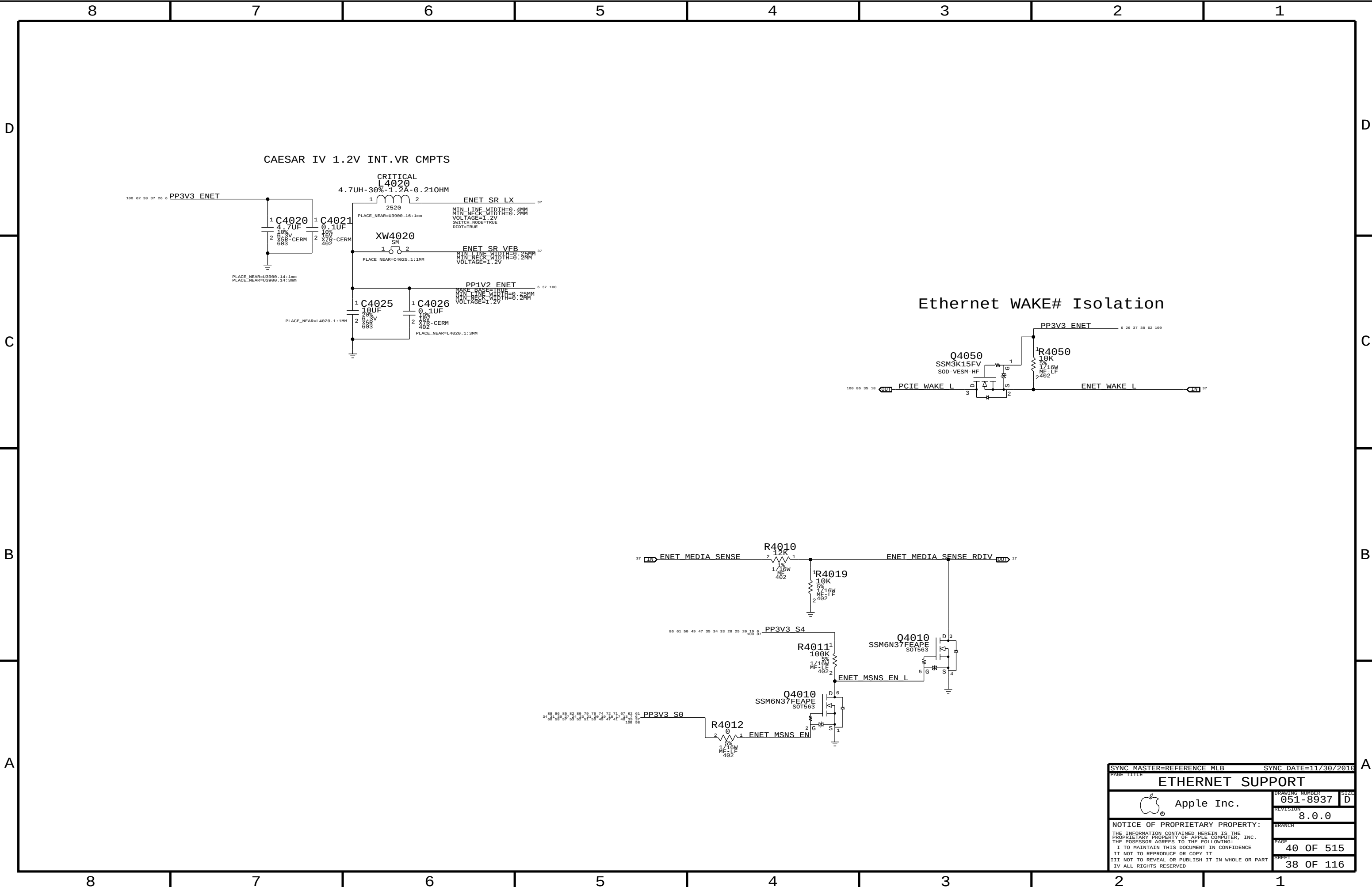


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|-----------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------|--|
| PAGE TITLE                                                                                                                  |  | PAGE TITLE                                                                                                                  |  |
| WIRELESS, BT CONNECTOR                                                                                                      |  | WIRELESS, BT CONNECTOR                                                                                                      |  |
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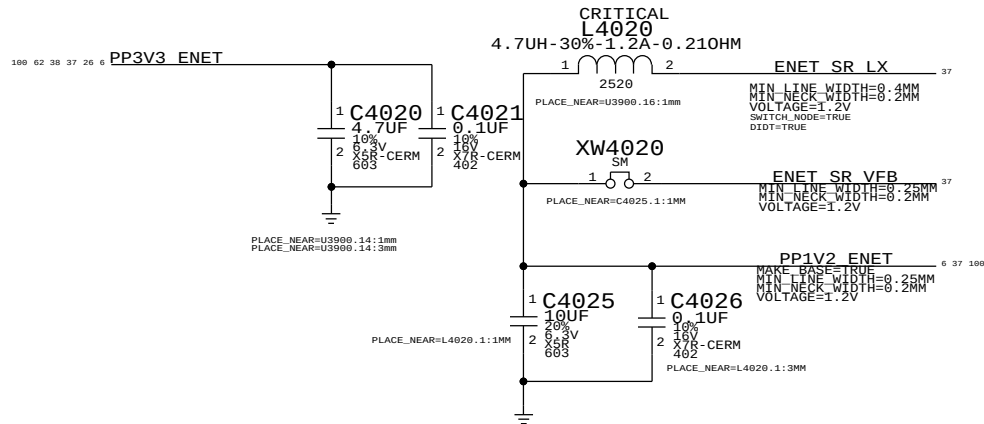




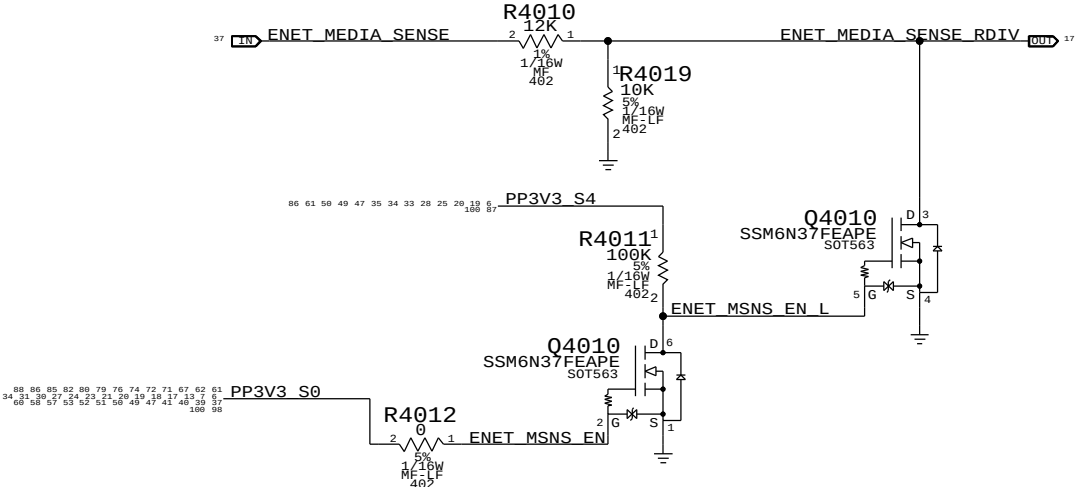
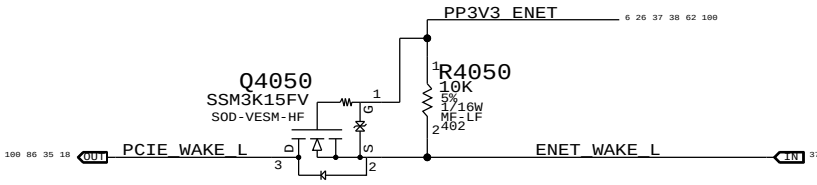
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| SYNC MASTER=REFERENCE_MLB                                                                                                   |  | SYNC DATE=11/23/2010 |           |
| PAGE TITLE                                                                                                                  |  | PAGE 116             |           |
| ETHERNET PHY (CAESAR IV)                                                                                                    |  | DRAWING NUMBER       | 051-8937  |
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CAESAR IV 1.2V INT.VR CMPTS

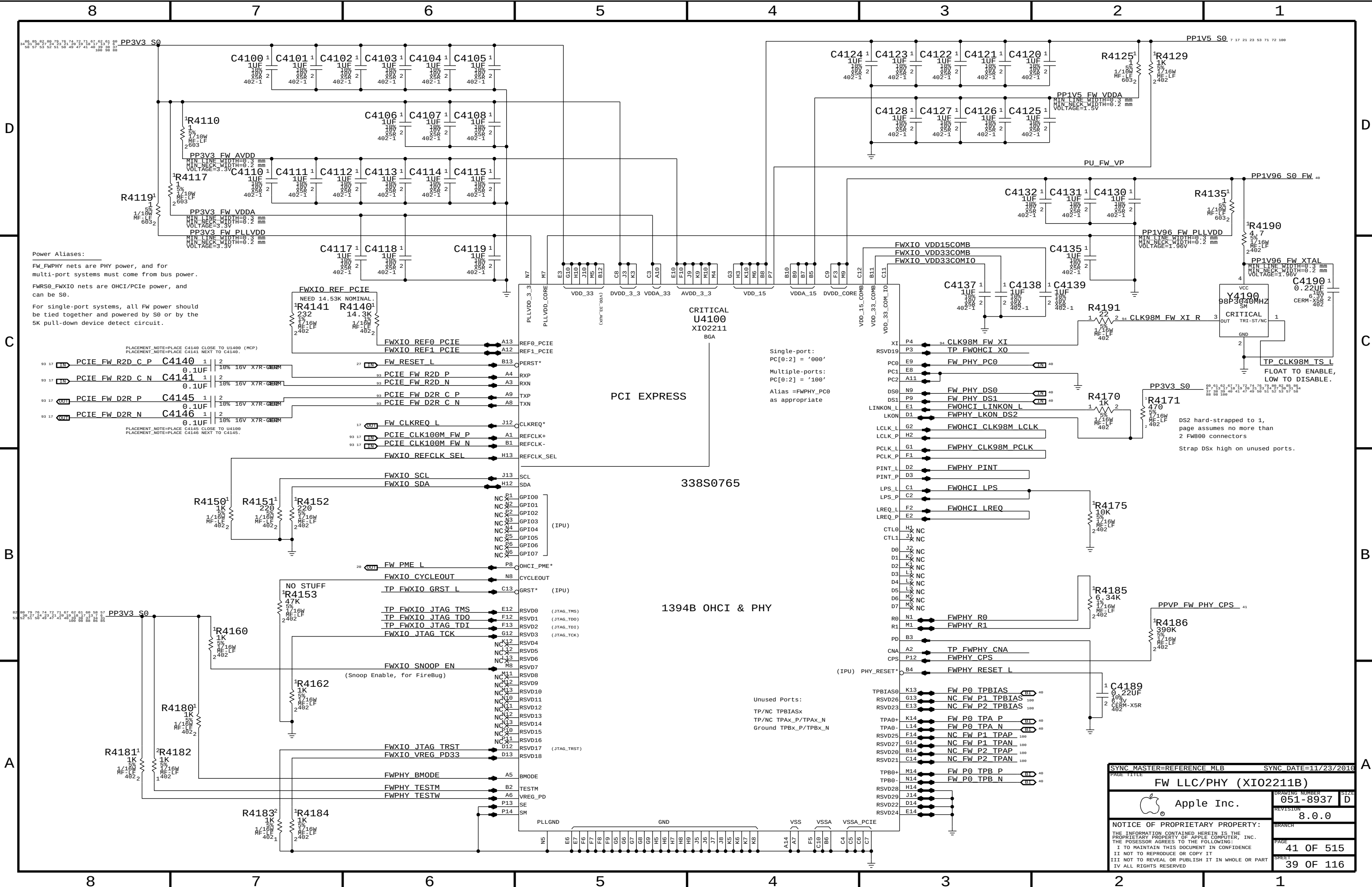


Ethernet WAKE# Isolation



|                                                                                                                             |  |                           |  |                      |  |
|-----------------------------------------------------------------------------------------------------------------------------|--|---------------------------|--|----------------------|--|
| PAGE TITLE                                                                                                                  |  | SYNC MASTER=REFERENCE MLB |  | SYNC DATE=11/30/2016 |  |
| ETHERNET SUPPORT                                                                                                            |  | DRAWING NUMBER            |  | SIZE                 |  |
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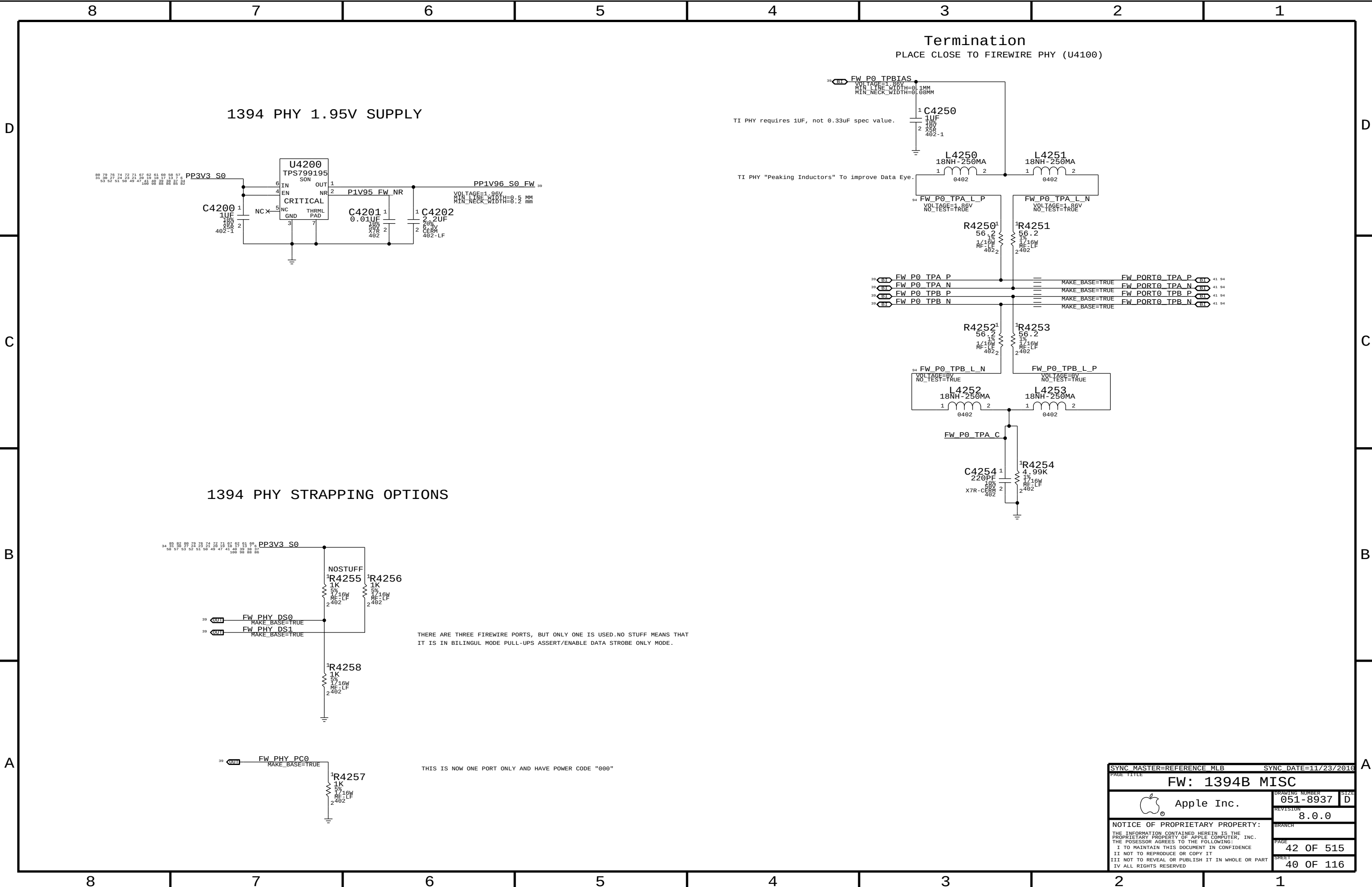


Power Aliases:  
FW\_FWPHY nets are PHY power, and for multi-port systems must come from bus power.  
FWRS0\_FWXIO nets are OHCI/PCIE power, and can be S0.  
For single-port systems, all FW power should be tied together and powered by S0 or by the 5K pull-down device detect circuit.

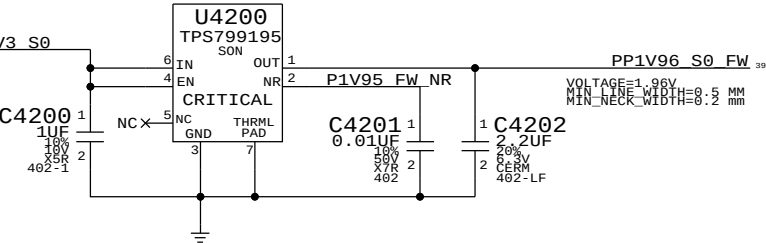
Single-port:  
PC[0:2] = '000'  
Multiple-ports:  
PC[0:2] = '100'  
Alias =FWPHY\_PC0 as appropriate

DS2 hard-strapped to 1, page assumes no more than 2 FW000 connectors  
Strap DSx high on unused ports.

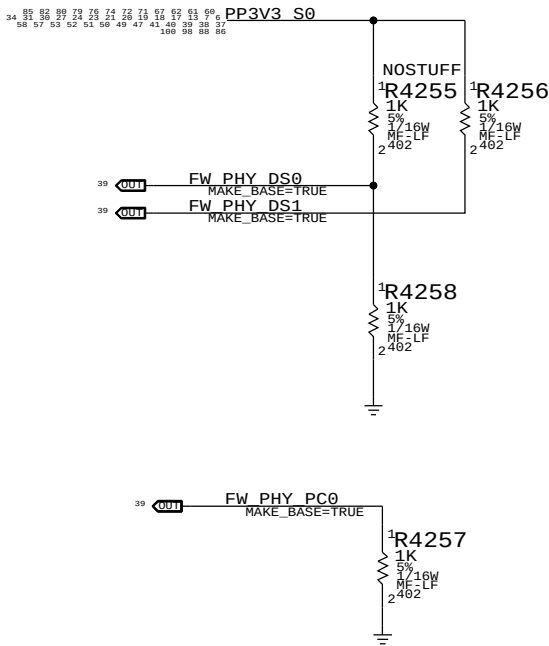
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| SYNC MASTER=REFERENCE MLB                                                                                                                                                                                                                                                                                                      |                | SYNC DATE=11/23/2016 |        |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                     |                |                      |        |
| FW LLC/PHY (XIO2211B)                                                                                                                                                                                                                                                                                                          |                |                      |        |
|  Apple Inc.                                                                                                                                                                                                                               | DRAWING NUMBER | 051-8937             | SIZE D |
|                                                                                                                                                                                                                                                                                                                                | REVISION       | 8.0.0                |        |
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| SHEET                                                                                                                                                                                                                                                                                                                          | 39             | OF                   | 116    |



1394 PHY 1.95V SUPPLY



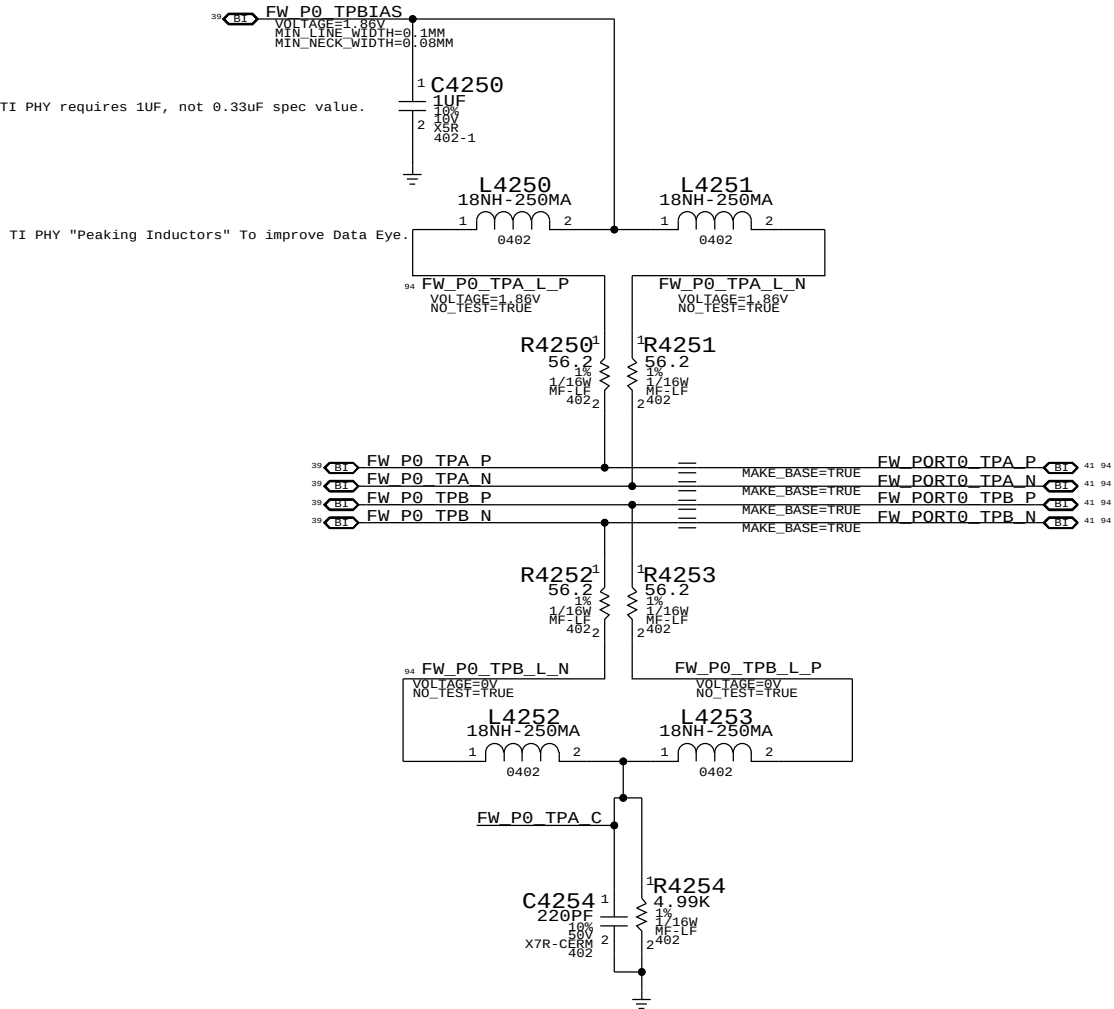
1394 PHY STRAPPING OPTIONS



THERE ARE THREE FIREWIRE PORTS, BUT ONLY ONE IS USED.NO STUFF MEANS THAT IT IS IN BILINGUL MODE PULL-UPS ASSERT/ENABLE DATA STROBE ONLY MODE.

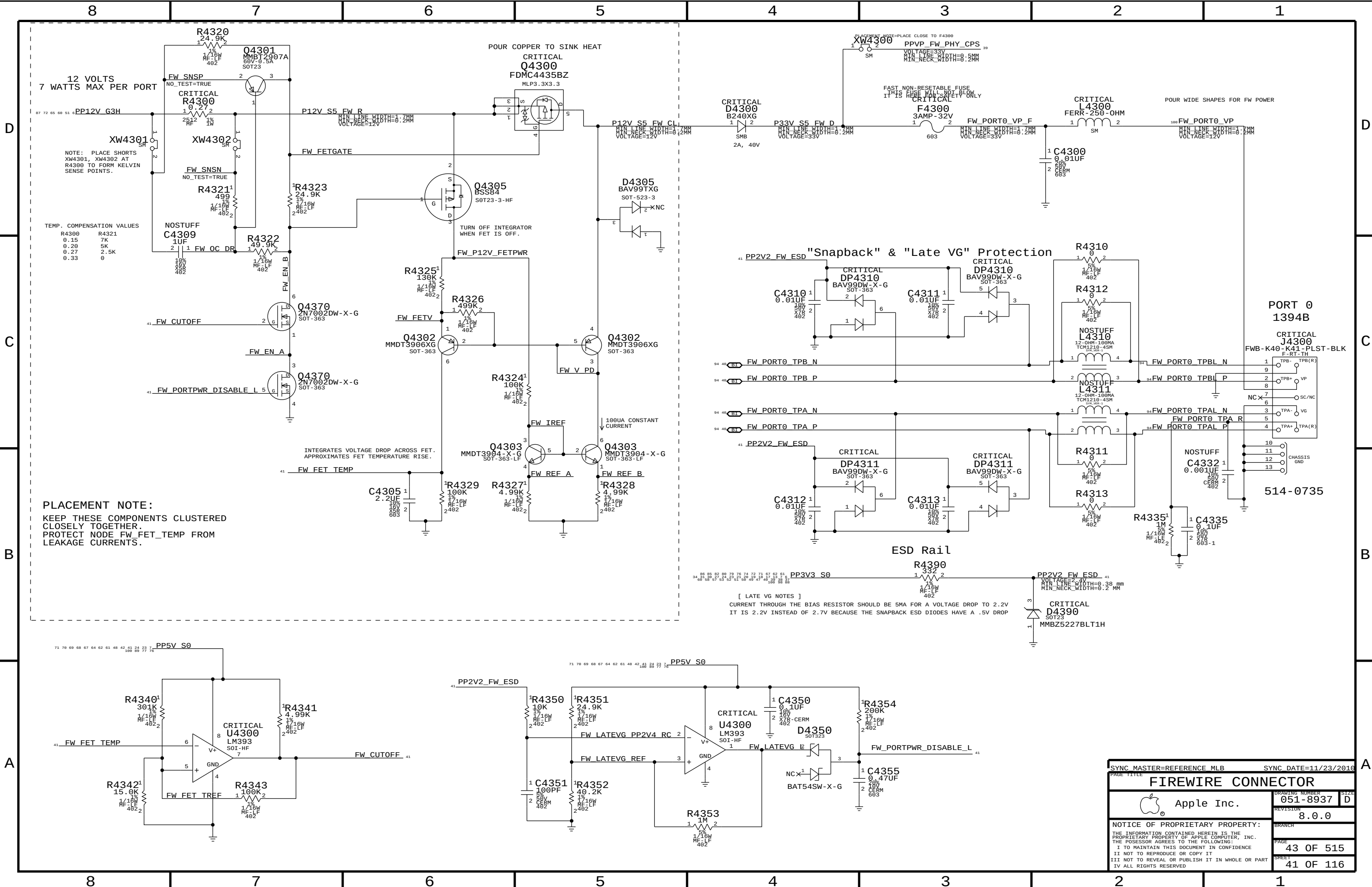
THIS IS NOW ONE PORT ONLY AND HAVE POWER CODE "000"

Termination  
PLACE CLOSE TO FIREWIRE PHY (U4100)



|                                                                                                                                                                                                                                                                                                                             |  |                           |  |                      |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------|--|----------------------|--|
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  | SYNC MASTER=REFERENCE MLB |  | SYNC DATE=11/23/2016 |  |
| FW: 1394B MISC                                                                                                                                                                                                                                                                                                              |  | DRAWING NUMBER            |  | SIZE                 |  |
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|                                                                                                                                                                                                                                                                                                                             |  | PAGE                      |  | SHEET                |  |
|                                                                                                                                                                                                                                                                                                                             |  | 42 OF 515                 |  | 40 OF 116            |  |





12 VOLTS  
7 WATTS MAX PER PORT

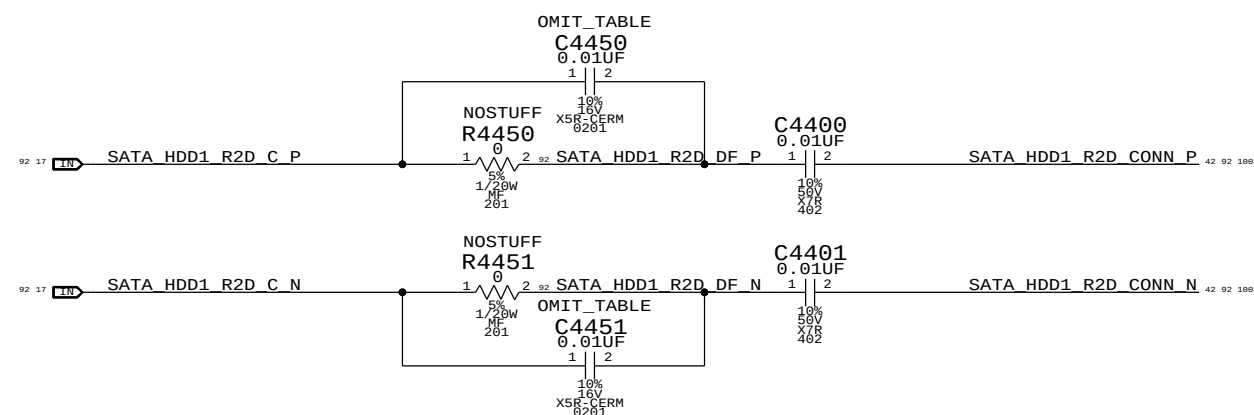
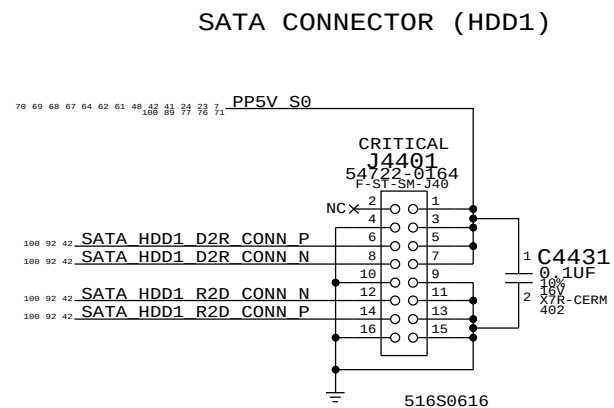
NOTE: PLACE SHORTS  
XW4301, XW4302 AT  
R4300 TO FORM KELVIN  
SENSE POINTS.

TEMP. COMPENSATION VALUES

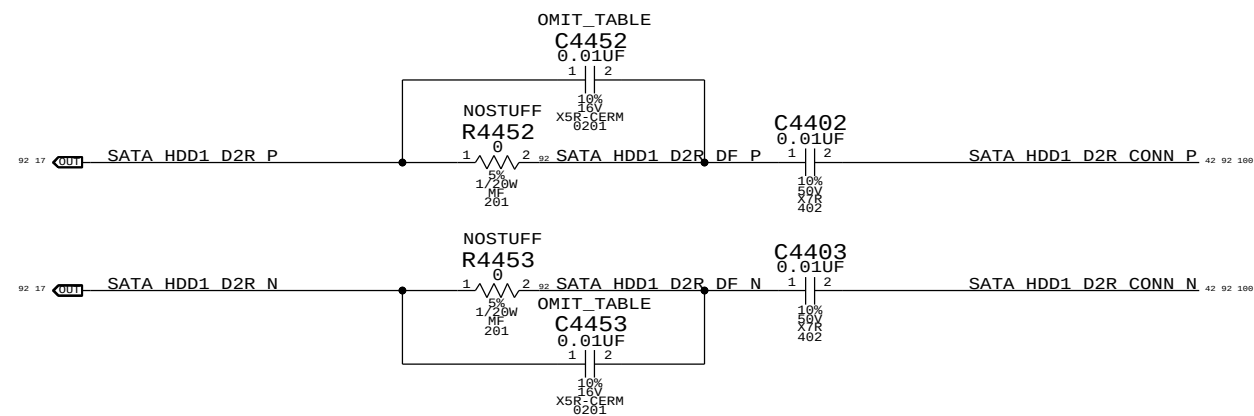
| R4300 | R4321 |
|-------|-------|
| 0.15  | 7K    |
| 0.20  | 5K    |
| 0.27  | 2.5K  |
| 0.33  | 0     |

PLACEMENT NOTE:  
KEEP THESE COMPONENTS CLUSTERED  
CLOSELY TOGETHER.  
PROTECT NODE FW\_FET\_TEMP FROM  
LEAKAGE CURRENTS.

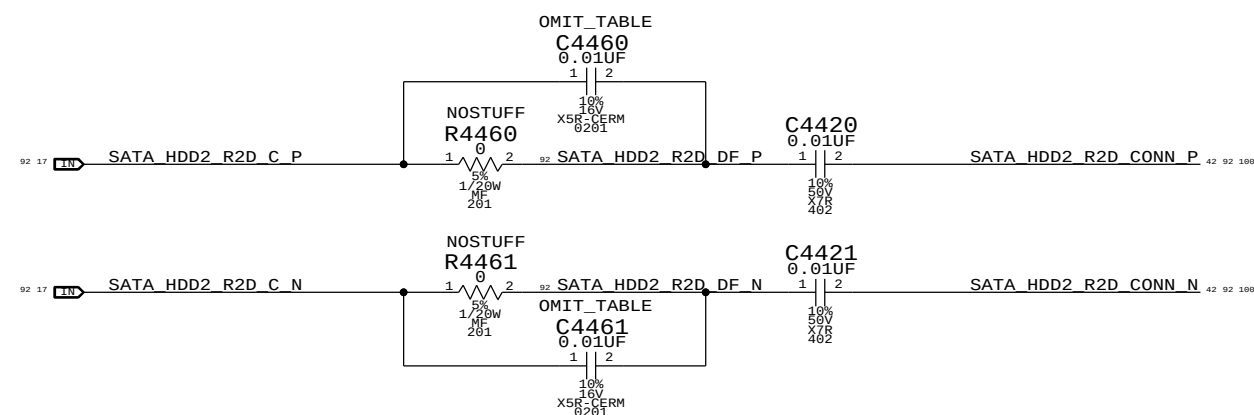
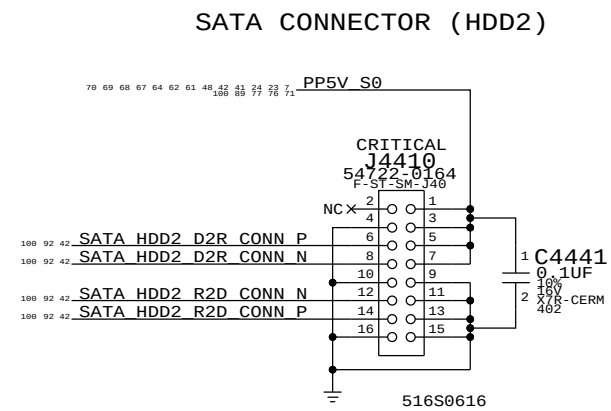
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|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=REFERENCE MLB                                                                                                   |  | SYNC DATE=11/23/2010 |           |
| PAGE TITLE                                                                                                                  |  | FIREWIRE CONNECTOR   |           |
| Apple Inc.                                                                                                                  |  | DRAWING NUMBER       | 051-8937  |
|                                                                                                                             |  | REVISION             | 8.0.0     |
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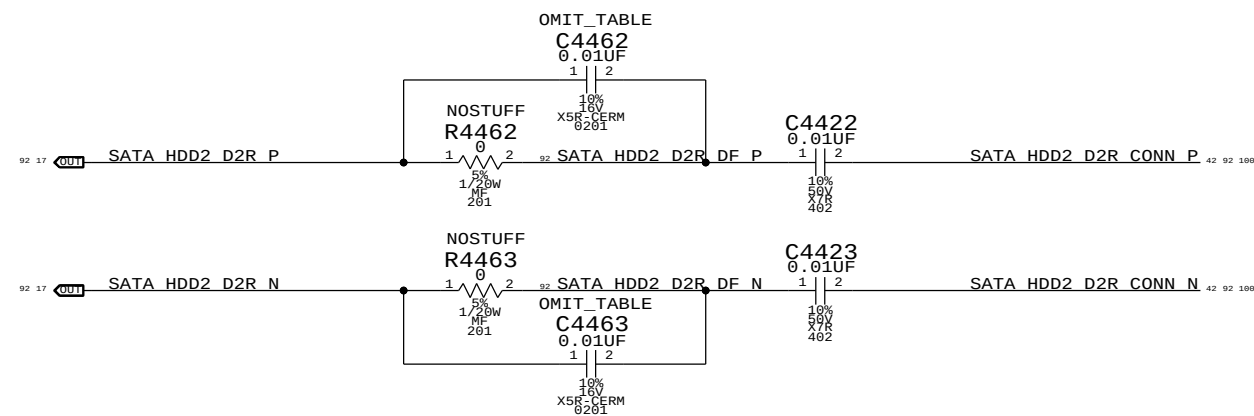
RESISTER ON CAP PADS!!



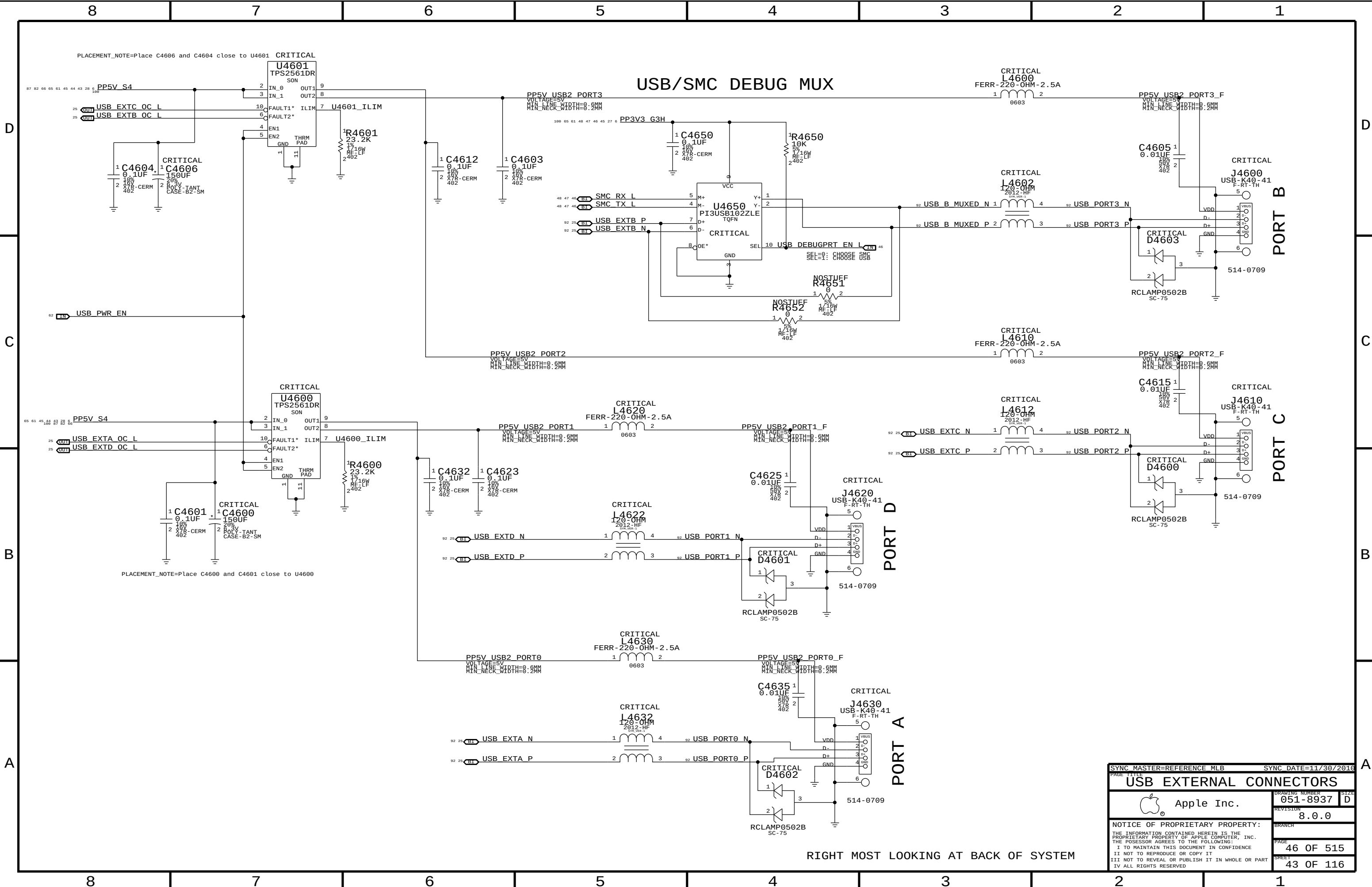
| PART#    | QTY | DESCRIPTION         | REFERENCE DESIGNATOR(S)    | BOM OPTION |
|----------|-----|---------------------|----------------------------|------------|
| 117S0002 | 4   | RES, ZERO OHM, 0201 | C4450, C4451, C4452, C4453 |            |
| 117S0002 | 4   | RES, ZERO OHM, 0201 | C4460, C4461, C4462, C4463 |            |



RESISTER ON CAP PADS!!

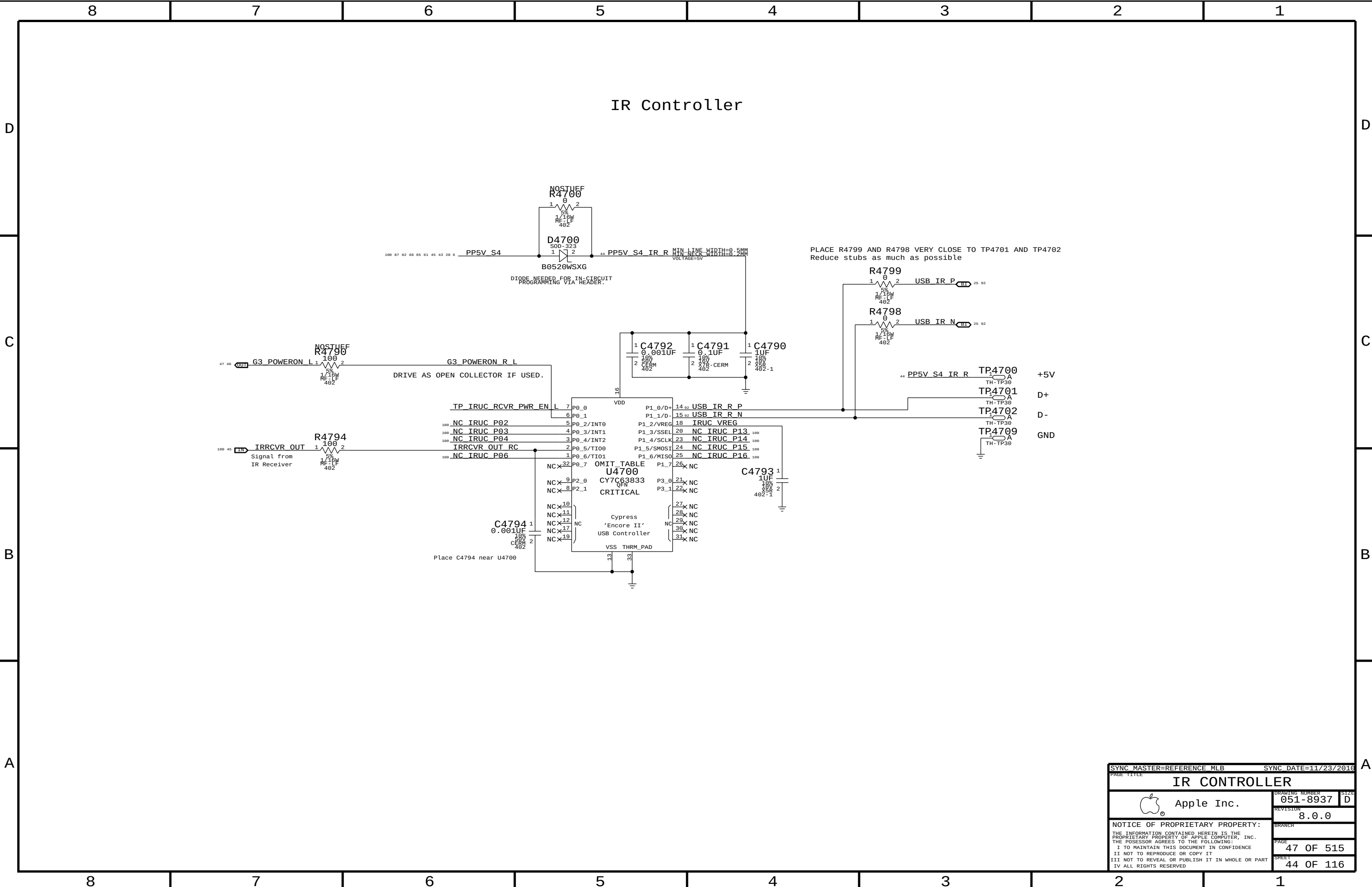


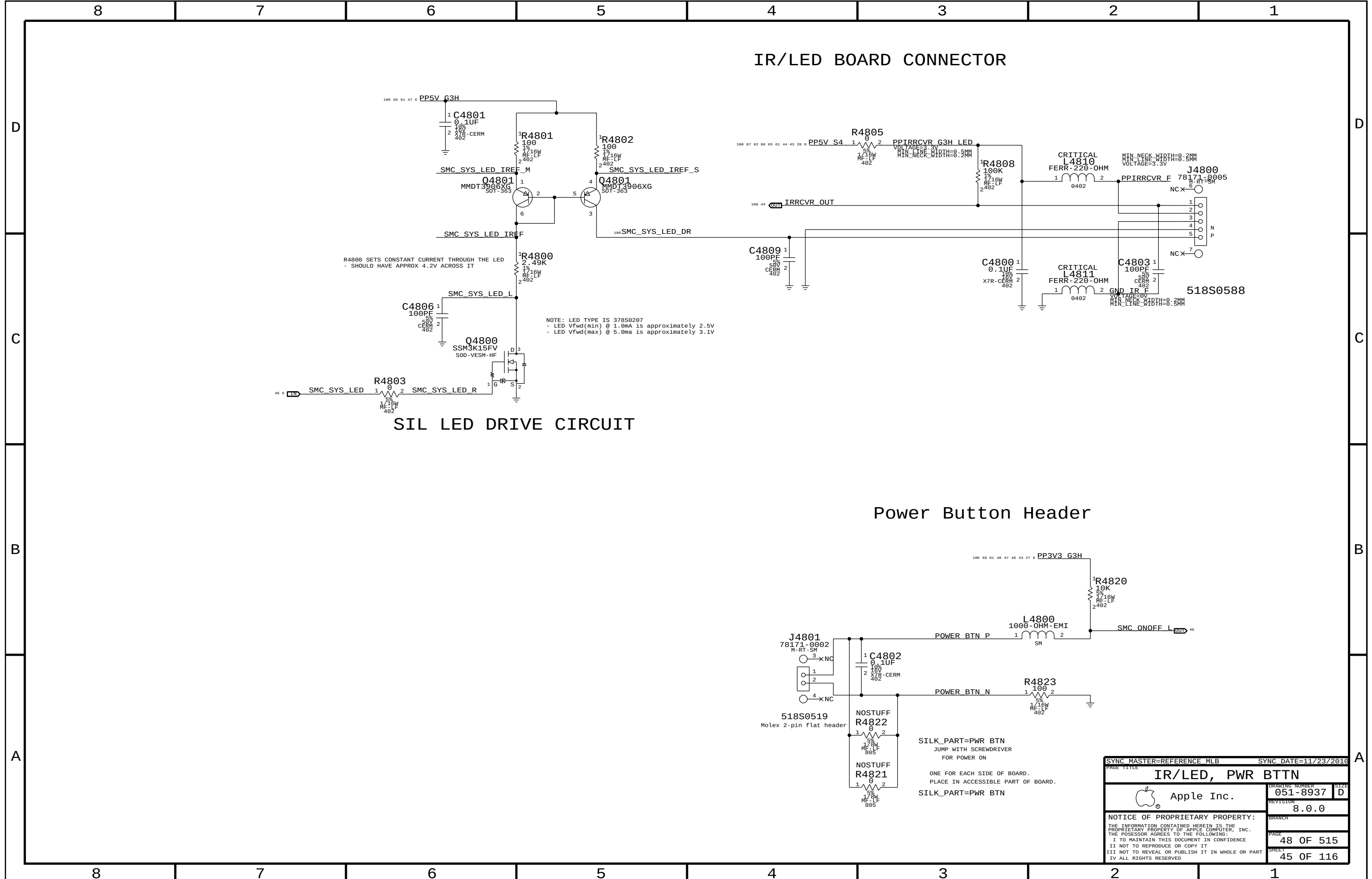
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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------|------|
| SYNC MASTER=REFERENCE MLB                                                                                                                                                                                                                                                                      |  | SYNC DATE=11/23/2010                            |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                     |  |                                                 |      |
| <b>HDD SATA CONNECTORS</b>                                                                                                                                                                                                                                                                     |  |                                                 |      |
| <br>Apple Inc.                                                                                                                                                                                            |  | DRAWING NUMBER                                  | SIZE |
|                                                                                                                                                                                                                                                                                                |  | 051-8937                                        | D    |
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|                                                                                                                             |  |                      |  |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|--|
| PAGE TITLE                                                                                                                  |  | SYNC DATE=11/30/2016 |  |
| USB EXTERNAL CONNECTORS                                                                                                     |  | DRAWING NUMBER       |  |
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|                                                                                                                             |  | 43 OF 116            |  |

RIGHT MOST LOOKING AT BACK OF SYSTEM



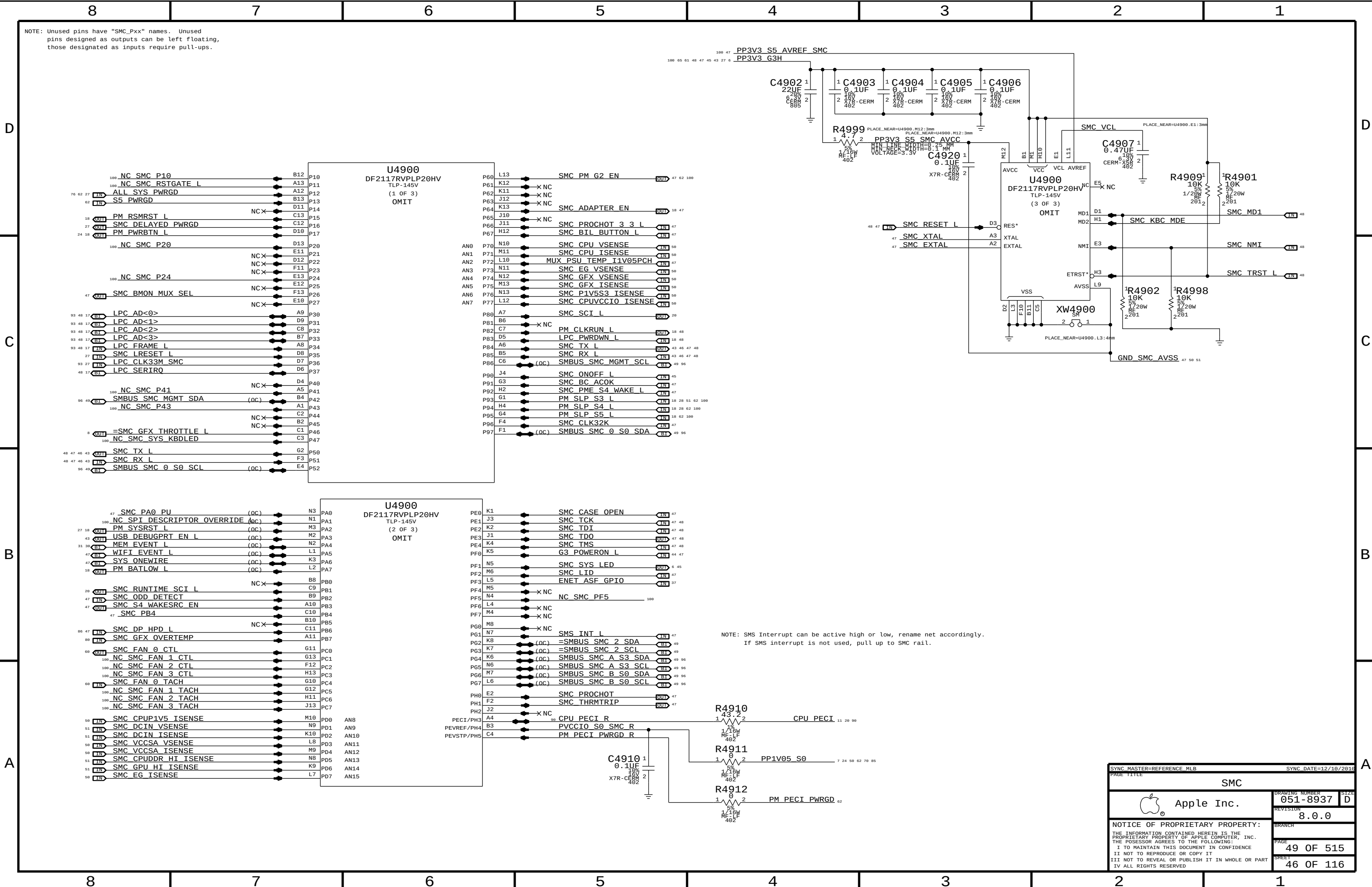


IR/LED BOARD CONNECTOR

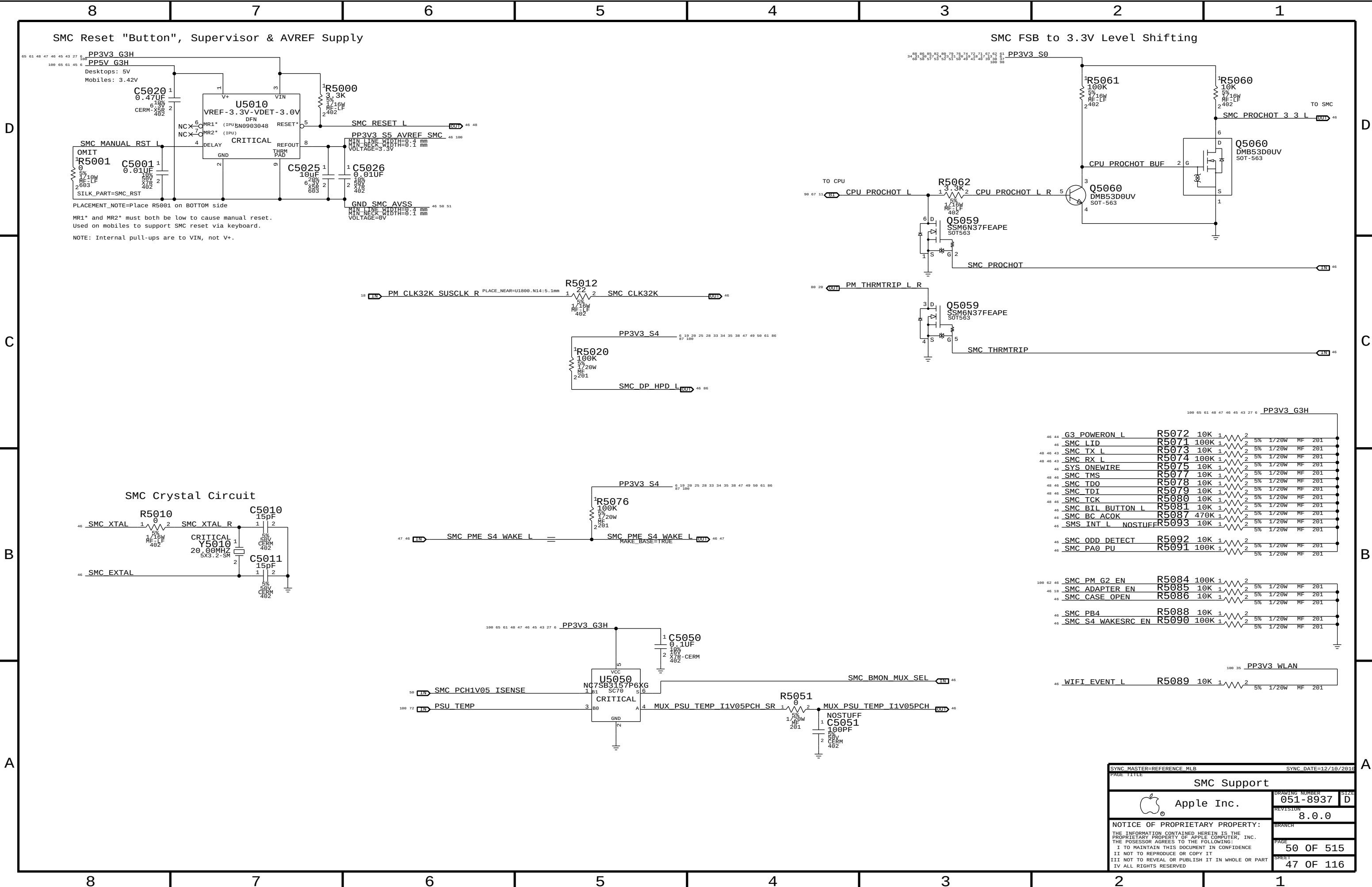
SIL LED DRIVE CIRCUIT

Power Button Header

|                                                                                                                             |  |                      |        |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|--------|
| SYNC MASTER=REFERENCE MLB                                                                                                   |  | SYNC DATE=11/23/2016 |        |
| PAGE TITLE                                                                                                                  |  | IR/LED, PWR BTTN     |        |
| DRAWING NUMBER                                                                                                              |  | 051-8937             | SIZE D |
| REVISION                                                                                                                    |  | 8.0.0                | BRANCH |
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SMC Reset "Button", Supervisor & AVREF Supply

SMC FSB to 3.3V Level Shifting

SMC Crystal Circuit

SYNC MASTER=REFERENCE\_MLB

SYNC DATE=12/18/2010

PAGE TITLE

SMC Support

 Apple Inc.

051-8937

REVISION

8.0.0

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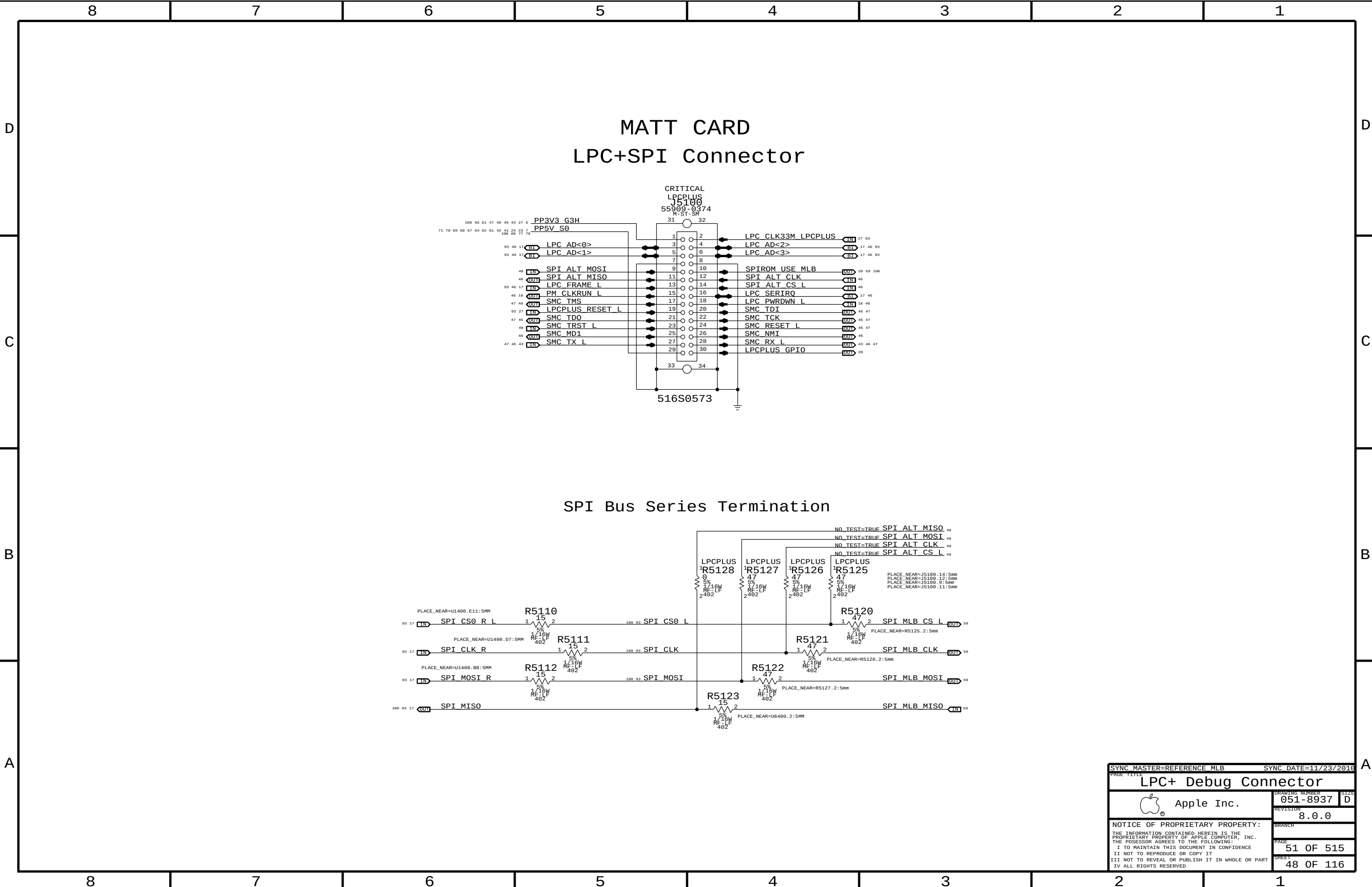
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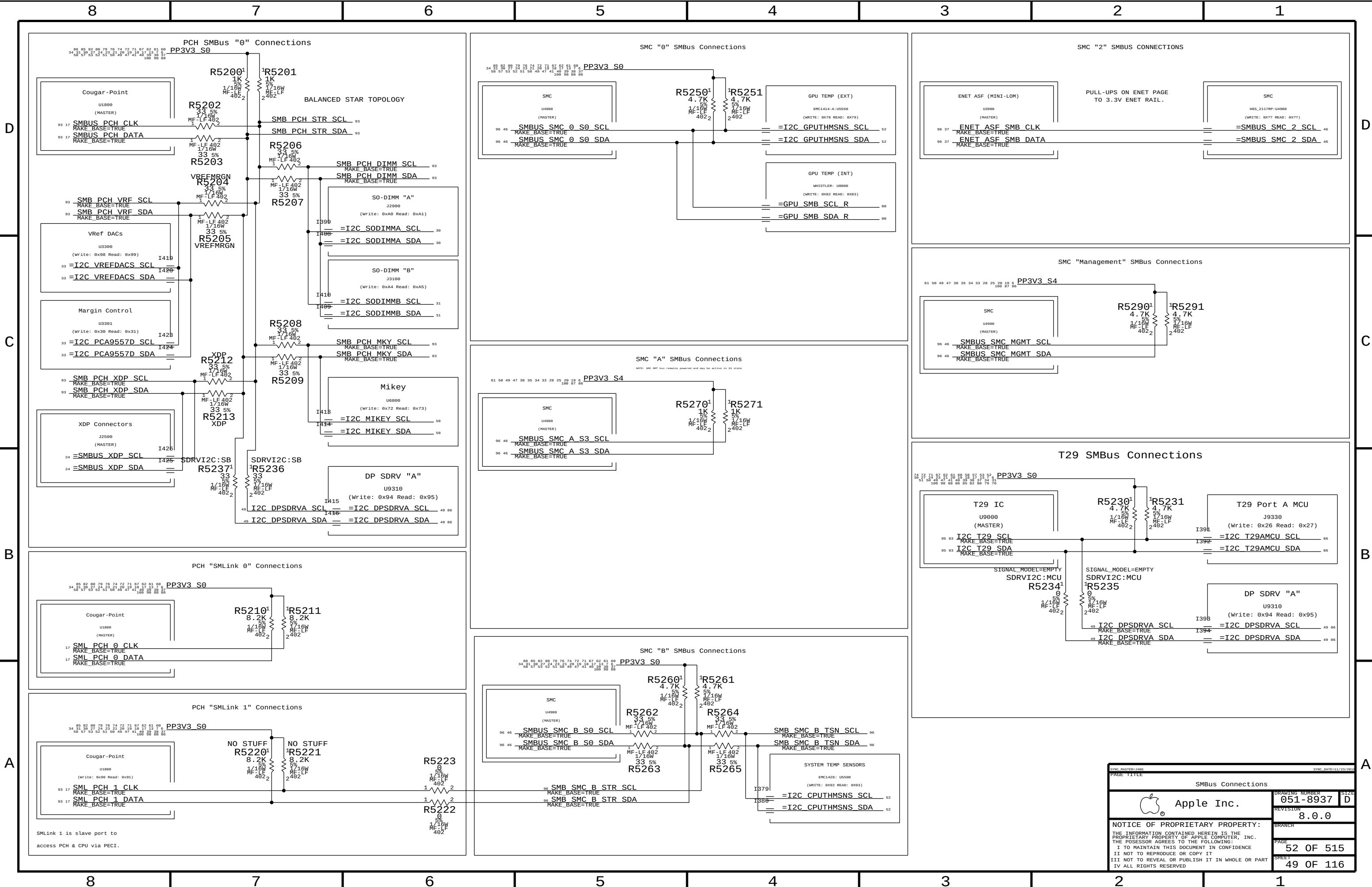
SHEET

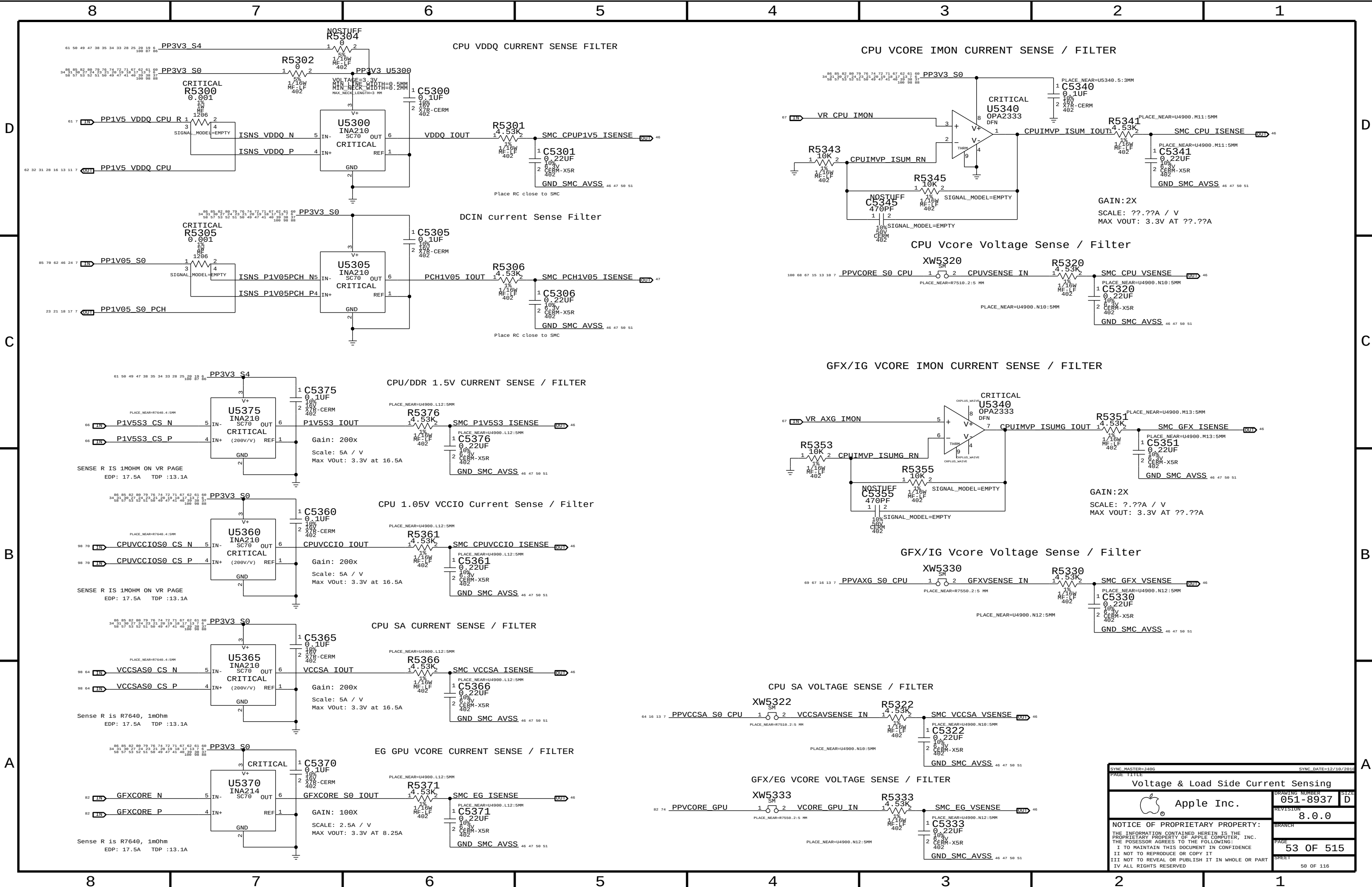
47 OF 116



|                                                                                                                                                                                                                                                                                                                             |  |                           |  |                      |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------|--|----------------------|--|
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  | SYNC MASTER=REFERENCE MLB |  | SYNC DATE=11/23/2016 |  |
| LPC+ Debug Connector                                                                                                                                                                                                                                                                                                        |  | DRAWING NUMBER            |  | 051-8937             |  |
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|                                                                                                                                                                                                                                                                                                                             |  |                           |  | 51 OF 515            |  |
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|                                                                                                                                                                                                                                                                                                                             |  |                           |  | 48 OF 116            |  |







|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| 8 | 7 | 6 | 5 | 4 | 3 | 2 | 1 |
|---|---|---|---|---|---|---|---|



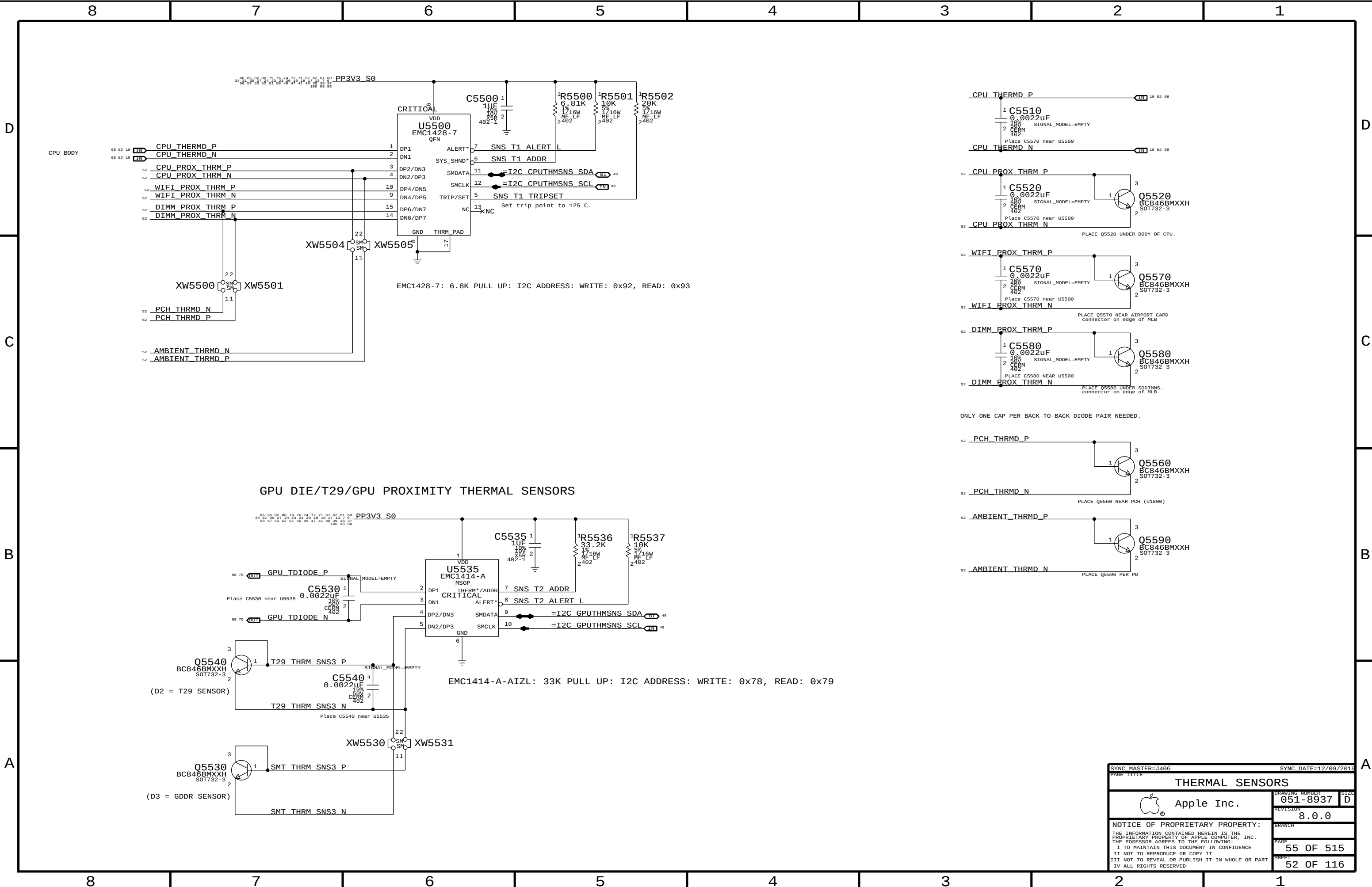
## A



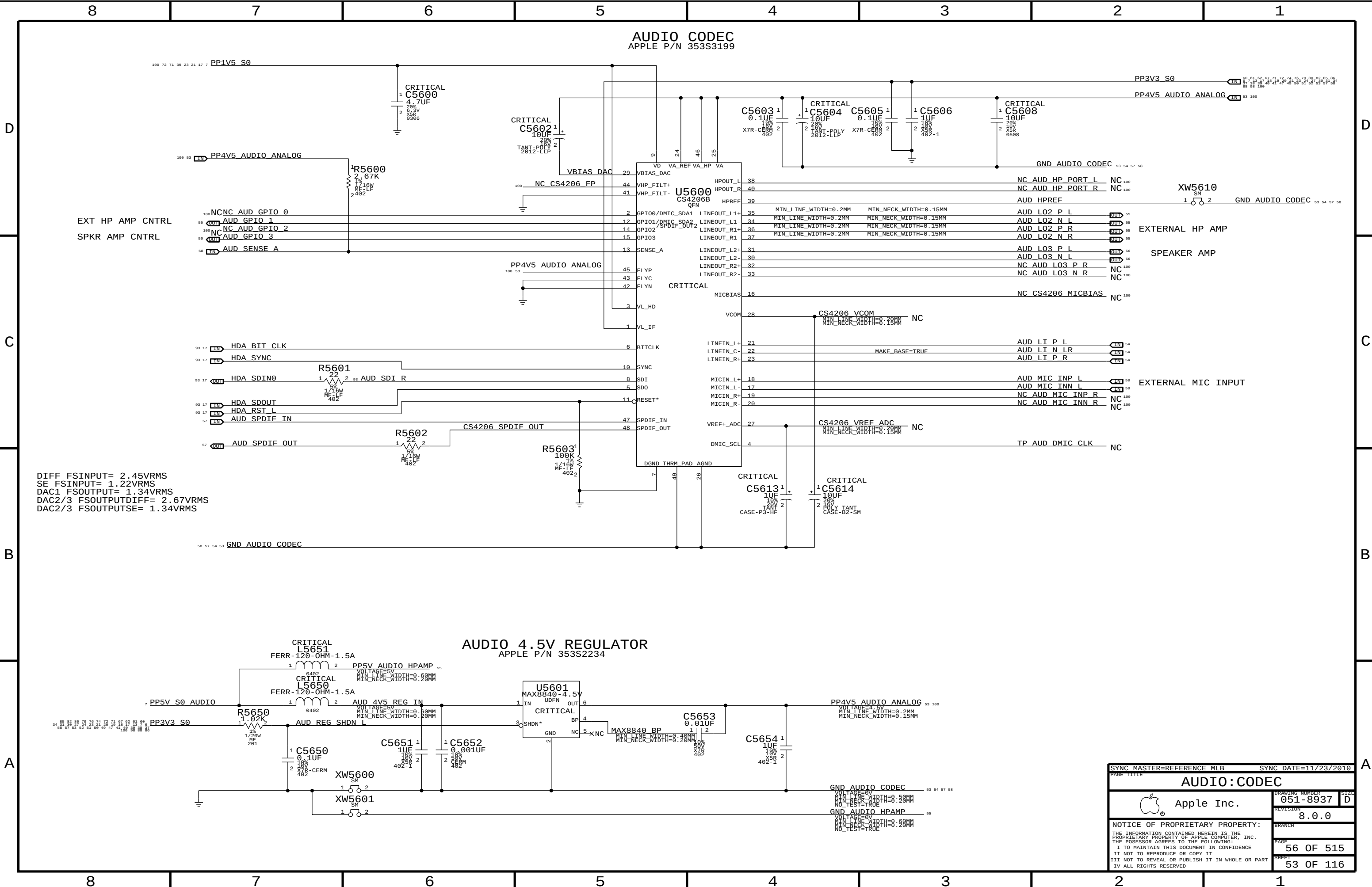
## D

C

A



|                                                                                                                                                                                                                                                                                                                             |  |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=J40G                                                                                                                                                                                                                                                                                                            |  | SYNC DATE=12/09/2010 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  | THERMAL SENSORS      |           |
|  Apple Inc.                                                                                                                                                                                                                            |  | DRAWING NUMBER       | 051-8937  |
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|                                                                                                                                                                                                                                                                                                                             |  | PAGE                 | 55 OF 515 |
|                                                                                                                                                                                                                                                                                                                             |  | SHEET                | 52 OF 116 |
|                                                                                                                                                                                                                                                                                                                             |  |                      |           |



DIFF FSINPUT= 2.45VRMS  
SE FSINPUT= 1.22VRMS  
DAC1 FSOUTPUT= 1.34VRMS  
DAC2/3 FSOUTPUTDIFF= 2.67VRMS  
DAC2/3 FSOUTPUTSE= 1.34VRMS

|                                                                                                                             |  |                      |  |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|--|
| PAGE TITLE                                                                                                                  |  | SYNC DATE=11/23/2010 |  |
| AUDIO:CODEC                                                                                                                 |  | DRAWING NUMBER       |  |
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|                                                                                                                             |  | 53 OF 116            |  |

D

C

B

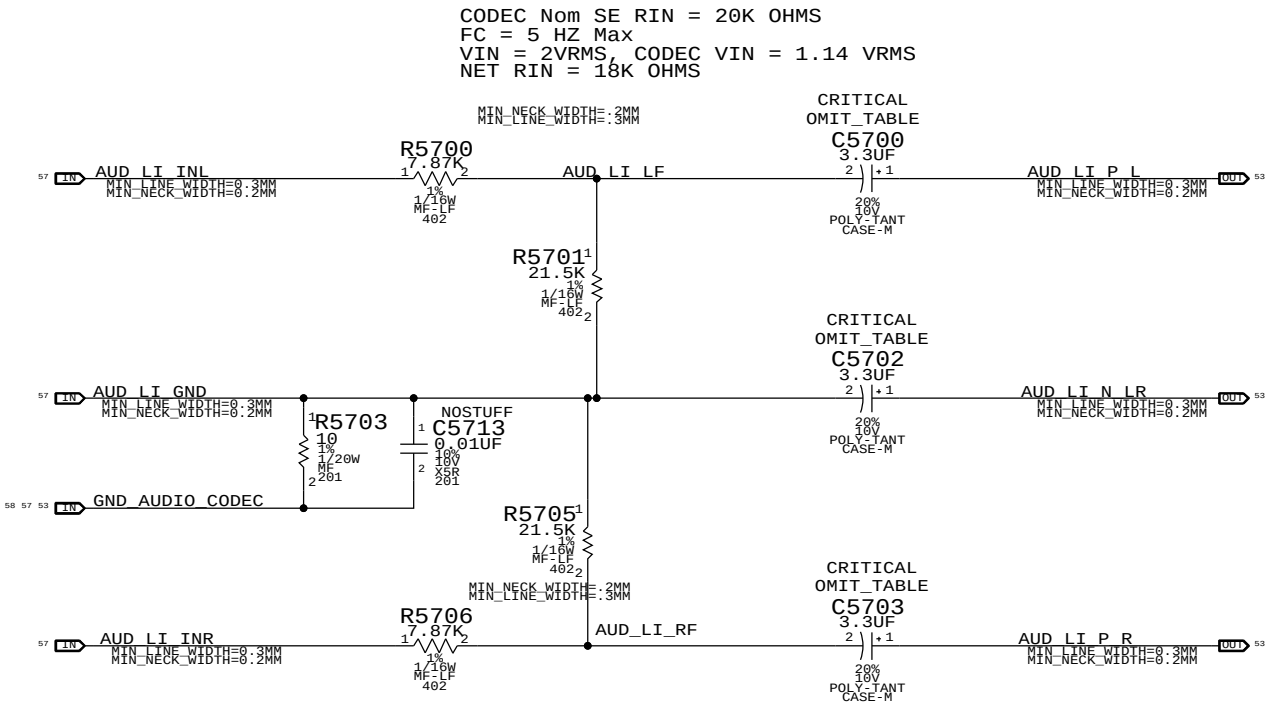
A

D

C

B

A



| PART#    | QTY | DESCRIPTION                           | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|----------|-----|---------------------------------------|-------------------------|----------|------------|
| 128S9399 | 3   | CAP, TANT, POLY, 4.7UF, 20%, 16V, SMD | C5700, C5702, C5703     | CRITICAL | ?          |

SYNC MASTER=REFERENCE MLB

SYNC DATE=11/23/2016

AUDIO: LINE - IN

 Apple Inc.

051-8937

D

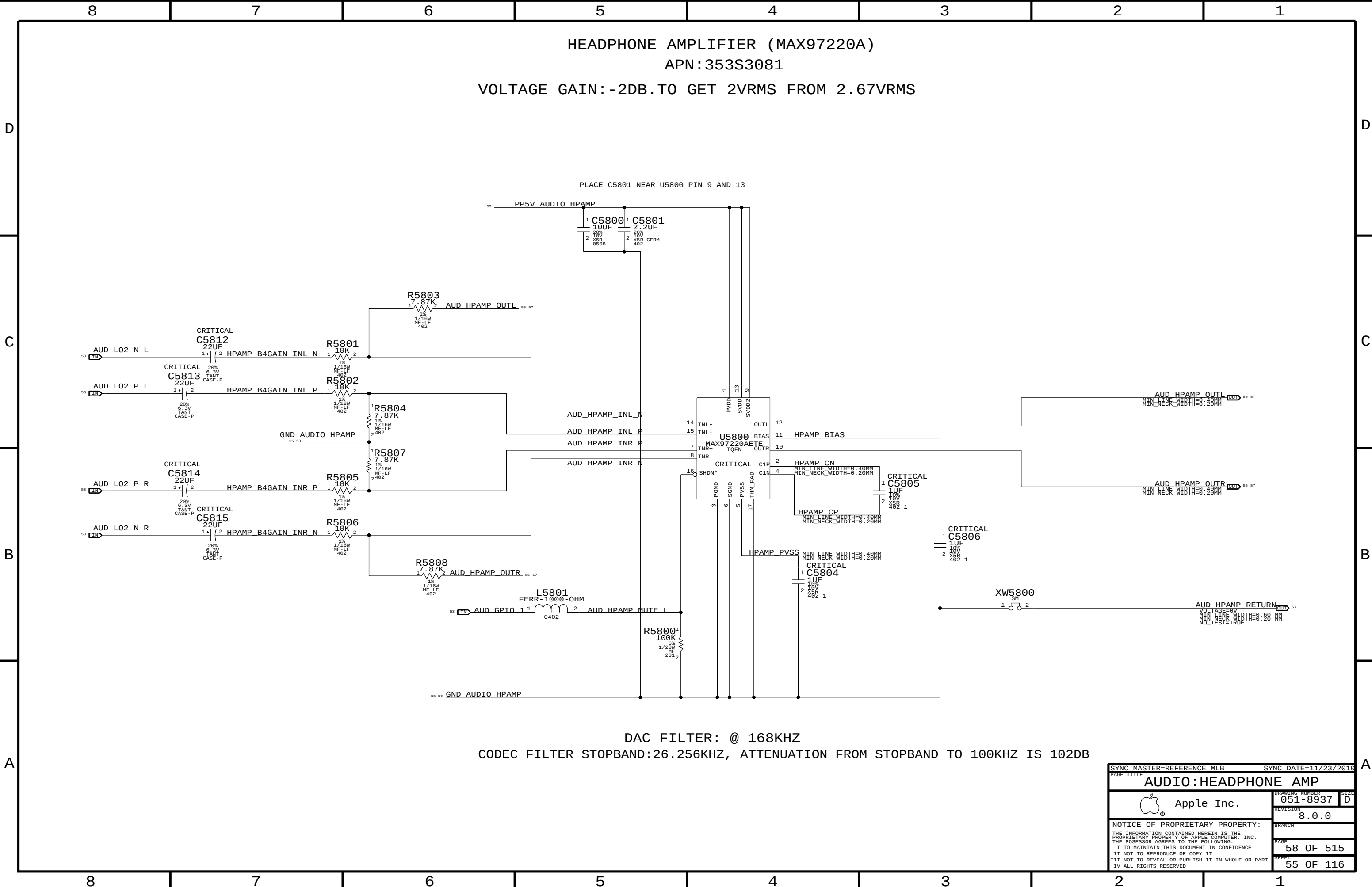
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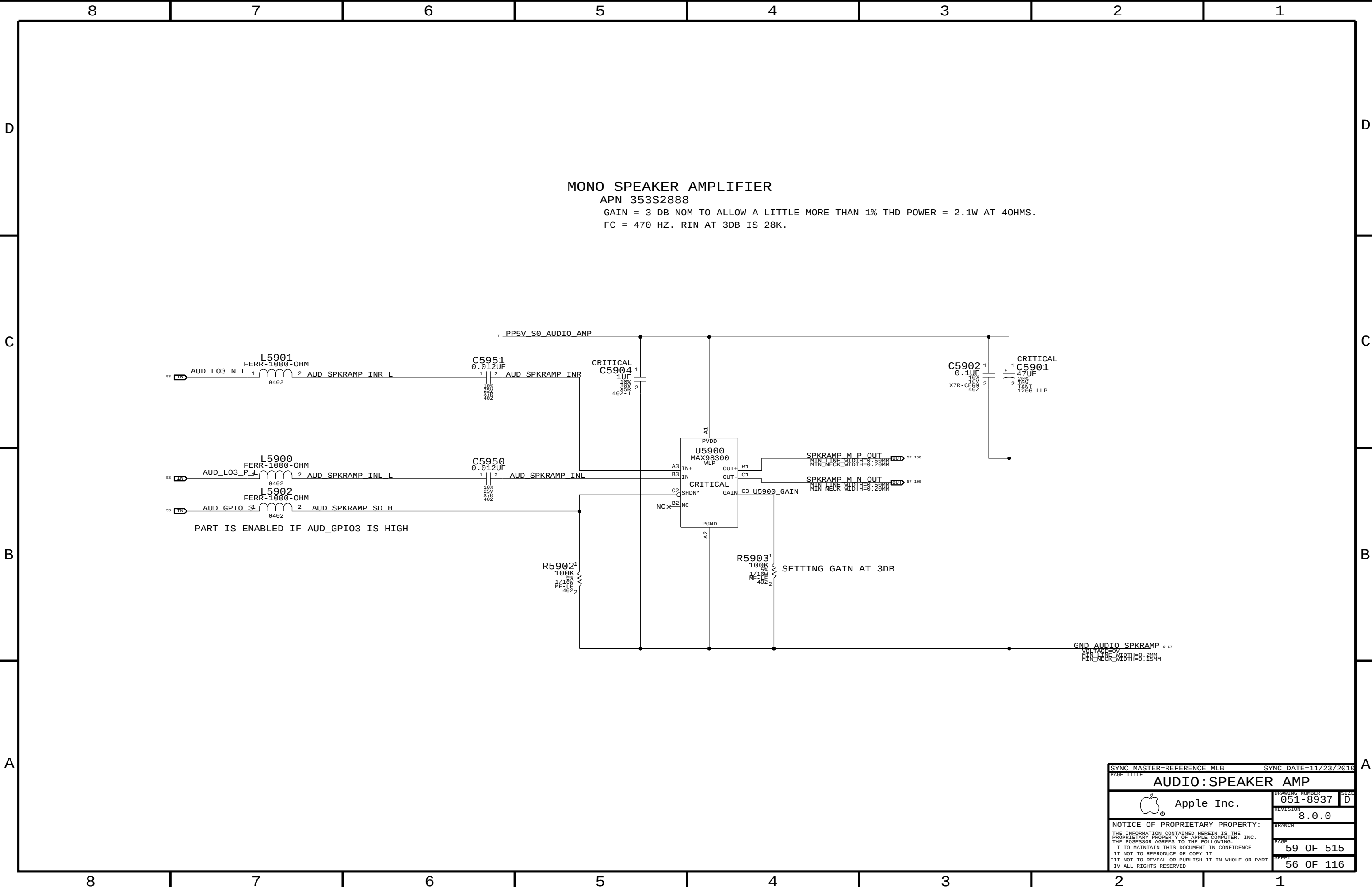
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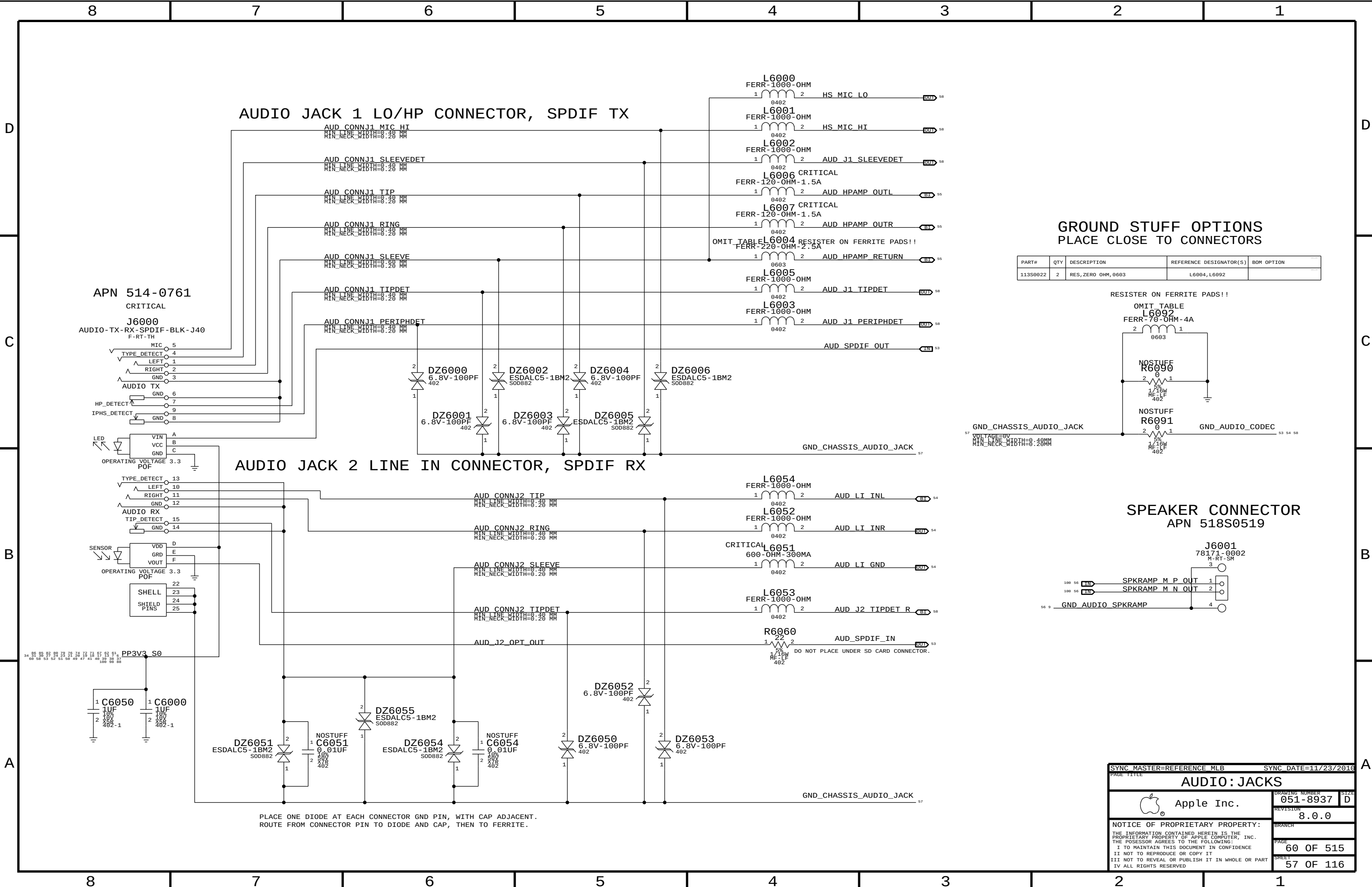




|                                                                                                                             |  |                           |  |                      |  |
|-----------------------------------------------------------------------------------------------------------------------------|--|---------------------------|--|----------------------|--|
| PAGE TITLE                                                                                                                  |  | SYNC MASTER=REFERENCE MLB |  | SYNC DATE=11/23/2016 |  |
| AUDIO:HEADPHONE AMP                                                                                                         |  | DRAWING NUMBER            |  | SIZE                 |  |
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GROUND STUFF OPTIONS  
PLACE CLOSE TO CONNECTORS

| PART#    | QTY | DESCRIPTION       | REFERENCE DESIGNATOR(S) | BOM OPTION |
|----------|-----|-------------------|-------------------------|------------|
| 113S0022 | 2   | RES,ZERO OHM,0603 | L6004,L6092             |            |

RESISTER ON FERRITE PADS!!

OMIT TABLE

L6092

FERR-70-OHM-4A

0603

NOSTUFF

R6090

0

1/10W

MF-LF

402

NOSTUFF

R6091

0

1/10W

MF-LF

402

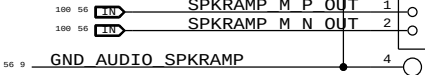
SPEAKER CONNECTOR  
APN 518S0519

J6001

78171-0002

M-RT-SM

3



SYNC MASTER=REFERENCE MLB SYNC DATE=11/23/2016

PAGE TITLE

AUDIO: JACKS



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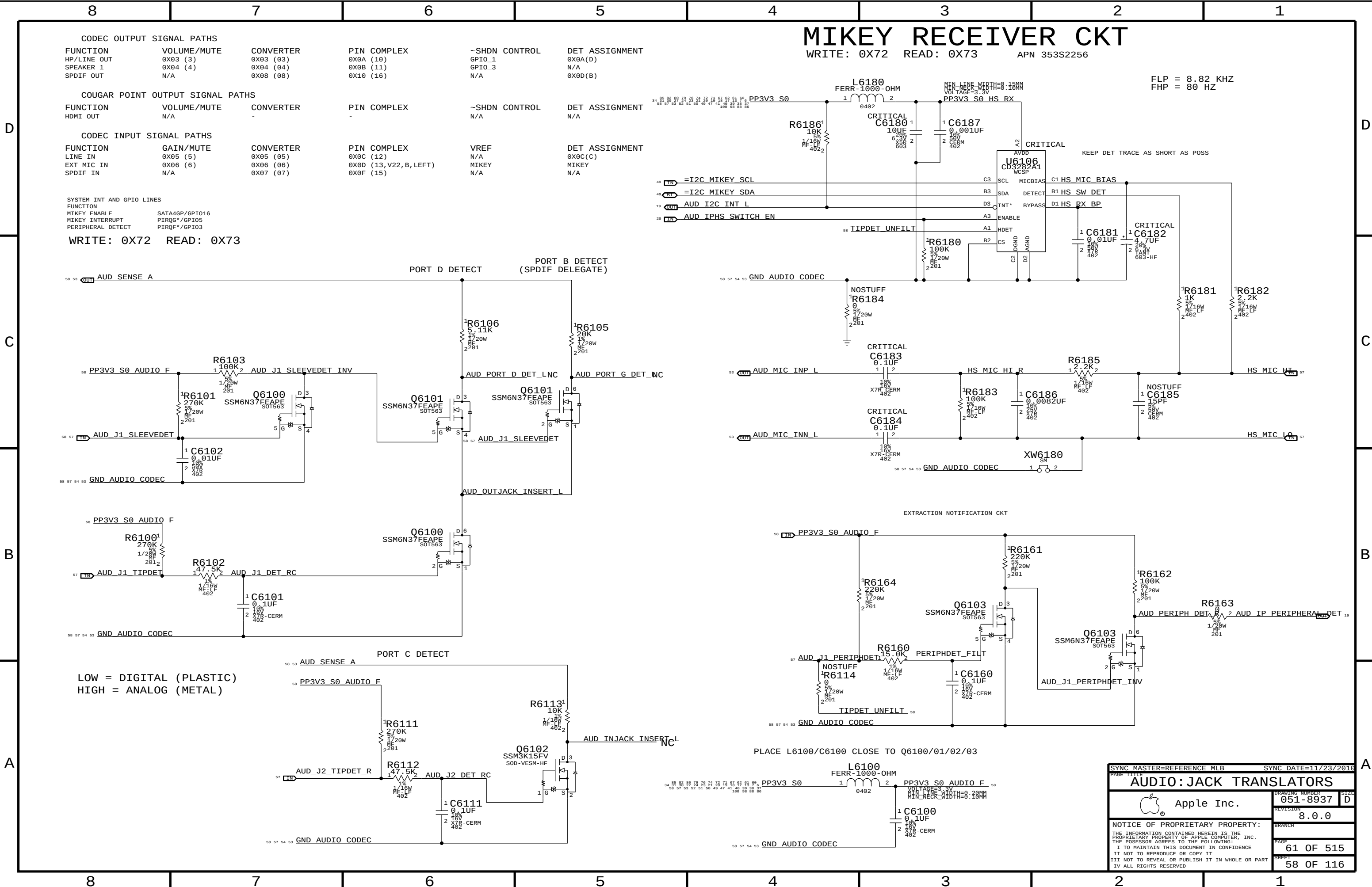
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MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S2256

FLP = 8.82 KHZ  
FHP = 80 HZ

CODEC OUTPUT SIGNAL PATHS

| FUNCTION    | VOLUME/MUTE | CONVERTER | PIN COMPLEX | ~SHDN CONTROL | DET ASSIGNMENT |
|-------------|-------------|-----------|-------------|---------------|----------------|
| HP/LINE OUT | 0X03 (3)    | 0X03 (03) | 0X0A (10)   | GPIO_1        | 0X0A(D)        |
| SPEAKER 1   | 0X04 (4)    | 0X04 (04) | 0X0B (11)   | GPIO_3        | N/A            |
| SPDIF OUT   | N/A         | 0X08 (08) | 0X10 (16)   | N/A           | 0X0D(B)        |

COUGAR POINT OUTPUT SIGNAL PATHS

| FUNCTION | VOLUME/MUTE | CONVERTER | PIN COMPLEX | ~SHDN CONTROL | DET ASSIGNMENT |
|----------|-------------|-----------|-------------|---------------|----------------|
| HDMI OUT | N/A         | -         | -           | N/A           | N/A            |

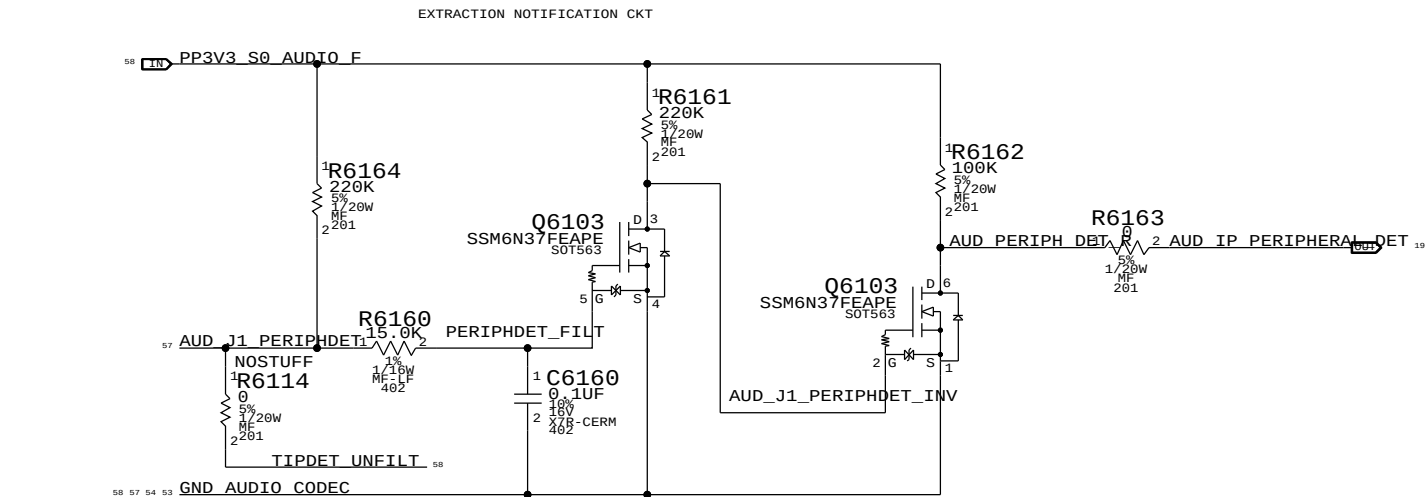
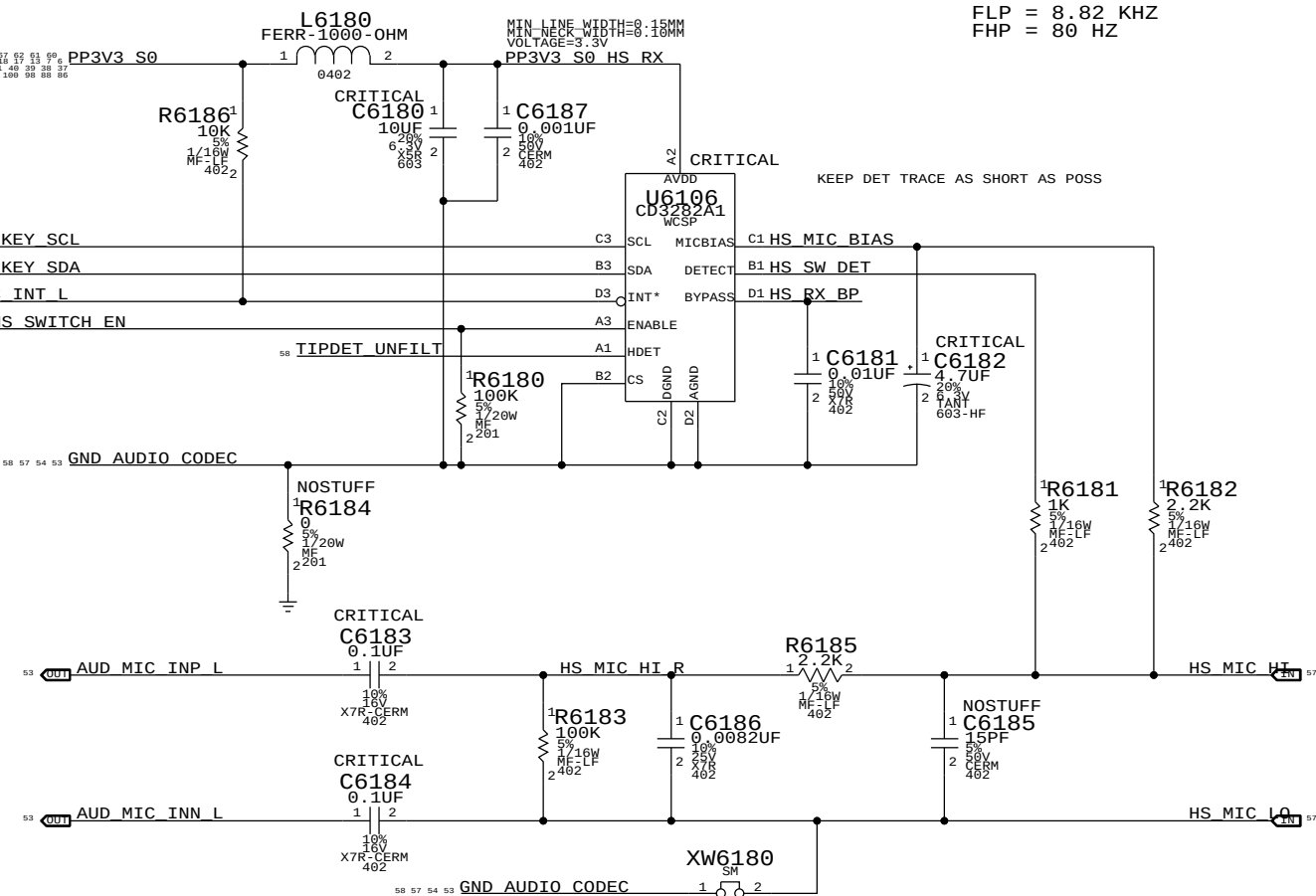
CODEC INPUT SIGNAL PATHS

| FUNCTION   | GAIN/MUTE | CONVERTER | PIN COMPLEX             | VREF  | DET ASSIGNMENT |
|------------|-----------|-----------|-------------------------|-------|----------------|
| LINE IN    | 0X05 (5)  | 0X05 (05) | 0X0C (12)               | N/A   | 0X0C(C)        |
| EXT MIC IN | 0X06 (6)  | 0X06 (06) | 0X0D (13, V22, B, LEFT) | MIKEY | MIKEY          |
| SPDIF IN   | N/A       | 0X07 (07) | 0X0F (15)               | N/A   | N/A            |

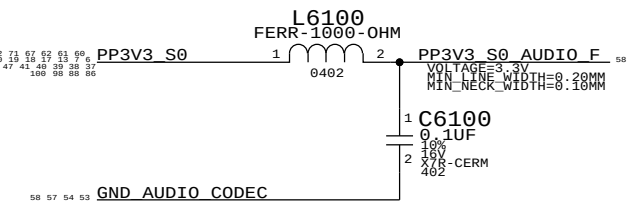
SYSTEM INT AND GPIO LINES

| FUNCTION          |                |
|-------------------|----------------|
| MIKEY ENABLE      | SATA4GP/GPIO16 |
| MIKEY INTERRUPT   | PIRQG*/GPIO5   |
| PERIPHERAL DETECT | PIRQF*/GPIO3   |

WRITE: 0X72 READ: 0X73



PLACE L6100/C6100 CLOSE TO Q6100/01/02/03



SYNC MASTER=REFERENCE MLB

SYNC DATE=11/23/2016

AUDIO: JACK TRANSLATORS

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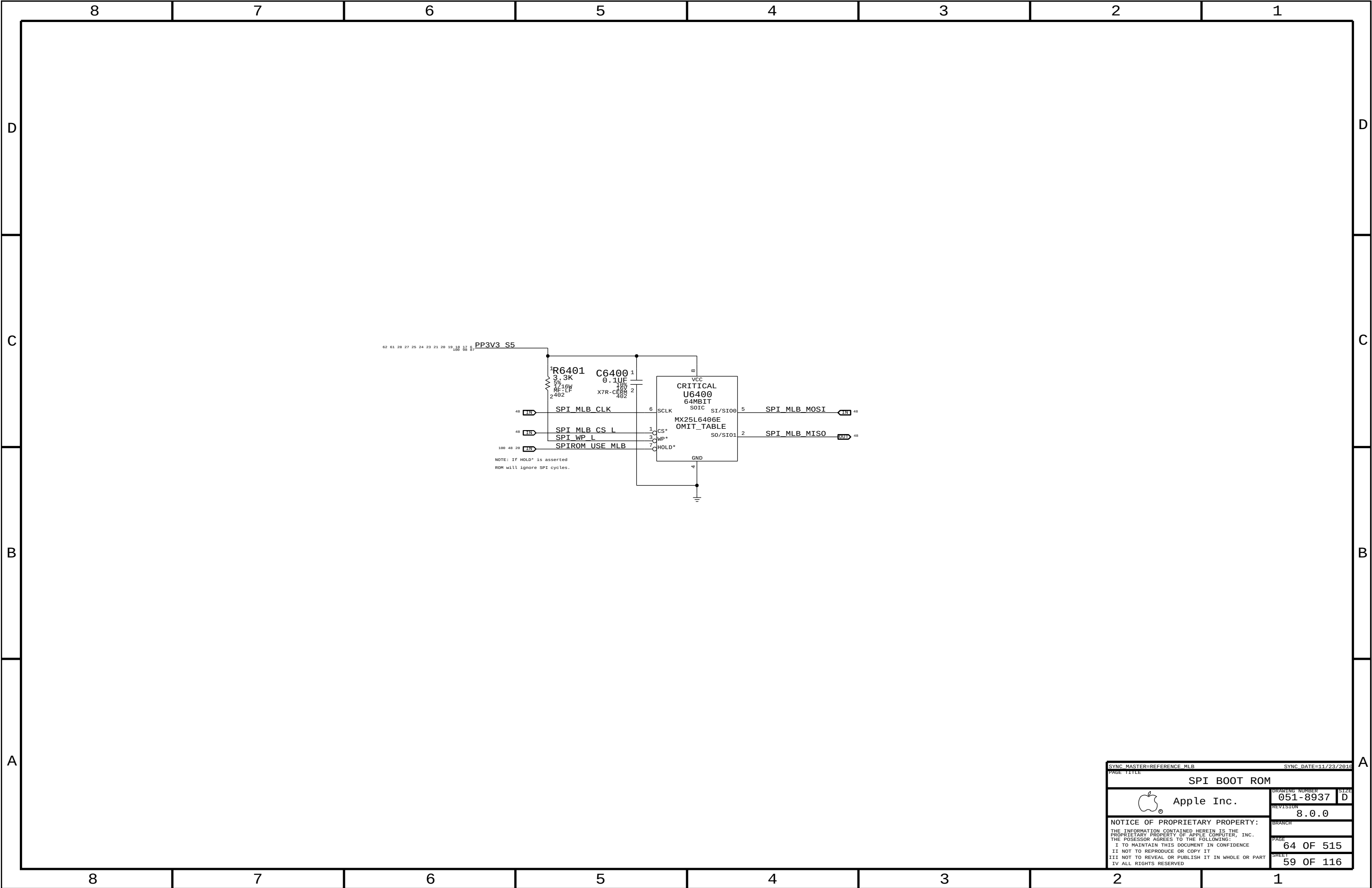
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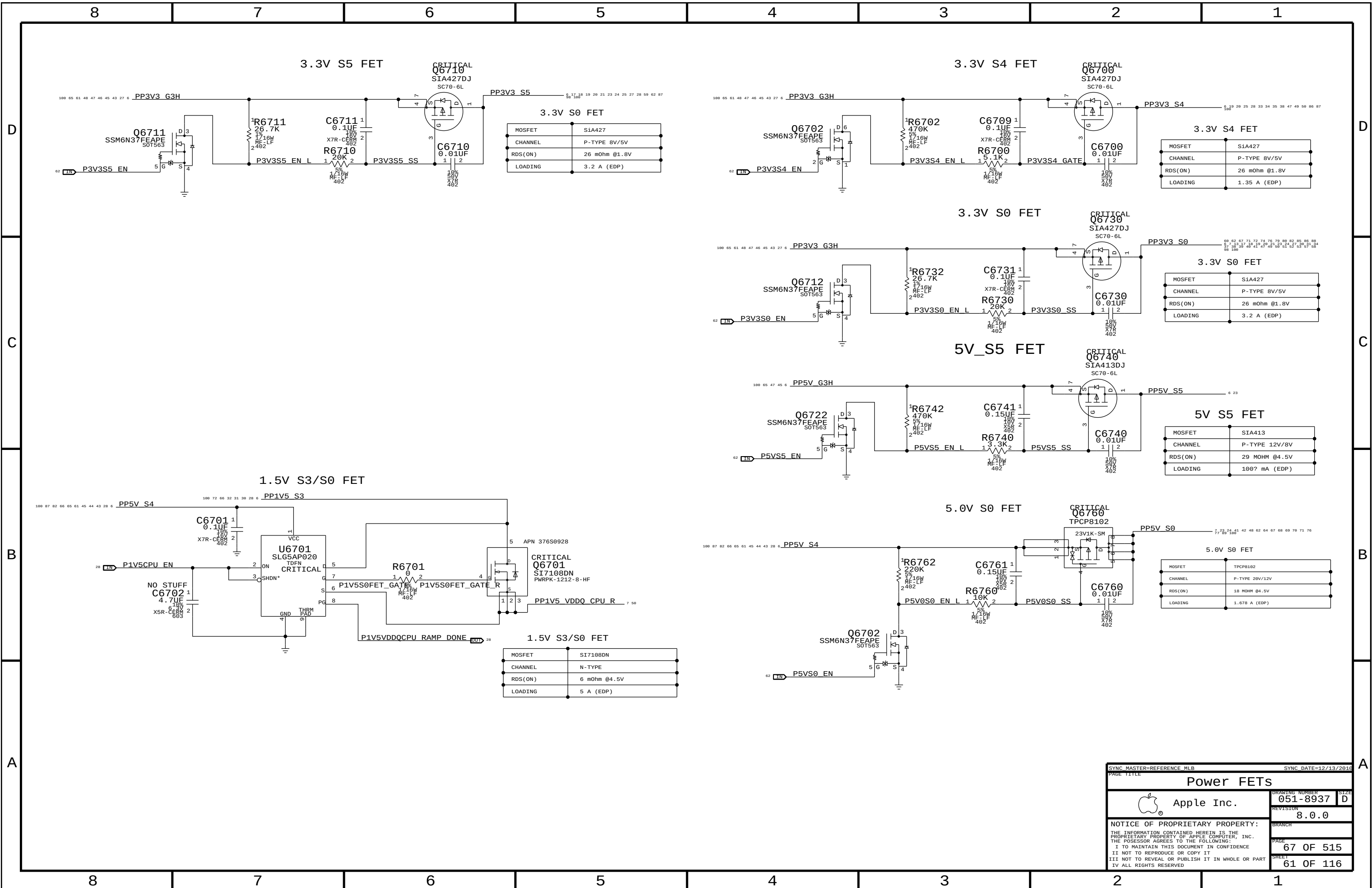
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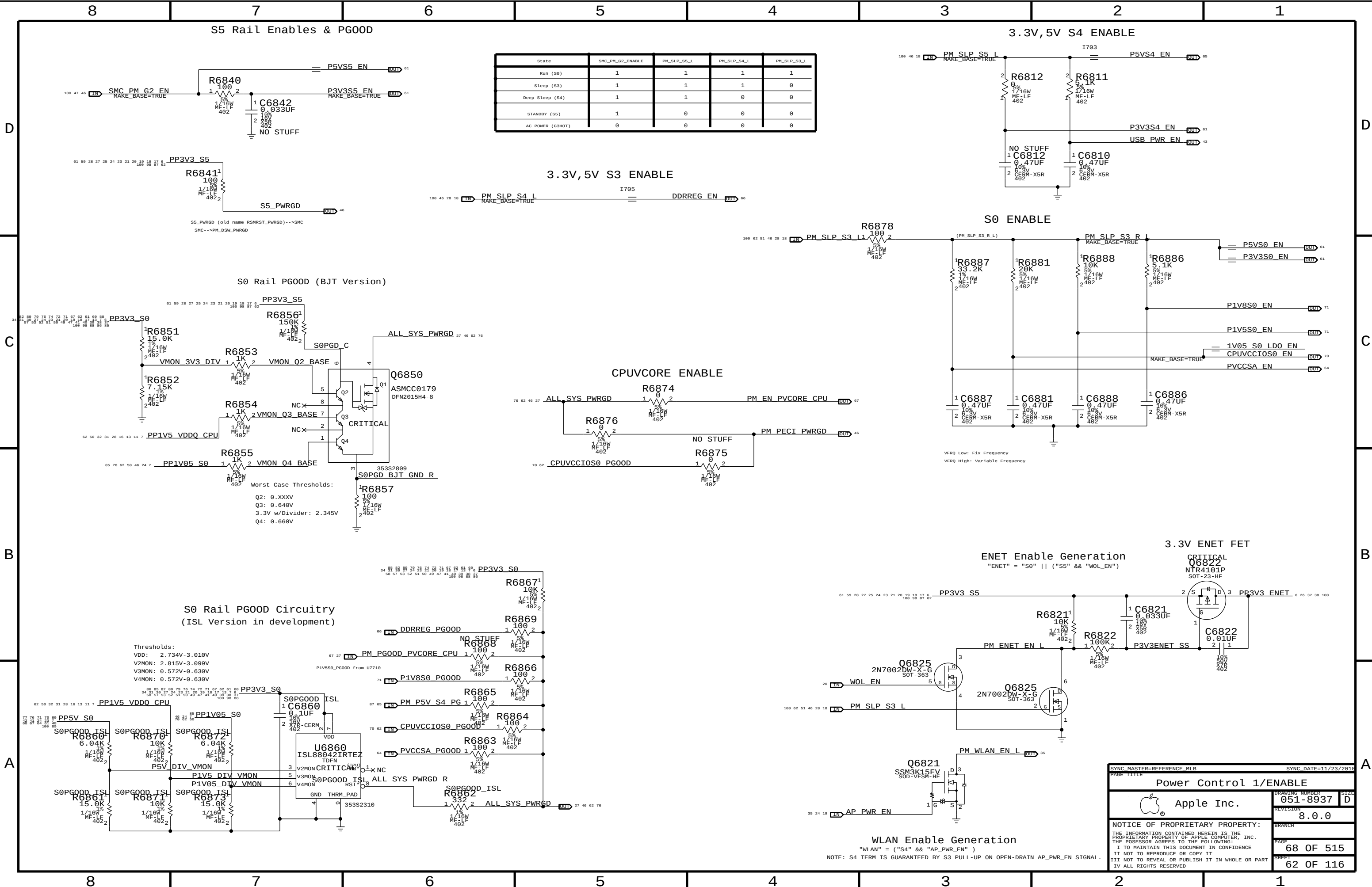
SHEET  
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SIZE  
D



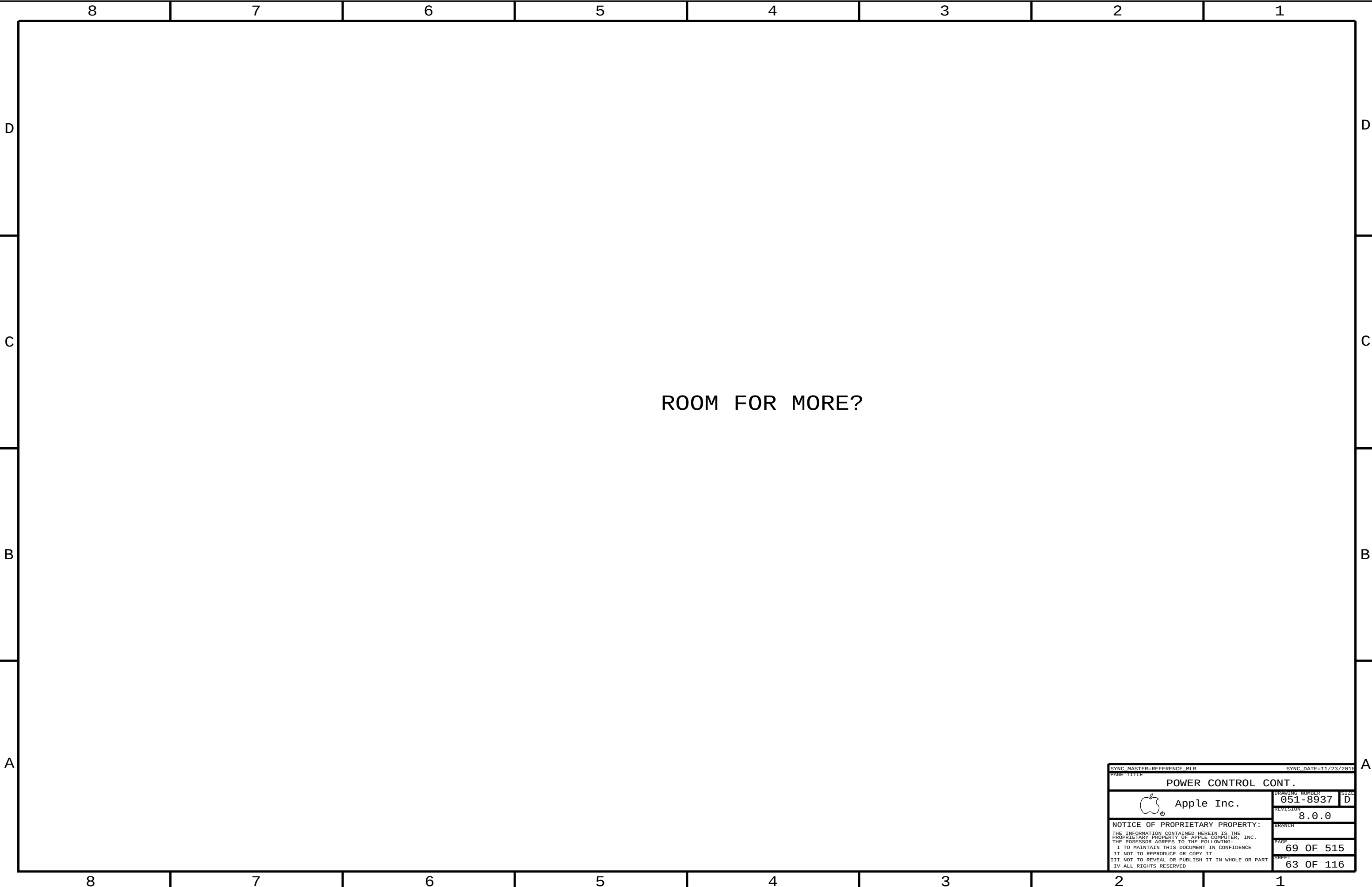






| State            | SMC_PM_G2_ENABLE | PM_SLP_S5_L | PM_SLP_S4_L | PM_SLP_S3_L |
|------------------|------------------|-------------|-------------|-------------|
| Run (S0)         | 1                | 1           | 1           | 1           |
| Sleep (S3)       | 1                | 1           | 1           | 0           |
| Deep Sleep (S4)  | 1                | 1           | 0           | 0           |
| STANDBY (S5)     | 1                | 0           | 0           | 0           |
| AC POWER (G3MGT) | 0                | 0           | 0           | 0           |





SYNC MASTER=REFERENCE\_MLB

SYNC DATE=11/23/2010

PAGE TITLE

POWER CONTROL CONT.

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051-8937

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8.0.0

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PAGE

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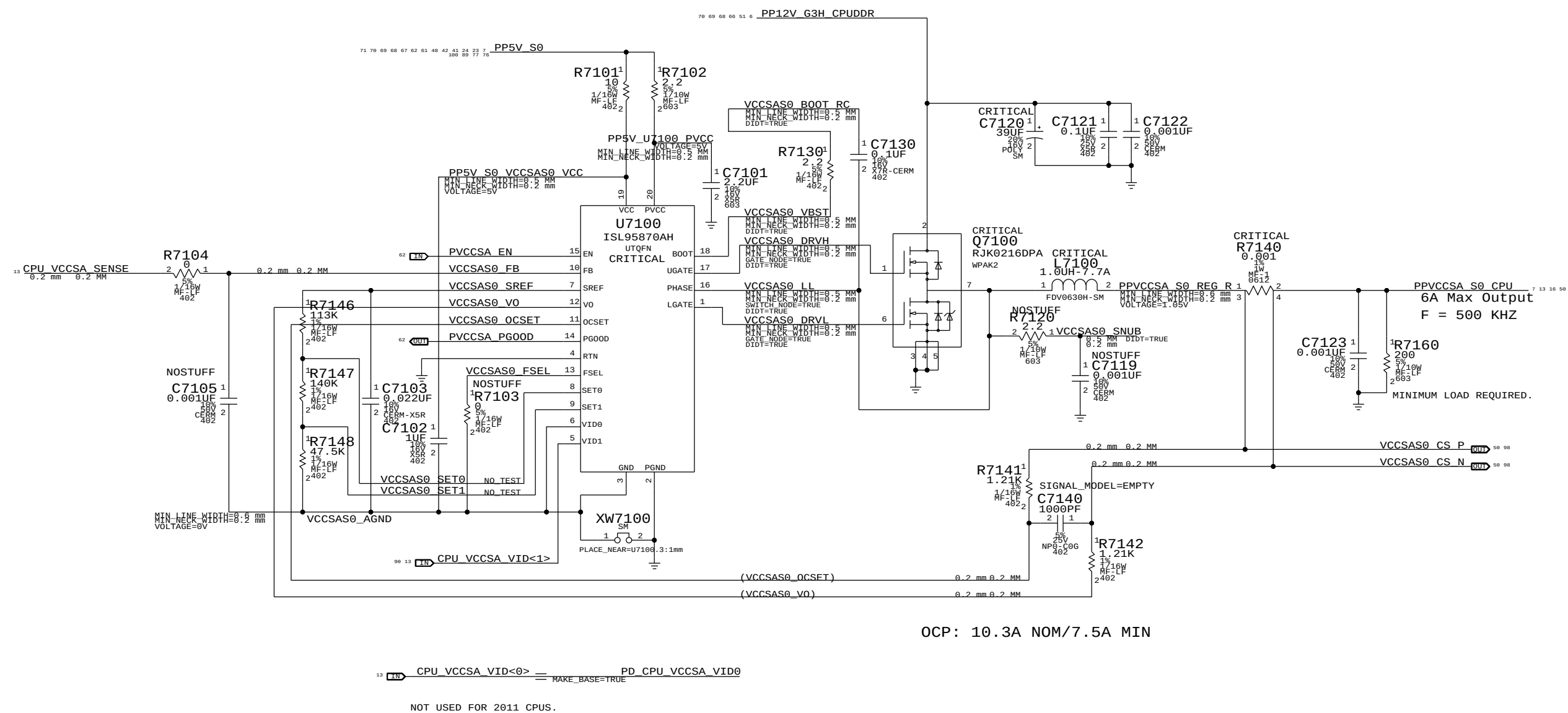
SHEET

63 OF 116

SIZE

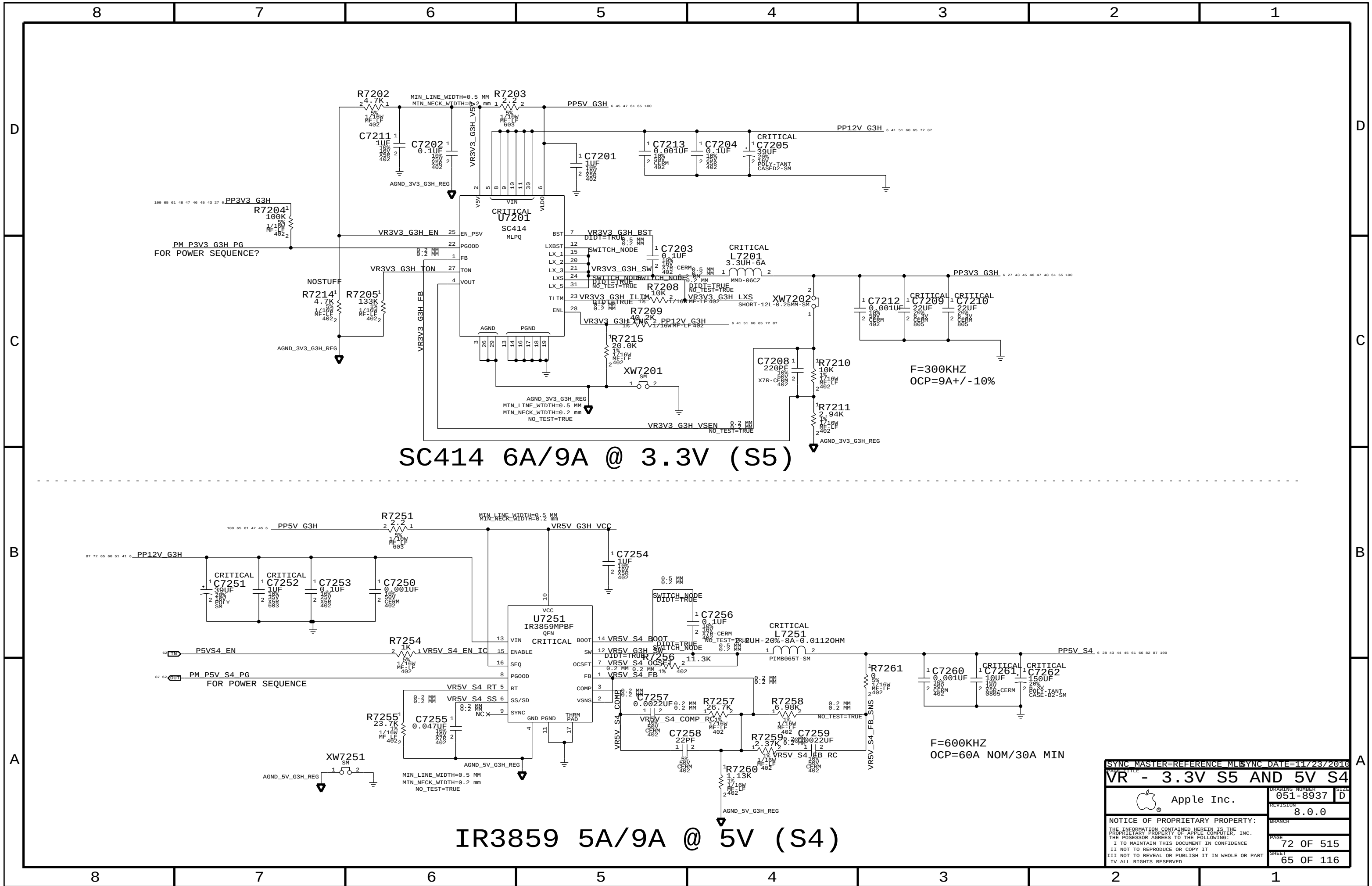
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# System Agent Power Supply

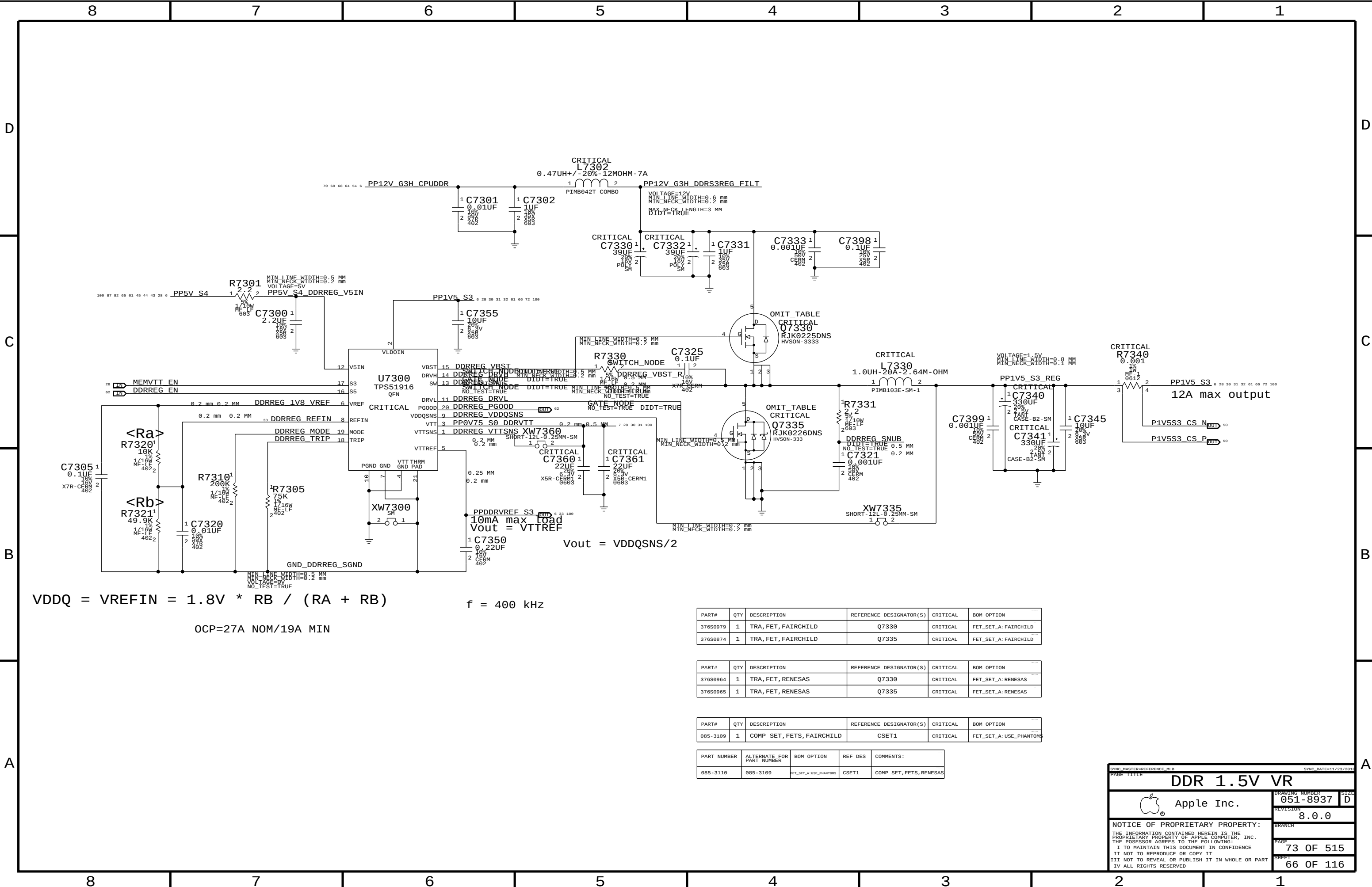


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|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=REFERENCE MLB                                                                                                   |  | SYNC DATE=11/23/2010 |           |
| PAGE TITLE                                                                                                                  |  |                      |           |
| VR - SYSTEM AGENT SUPPLY                                                                                                    |  |                      |           |
|  Apple Inc.                            |  | DRAWING NUMBER       | 051-8937  |
|                                                                                                                             |  | SIZE                 | D         |
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|                                                                                                                                                                                                                                                                                                                                   |  |                |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------|--|
| SYNC MASTER=REFERENCE ML\$SYNC DATE=11/23/2016                                                                                                                                                                                                                                                                                    |  |                |  |
| PAGE: TITLE                                                                                                                                                                                                                                                                                                                       |  |                |  |
| VR - 3.3V S5 AND 5V S4                                                                                                                                                                                                                                                                                                            |  |                |  |
|  Apple Inc.                                                                                                                                                                                                                                  |  | DRAWING NUMBER |  |
|                                                                                                                                                                                                                                                                                                                                   |  | 051-8937       |  |
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|                                                                                                                                                                                                                                                                                                                                   |  | 65 OF 116      |  |



$$VDDQ = VREFIN = 1.8V * RB / (RA + RB)$$
$$f = 400 \text{ kHz}$$
$$OCP=27A \text{ NOM}/19A \text{ MIN}$$

| PART#    | QTY | DESCRIPTION         | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION          |
|----------|-----|---------------------|-------------------------|----------|---------------------|
| 376S9979 | 1   | TRA, FET, FAIRCHILD | Q7330                   | CRITICAL | FET_SET_A:FAIRCHILD |
| 376S9874 | 1   | TRA, FET, FAIRCHILD | Q7335                   | CRITICAL | FET_SET_A:FAIRCHILD |

| PART#    | QTY | DESCRIPTION       | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION        |
|----------|-----|-------------------|-------------------------|----------|-------------------|
| 376S9964 | 1   | TRA, FET, RENESAS | Q7330                   | CRITICAL | FET_SET_A:RENESAS |
| 376S9965 | 1   | TRA, FET, RENESAS | Q7335                   | CRITICAL | FET_SET_A:RENESAS |

| PART#    | QTY | DESCRIPTION               | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION             |
|----------|-----|---------------------------|-------------------------|----------|------------------------|
| 085-3109 | 1   | COMP SET, FETS, FAIRCHILD | CSET1                   | CRITICAL | FET_SET_A:USE_PHANTOMS |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION             | REF DES | COMMENTS:               |
|-------------|---------------------------|------------------------|---------|-------------------------|
| 085-3110    | 085-3109                  | FET_SET_A:USE_PHANTOMS | CSET1   | COMP SET, FETS, RENESAS |

SYNC MASTER-REFERENCE: HLB

SYNC DATE=11/23/2011

DDR 1.5V VR

Apple Inc.

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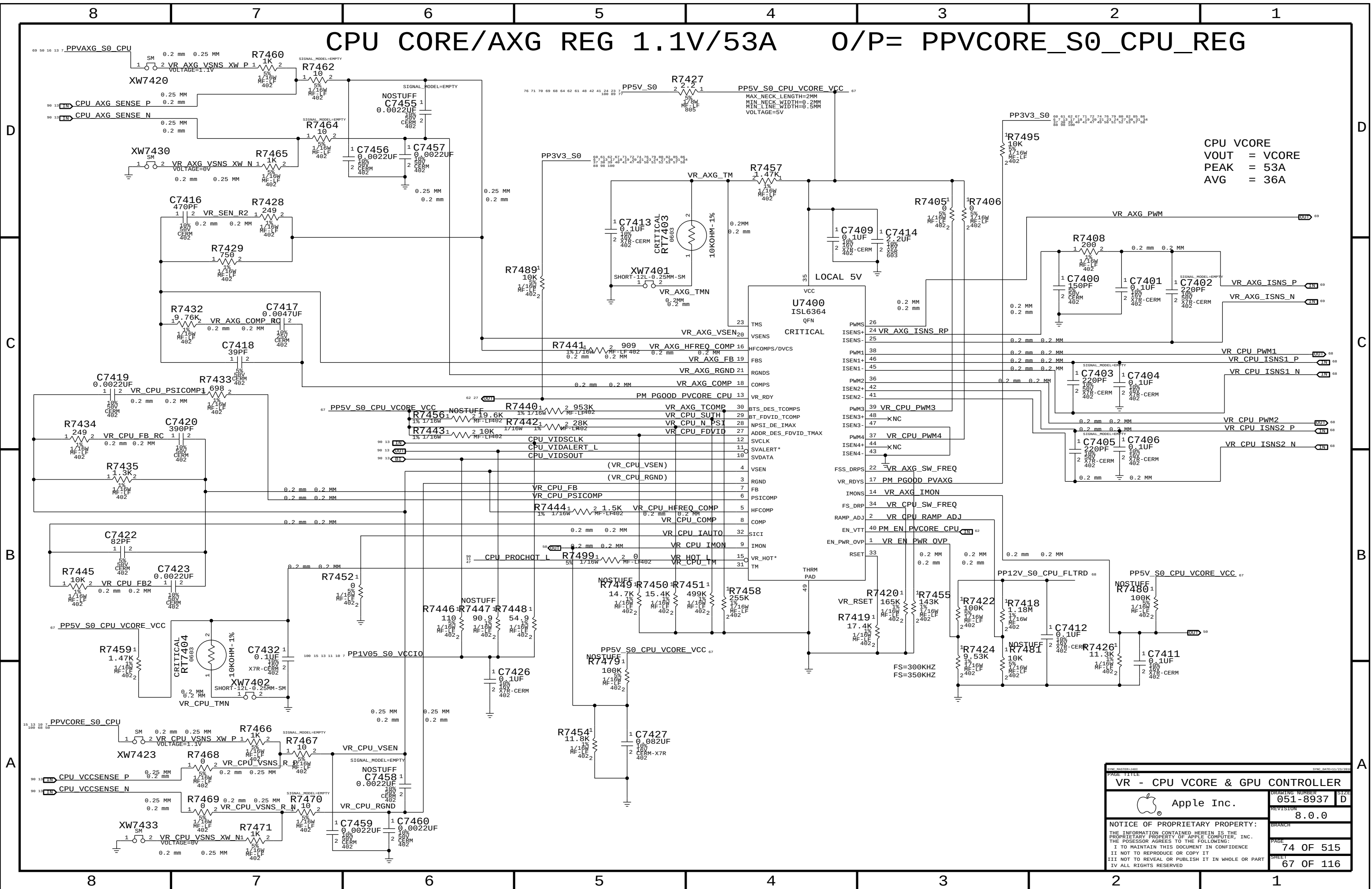
BRANCH

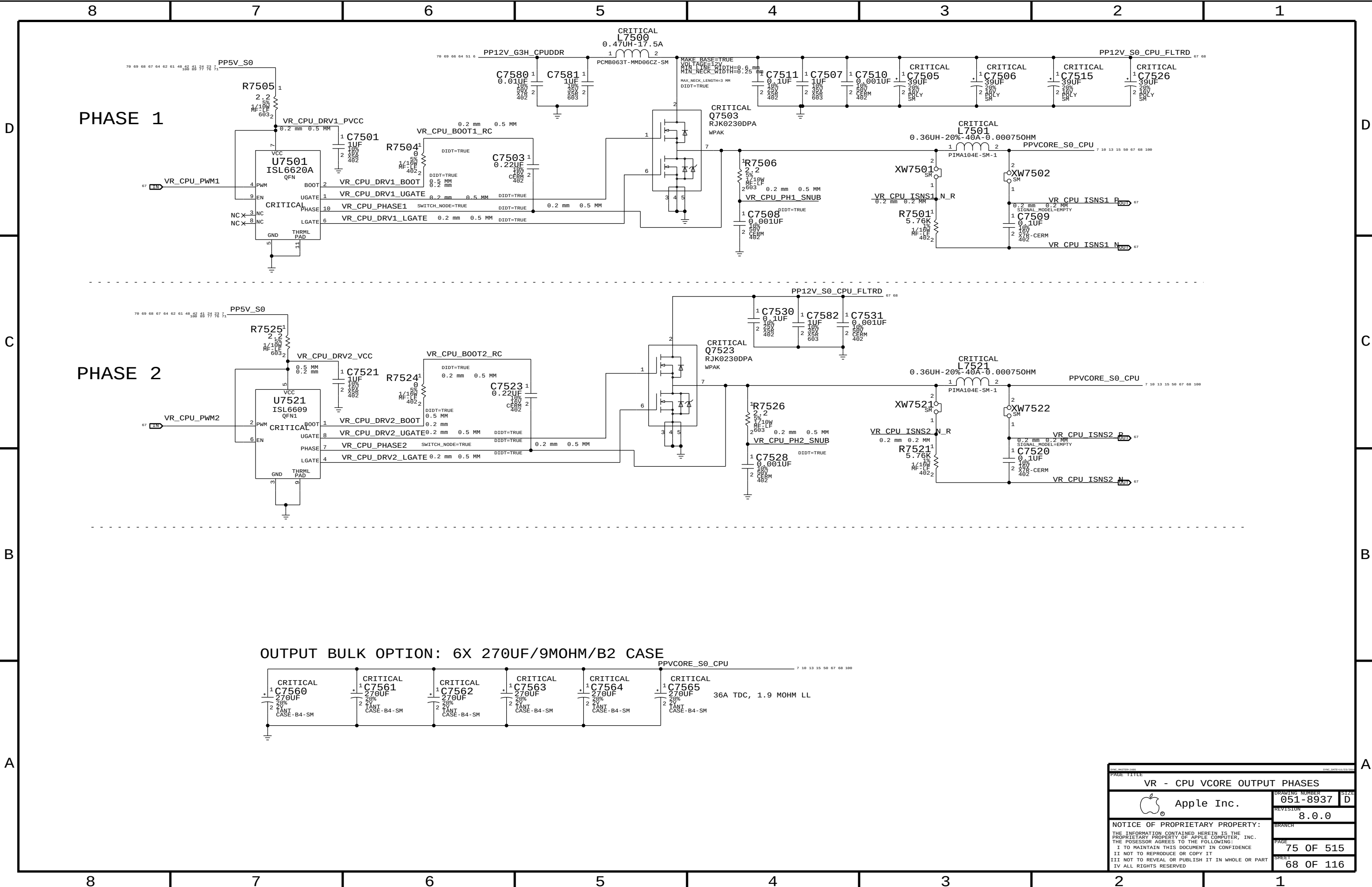
PAGE

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SHEET

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|                                                                                                                             |                |  |           |  |  |
|-----------------------------------------------------------------------------------------------------------------------------|----------------|--|-----------|--|--|
| VR - CPU Vcore Output Phases                                                                                                |                |  |           |  |  |
|  Apple Inc.                            | DRAWING NUMBER |  | 051-8937  |  |  |
|                                                                                                                             | REVISION       |  | 8.0.0     |  |  |
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AXG PHASE (8A AVG/15A PEAK, 3.9MOHM LL)

VR - GPU

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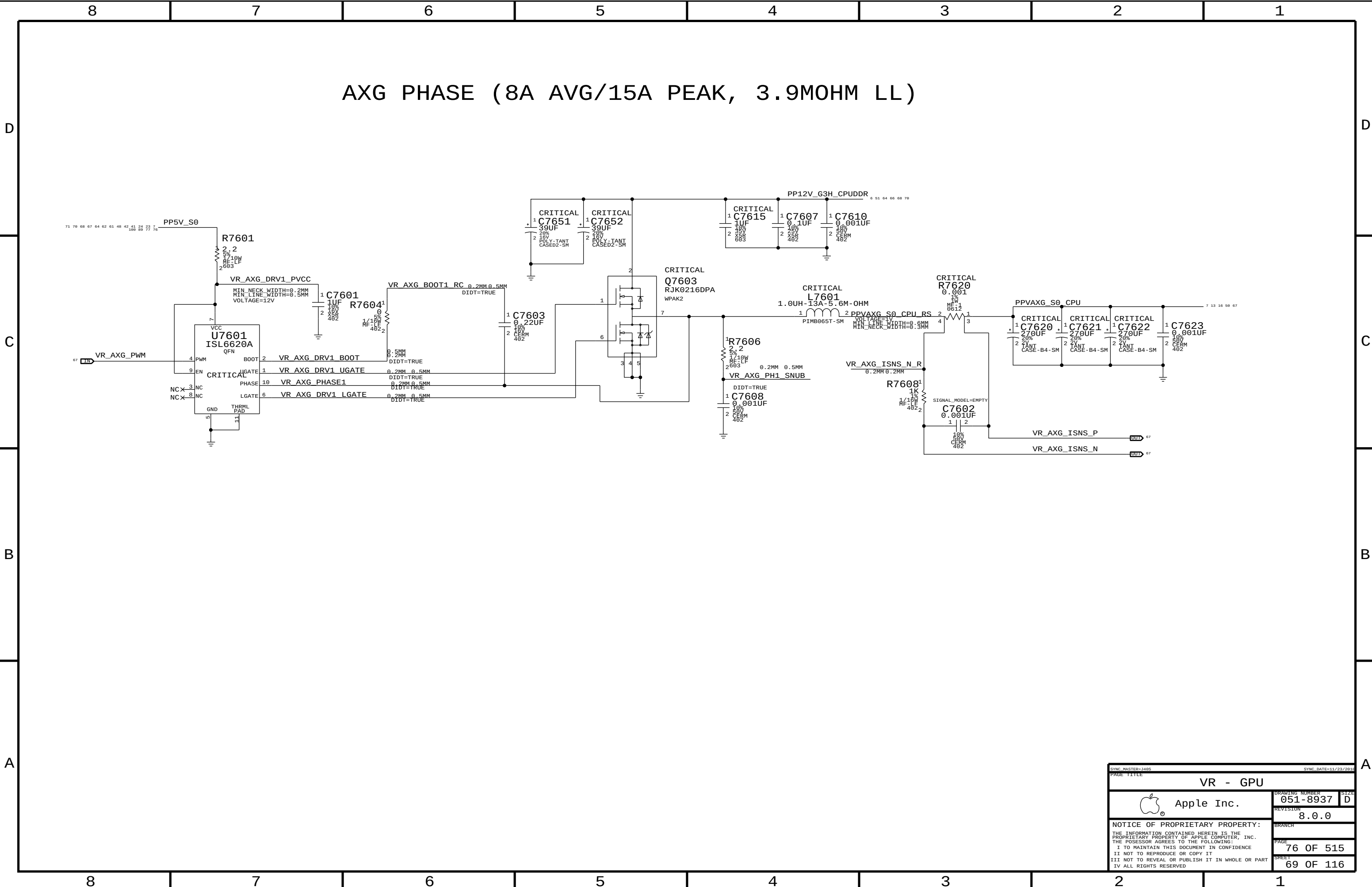
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DRAWING NUMBER: 051-8937

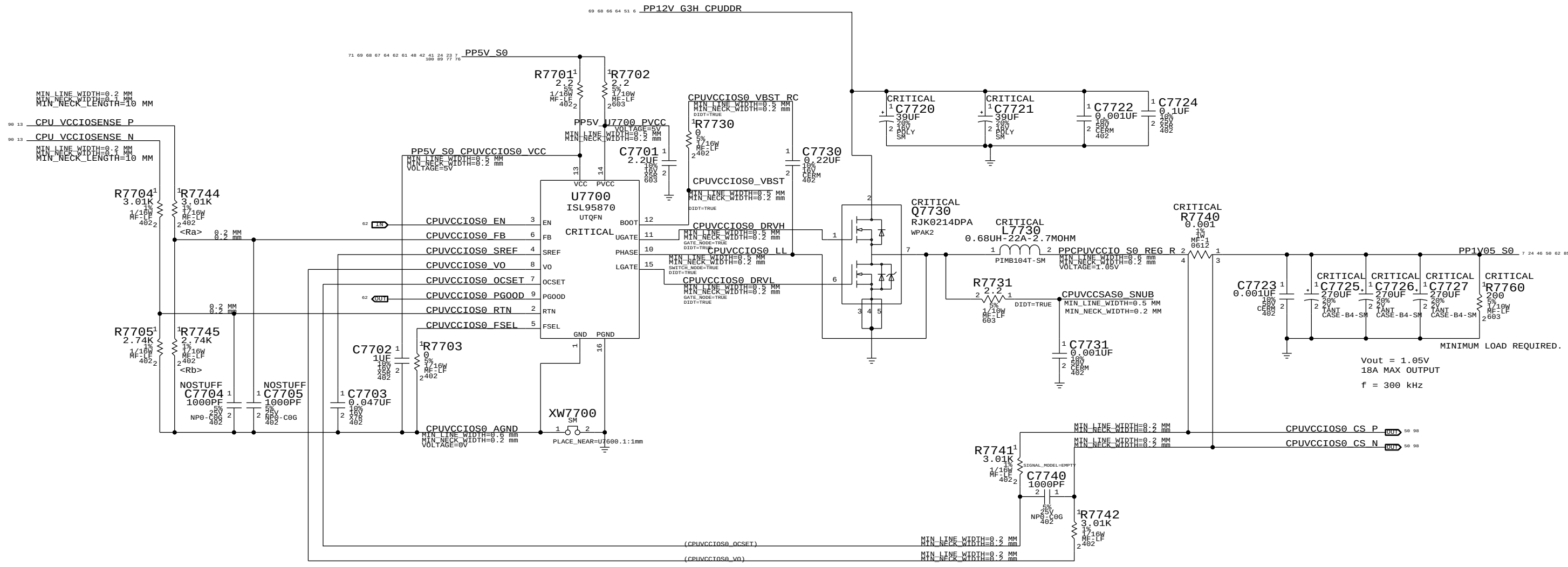
REVISION: 8.0.0

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CPU VCCIO (1.05V S0) REGULATOR @ 18A/22A

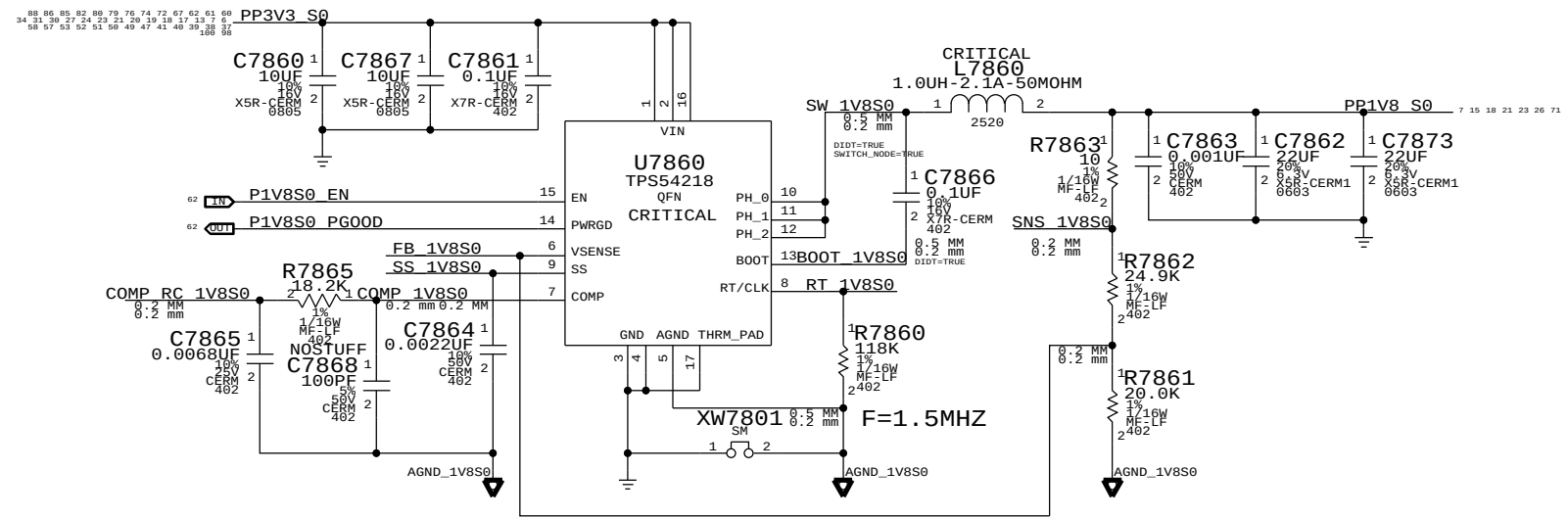


OCP =  $R7641 \times 8.5\mu A / R7640$   
OCP = 25.6A NOM/21.3A MIN  
 $V_{out} = 0.5V \times (1 + R_a / R_b)$

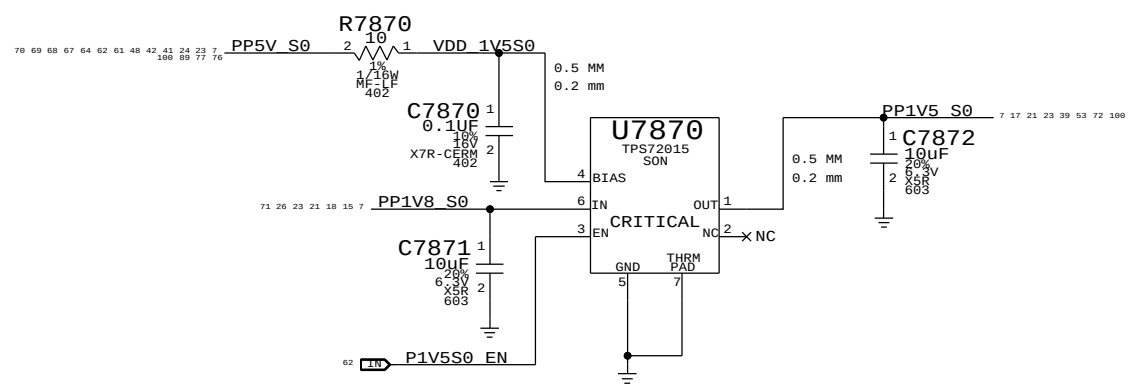
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|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=J49G                                                                                                            |  | SYNC DATE=12/02/2016 |           |
| PAGE TITLE                                                                                                                  |  | VR - 1.05V S0        |           |
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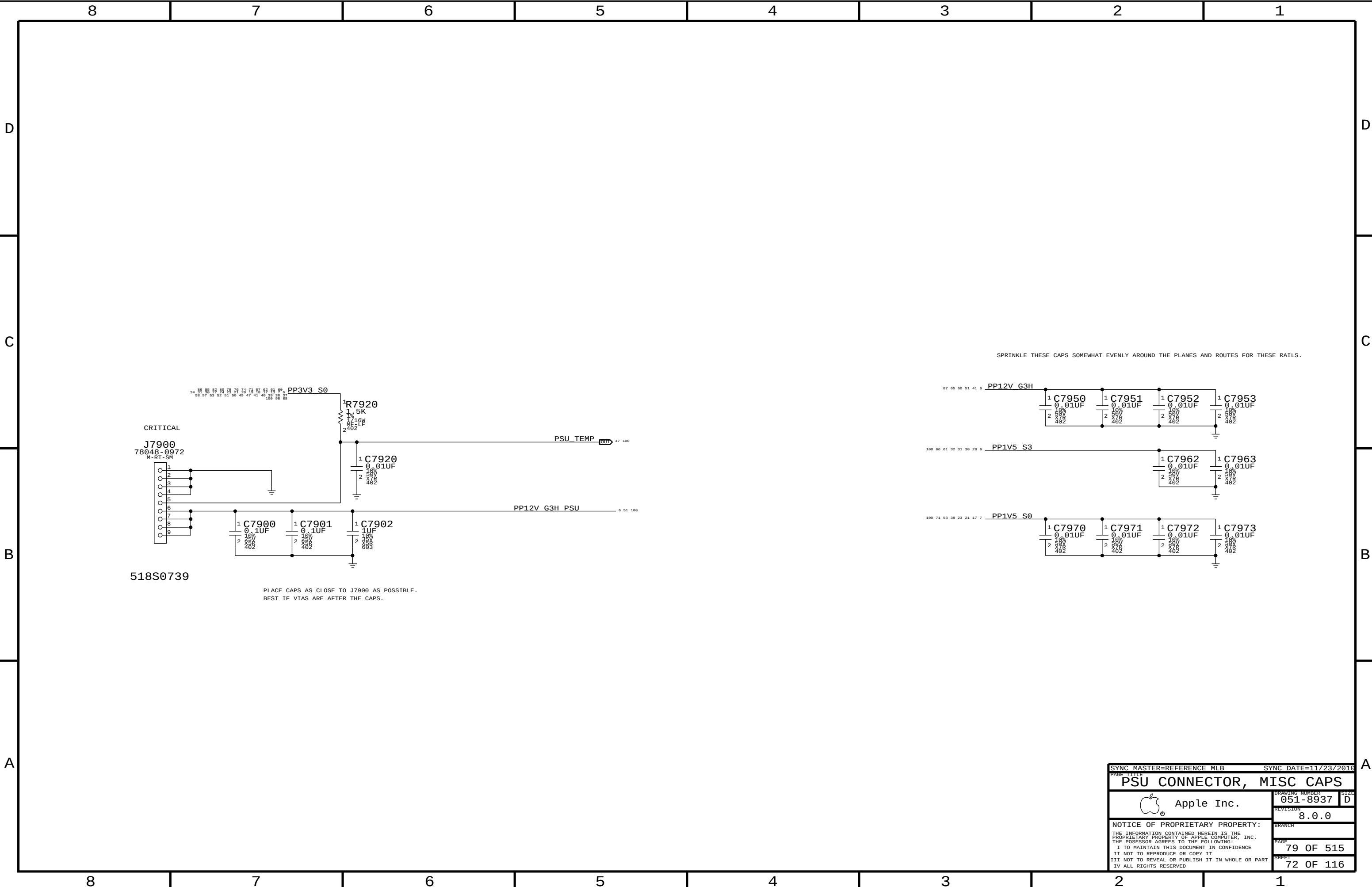


1.8V S0 SWITCHER @ 1.7A/2A



1.5V S0 @ 0.3A/0.5A



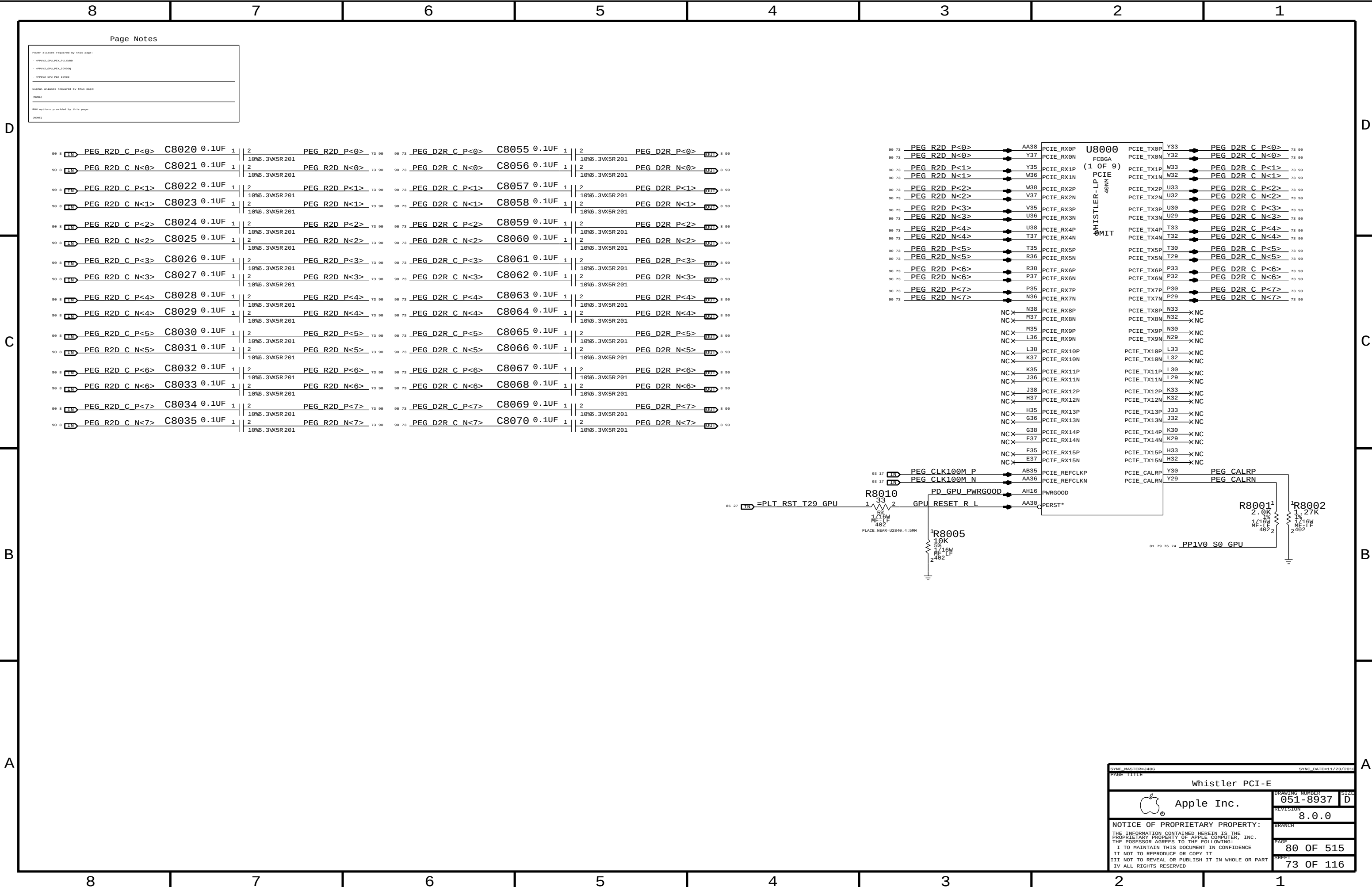


SYNC MASTER=REFERENCE MLB

SYNC DATE=11/23/2016

|                                                                                                                             |  |                          |           |
|-----------------------------------------------------------------------------------------------------------------------------|--|--------------------------|-----------|
| PAGE TITLE                                                                                                                  |  | PSU CONNECTOR, MISC CAPS |           |
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Page Notes

Power aliases required by this page:

- ~PP1V0\_GPU\_PEX\_PL1000
- ~PP1V0\_GPU\_PEX\_PL1000Q
- ~PP1V0\_GPU\_PEX\_PL1000Q

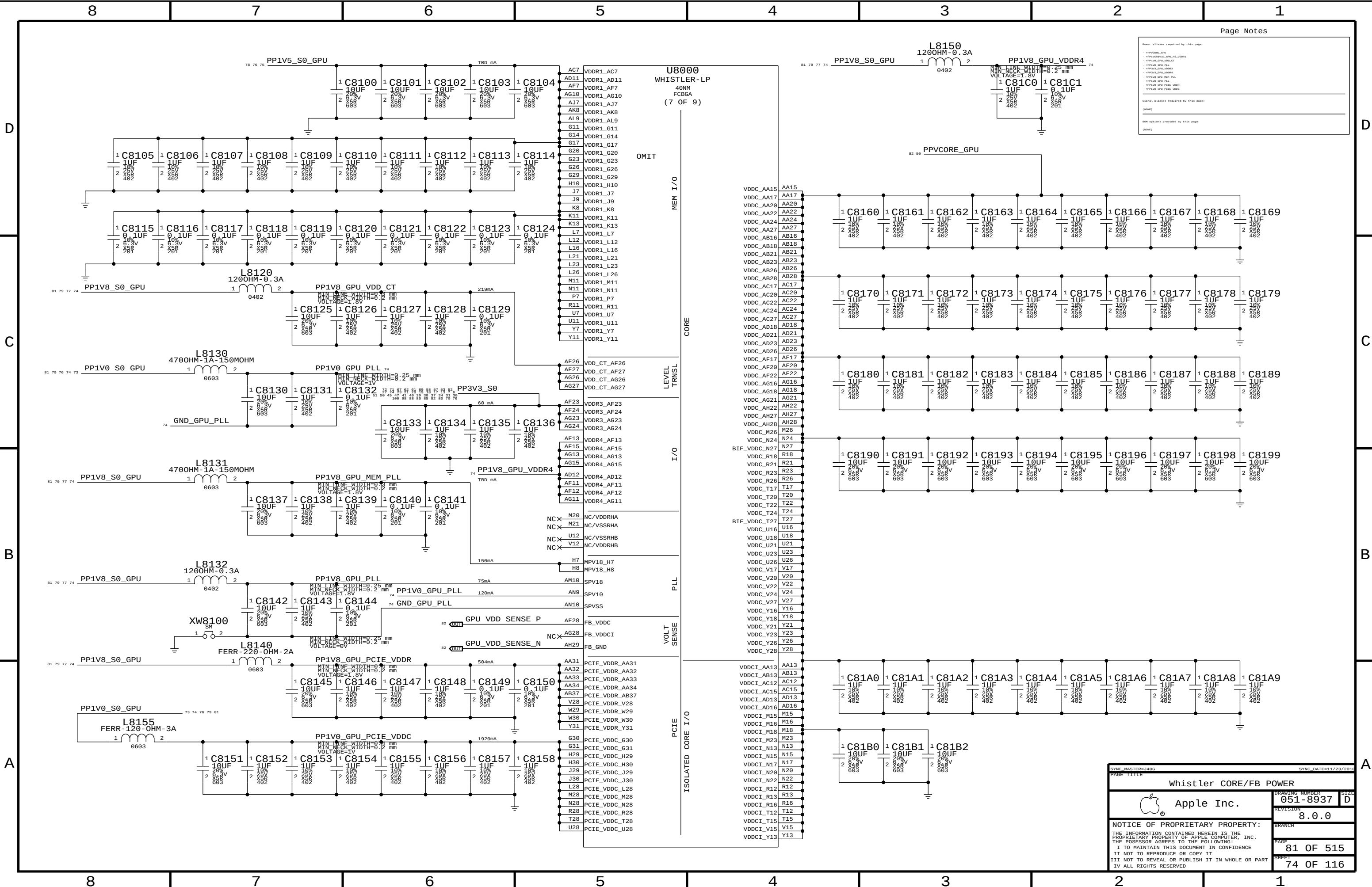
Signal aliases required by this page:

(NONE)

SDR options provided by this page:

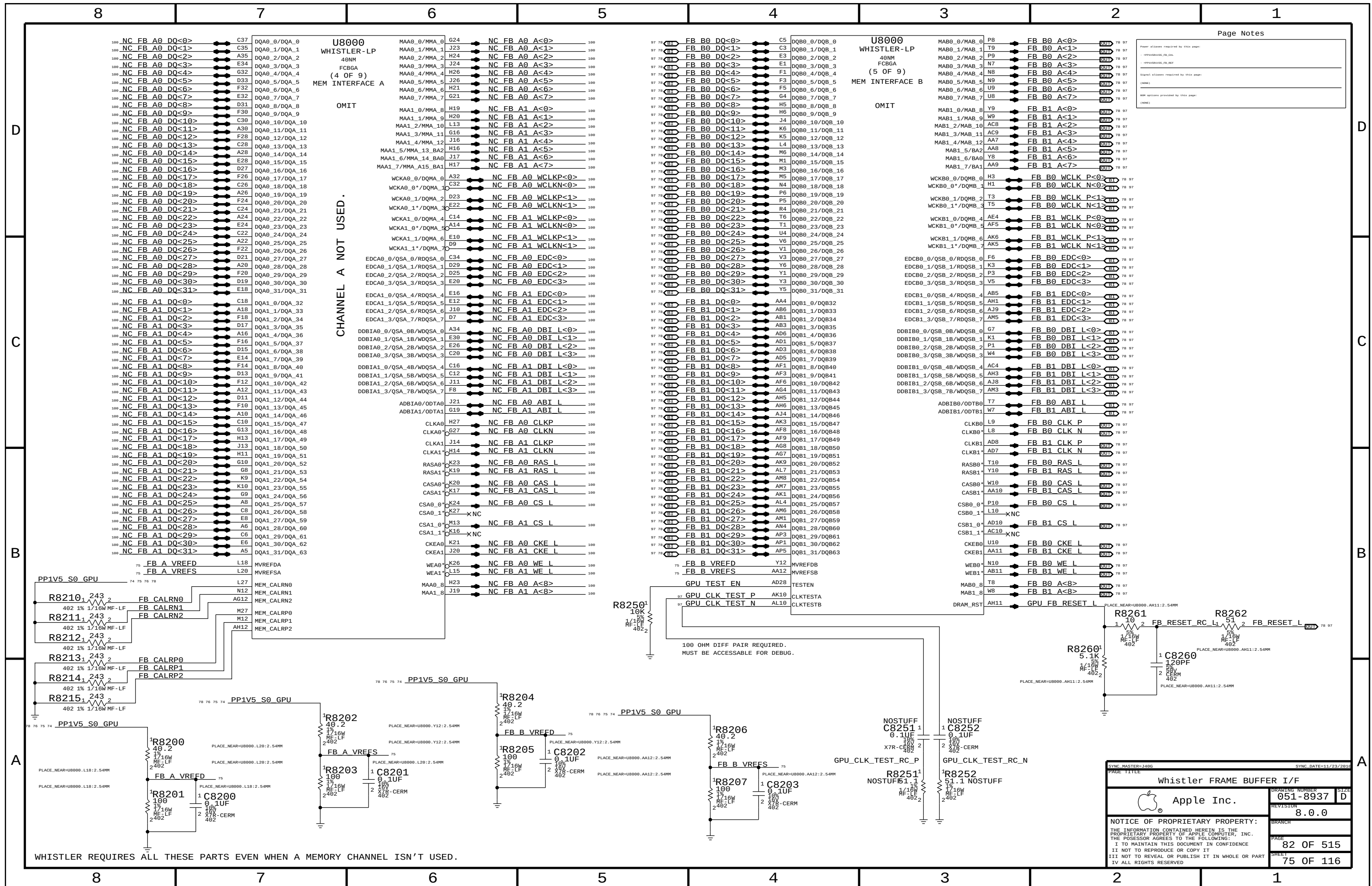
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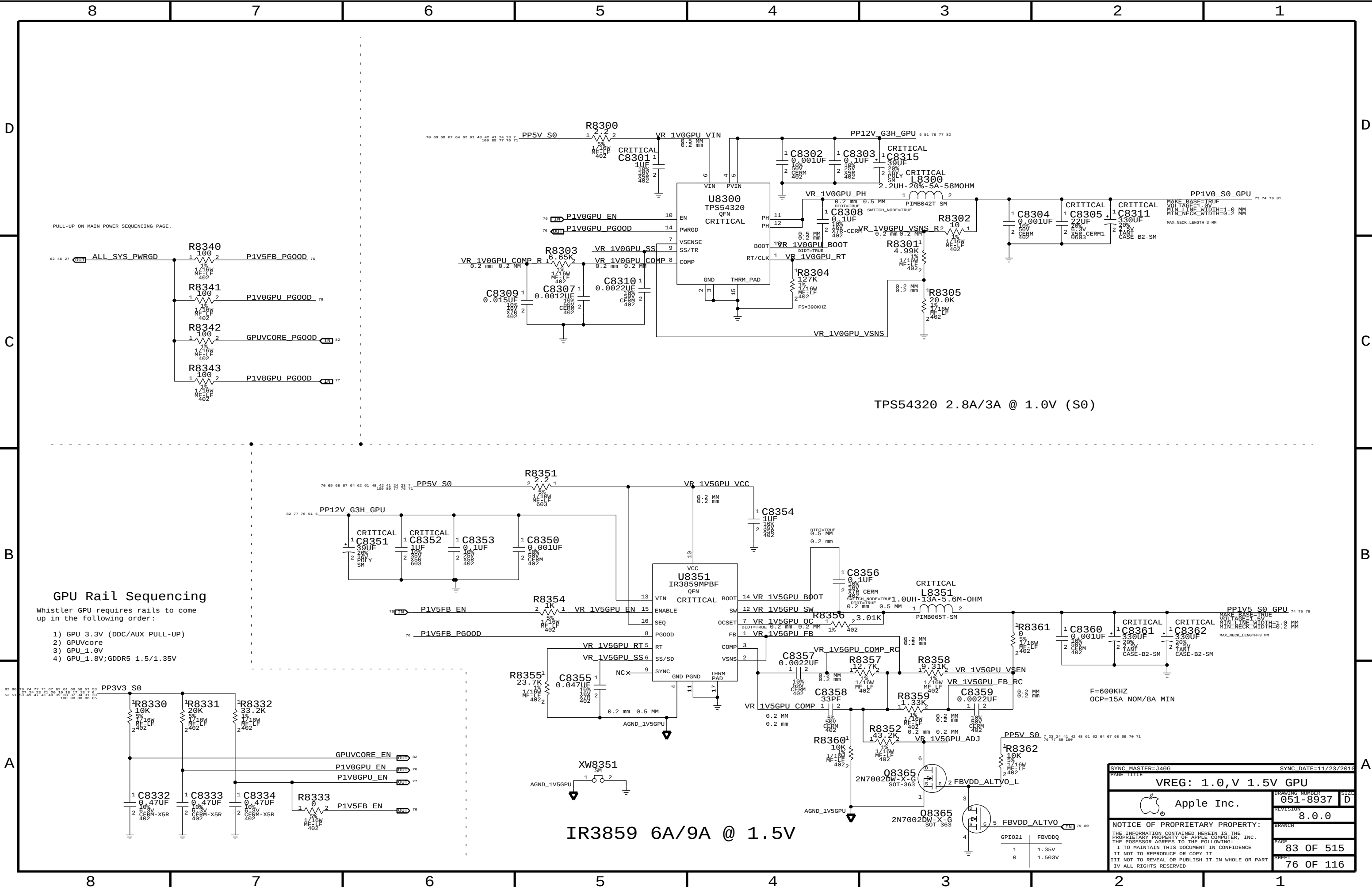
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| SYNC MASTER=1486                                                                                                                                                                                                                                                                                                            |  | SYNC DATE=11/23/2016 |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                  |  |                      |      |
| Whistler PCI-E                                                                                                                                                                                                                                                                                                              |  | DRAWING NUMBER       | SIZE |
|  Apple Inc.                                                                                                                                                                                                                            |  | 051-8937             | D    |
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| PAGE                                                                                                                                                                                                                                                                                                                        |  | 80 OF 515            |      |
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| Page Notes                           |  |
|--------------------------------------|--|
| Power planes required by this page:  |  |
| - PPVCORE_GPU                        |  |
| - PPVCORE_GPU_FB_VDD01               |  |
| - PPVCORE_GPU_VDD01                  |  |
| - PPVCORE_GPU_VDD02                  |  |
| - PPVCORE_GPU_VDD03                  |  |
| - PPVCORE_GPU_VDD04                  |  |
| - PPVCORE_GPU_VDD05                  |  |
| - PPVCORE_GPU_VDD06                  |  |
| - PPVCORE_GPU_VDD07                  |  |
| - PPVCORE_GPU_VDD08                  |  |
| - PPVCORE_GPU_VDD09                  |  |
| - PPVCORE_GPU_VDD10                  |  |
| - PPVCORE_GPU_VDD11                  |  |
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| - PPVCORE_GPU_VDD98                  |  |
| - PPVCORE_GPU_VDD99                  |  |
| - PPVCORE_GPU_VDD100                 |  |
| Signal planes required by this page: |  |
| (NONE)                               |  |
| BOM options provided by this page:   |  |
| (NONE)                               |  |

|                                                                                                                             |  |             |  |
|-----------------------------------------------------------------------------------------------------------------------------|--|-------------|--|
| PAGE TITLE                                                                                                                  |  | PAGE NUMBER |  |
| Whistler CORE/FB POWER                                                                                                      |  | 051-8937    |  |
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PULL-UP ON MAIN POWER SEQUENCING PAGE.

TPS54320 2.8A/3A @ 1.0V (S0)

### GPU Rail Sequencing

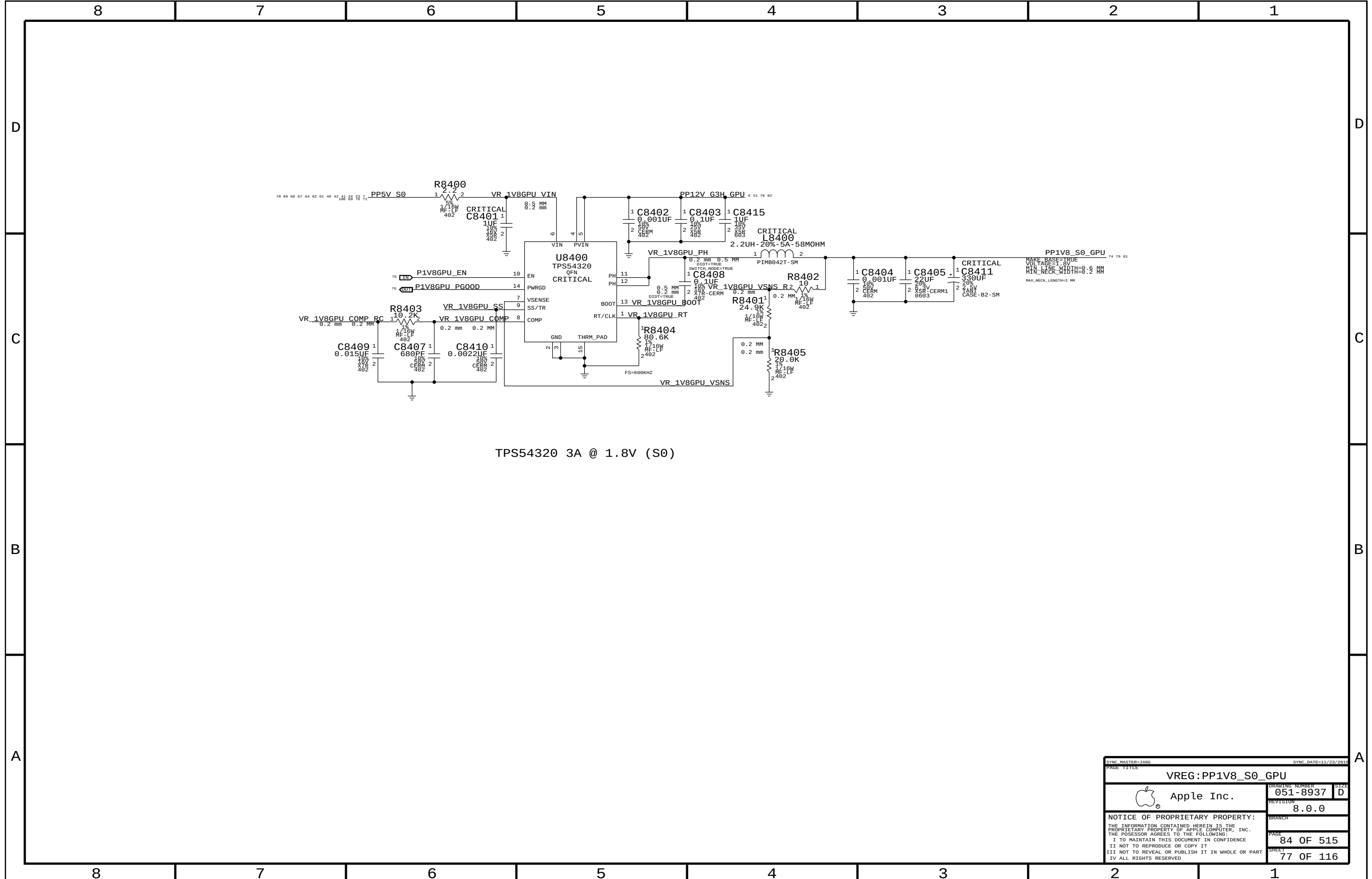
Whistler GPU requires rails to come up in the following order:

- 1) GPU\_3.3V (D/C/AUX PULL-UP)
- 2) GPUVcore
- 3) GPU\_1.0V
- 4) GPU\_1.8V;GDDR5 1.5/1.35V

IR3859 6A/9A @ 1.5V

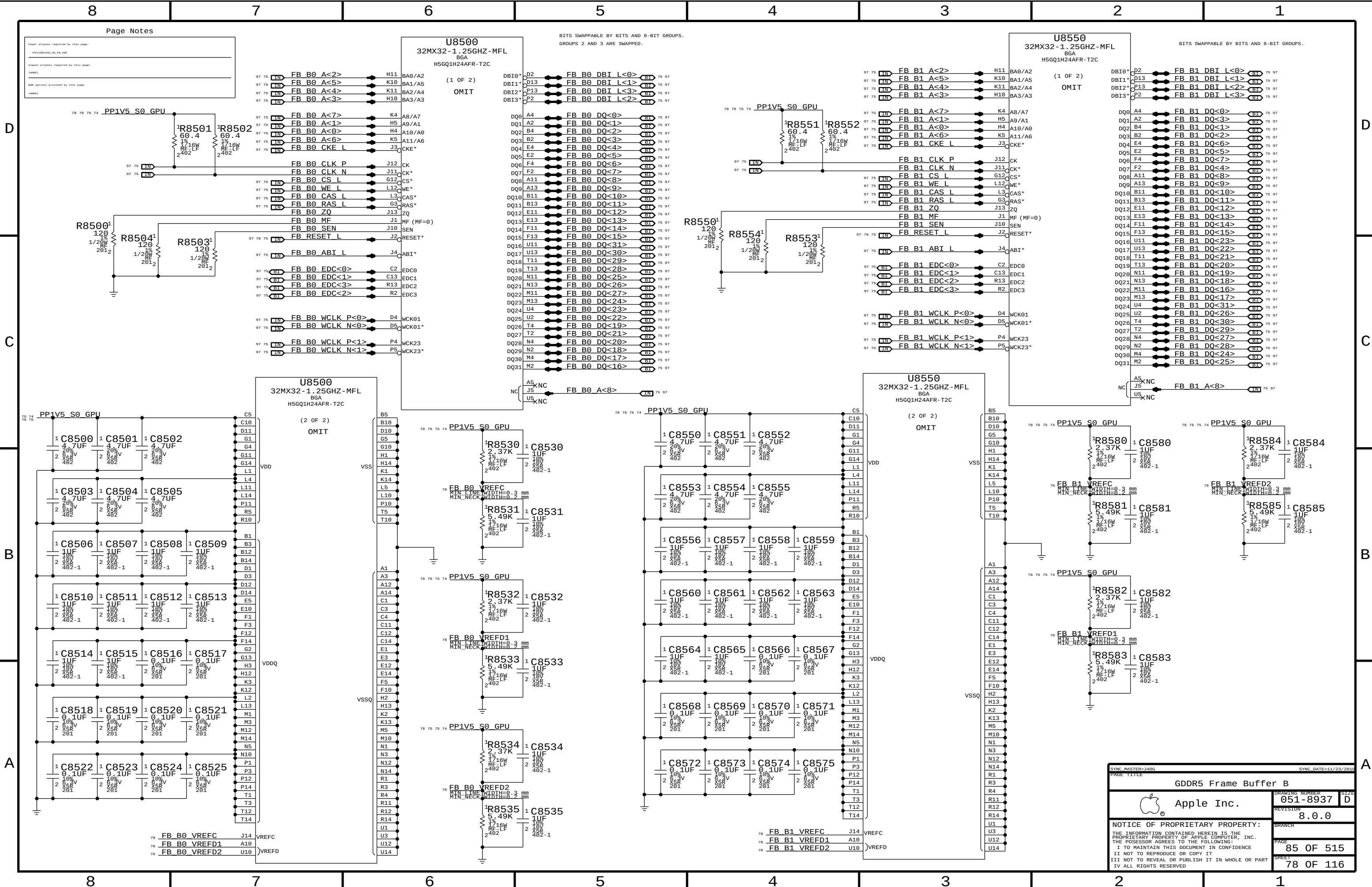
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| SYNC MASTER=J40G                                                                                                            |  | SYNC DATE=11/23/2016 |           |
| PAGE TITLE                                                                                                                  |  | VREG: 1.0,V 1.5V GPU |           |
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|        |        |
|--------|--------|
| GPIO21 | FBVDDQ |
| 1      | 1.35V  |
| 0      | 1.503V |



TPS54320 3A @ 1.8V (S0)

|                                                                                                                             |            |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------|------------|----------------------|-----------|
| SYNC_MASTER=1486                                                                                                            |            | SYNC_DATE=11/23/2016 |           |
| PAGE TITLE                                                                                                                  |            |                      |           |
| VREG:PP1V8_S0_GPU                                                                                                           |            |                      |           |
|                                        | Apple Inc. | DRAWING NUMBER       | 051-8937  |
|                                                                                                                             |            | SIZE                 | D         |
|                                                                                                                             |            | REVISION             | 8.0.0     |
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|                                                                                                                             |            | PAGE                 | 84 OF 515 |
|                                                                                                                             |            | SHEET                | 77 OF 116 |



Page Notes

Power classes required by this page:

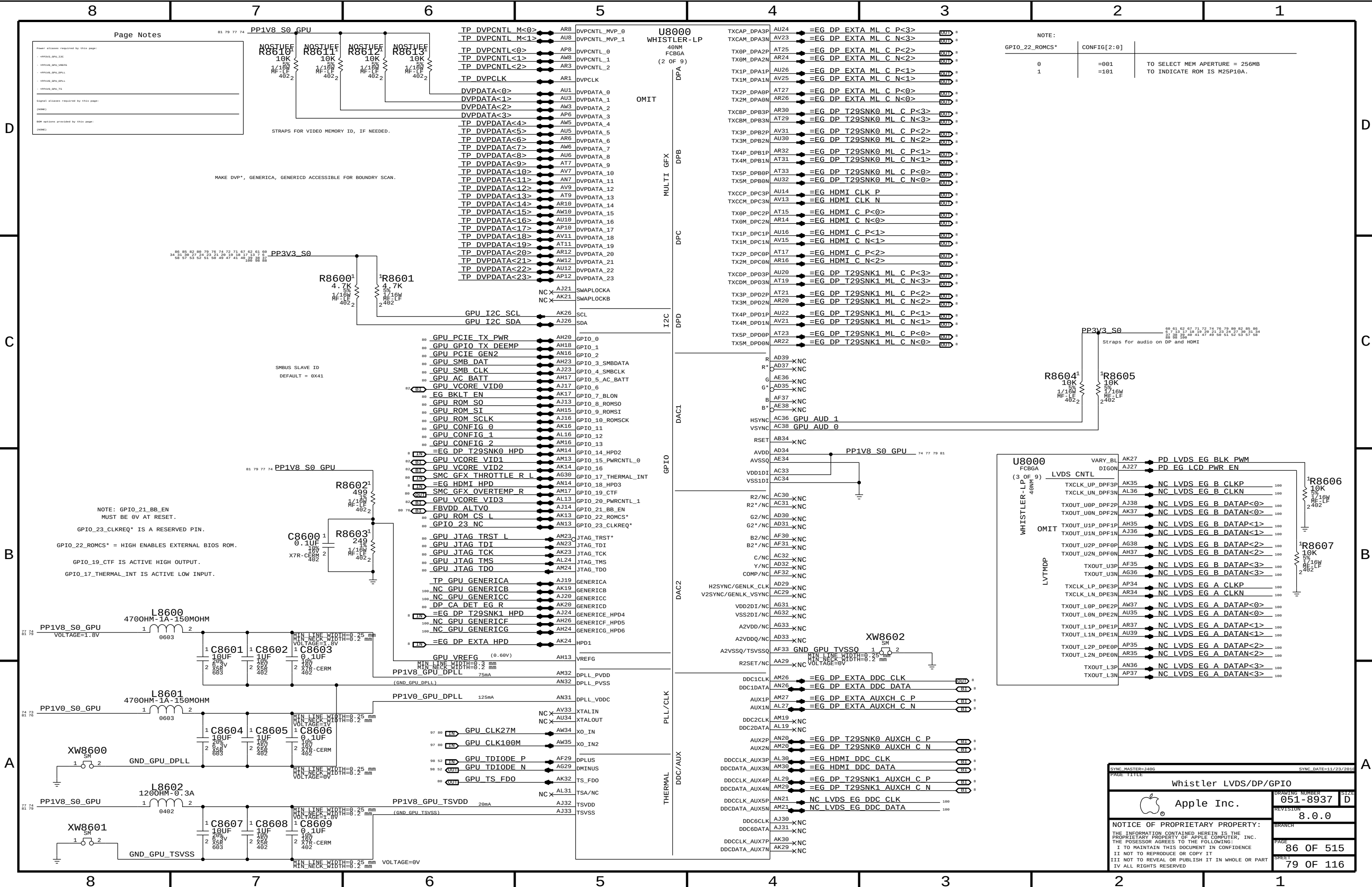
Signal classes required by this page:

SDM options provided by this page:

BITS SWAPPABLE BY BITS AND 8-BIT GROUPS.  
GROUPS 2 AND 3 ARE SWAPPED.

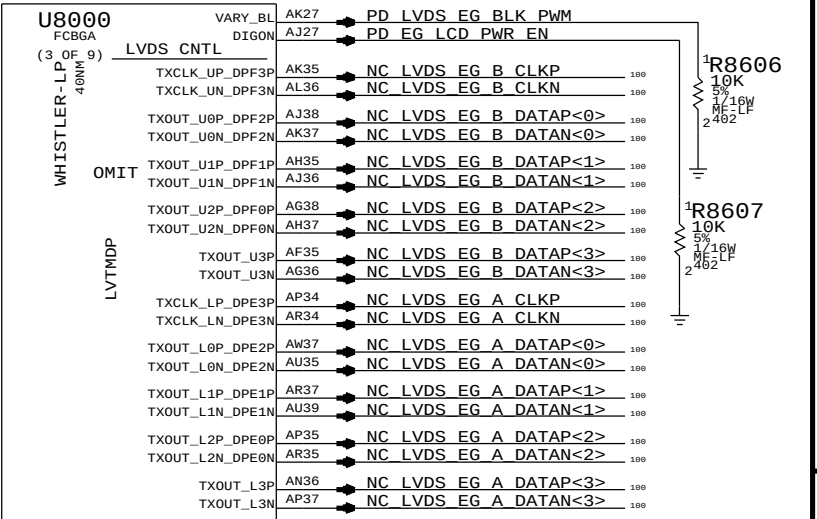
BITS SWAPPABLE BY BITS AND 8-BIT GROUPS.





NOTE:

| GPIO_22_ROMCS* | CONFIG[2:0] |                                |
|----------------|-------------|--------------------------------|
| 0              | =001        | TO SELECT MEM APERTURE = 256MB |
| 1              | =101        | TO INDICATE ROM IS M25P10A.    |



SYNC MASTER=1490

PAGE TITLE

Whistler LVDS/DP/GPIO

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DRAWING NUMBER

051-8937

REVISION

8.0.0

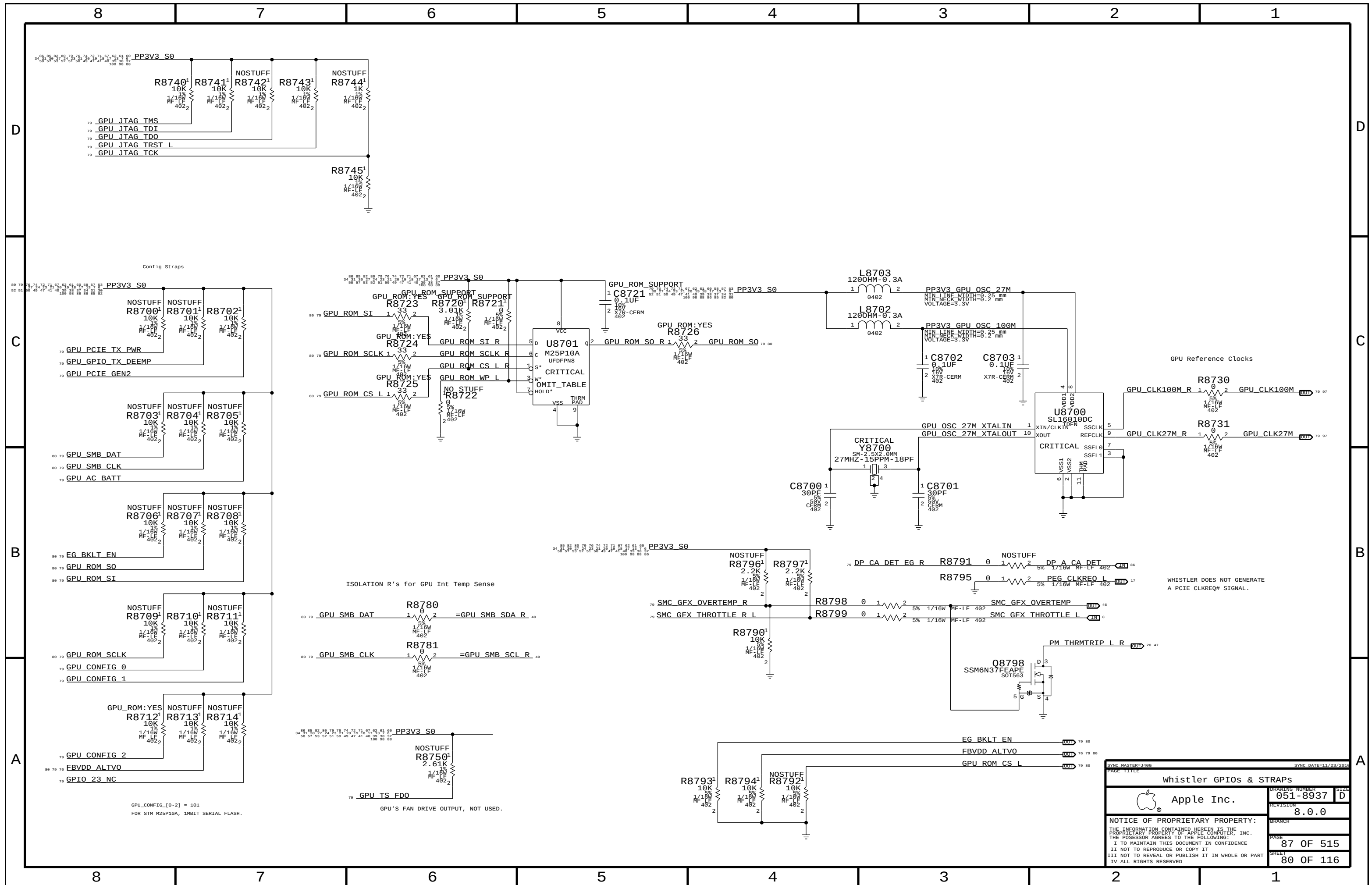
BRANCH

PAGE

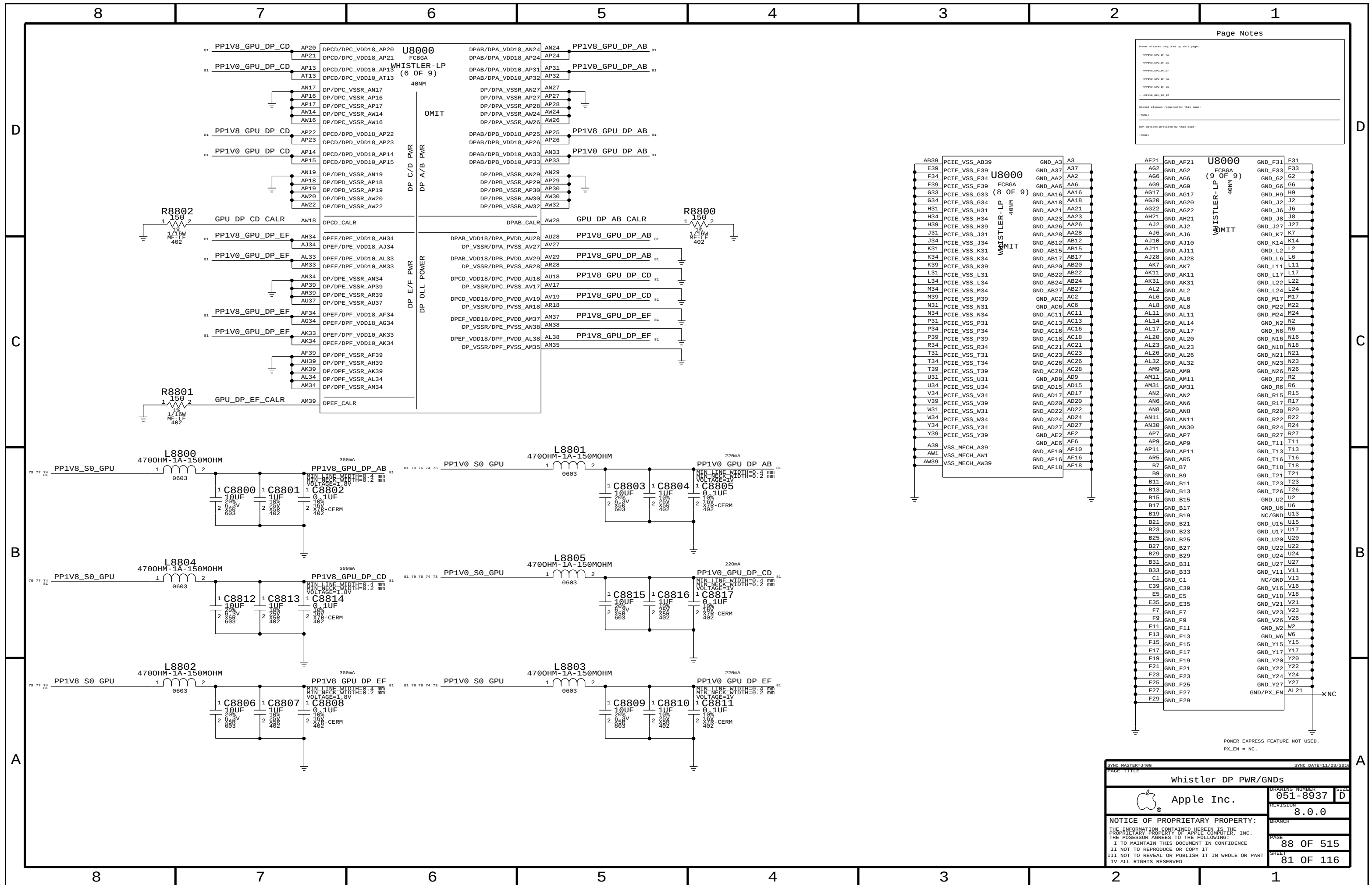
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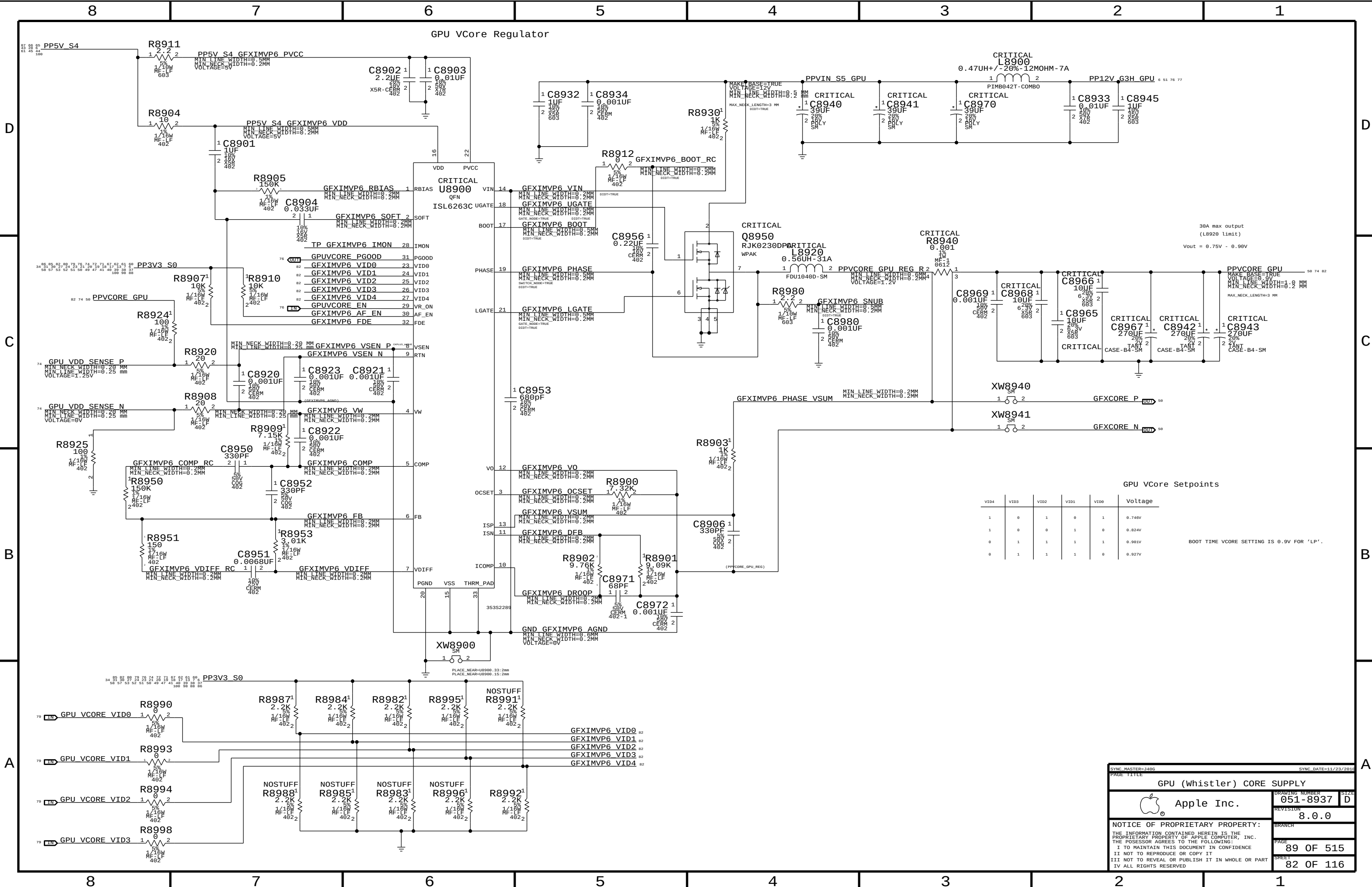
SHEET

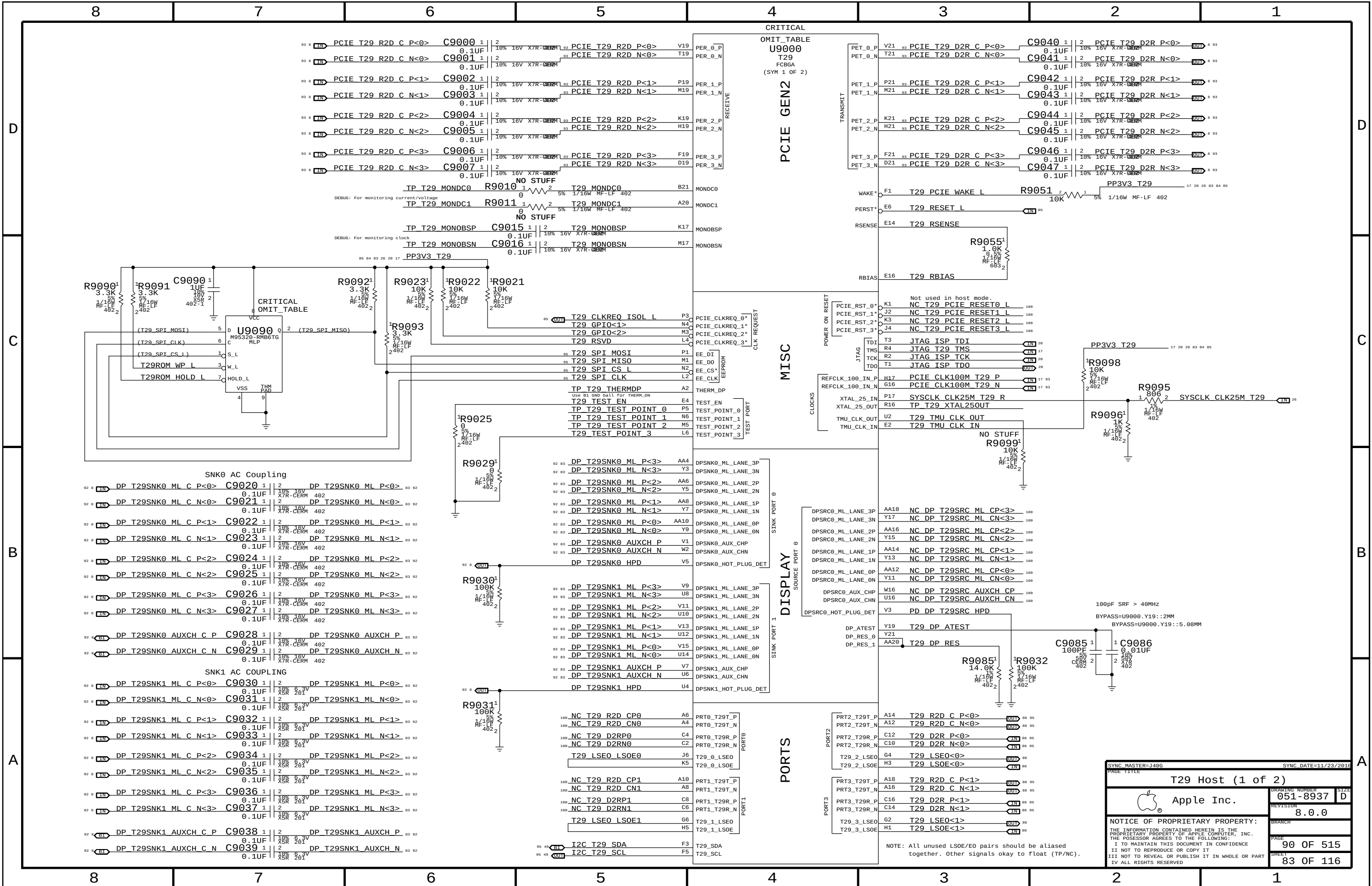
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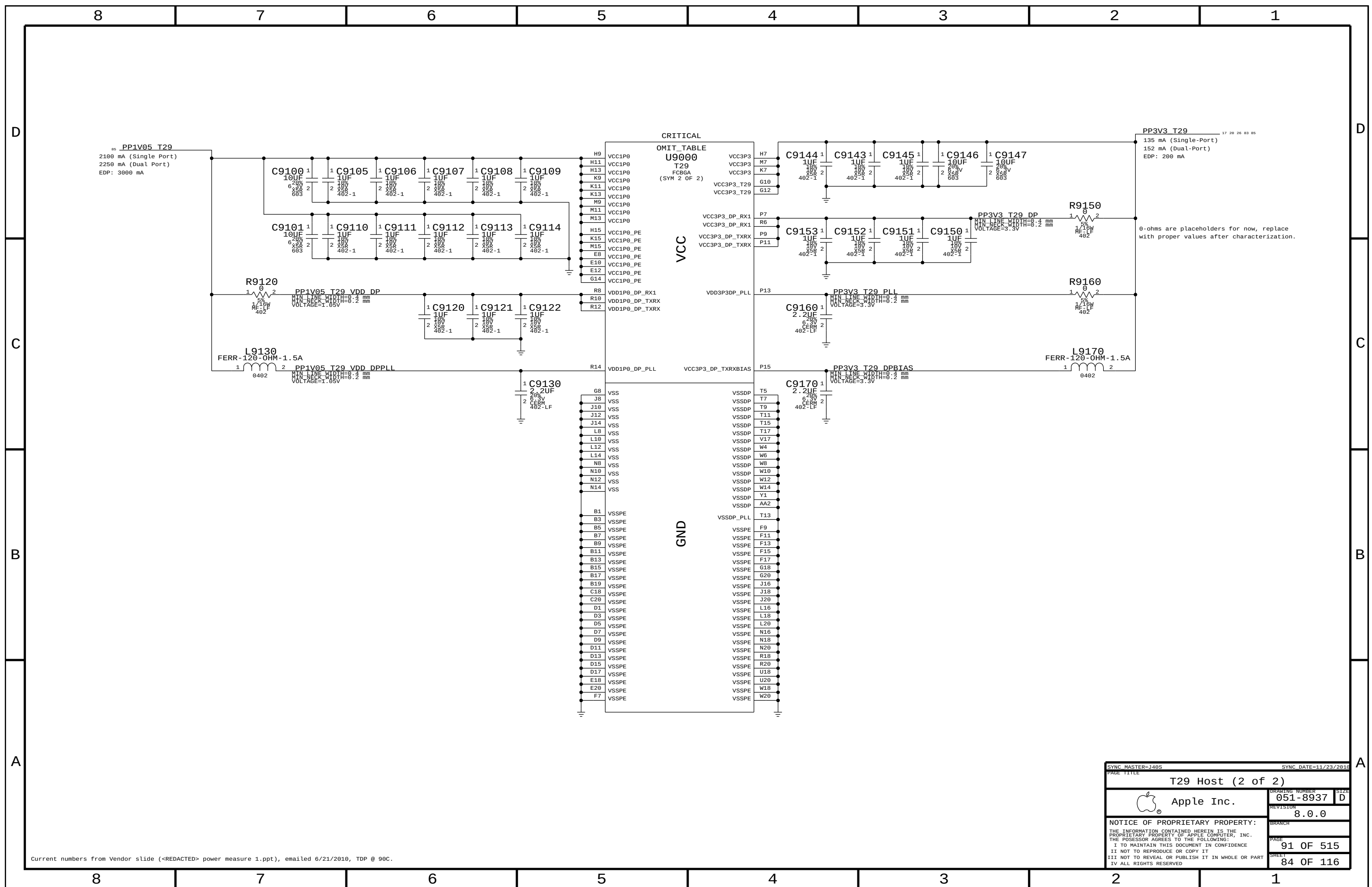












## Page Notes

Power aliases required by this page:

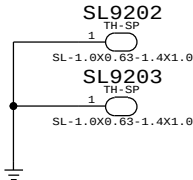
- PPVIN\_SW\_T29BST (8-13V Boost Input)
- PP18V\_T29\_REG (18V Boost Output)
- PP3V3\_T29\_P3V3T29FET (3.3V FET Input)
- PP3V3\_T29\_FET (3.3V FET Output)
- PP3V3\_S0\_T29PWRCTL
- PP1V05\_T29\_P1V05T29FET (1.05V FET Input)
- PP1V05\_T29\_FET (1.05V FET Output)

Signal aliases required by this page:

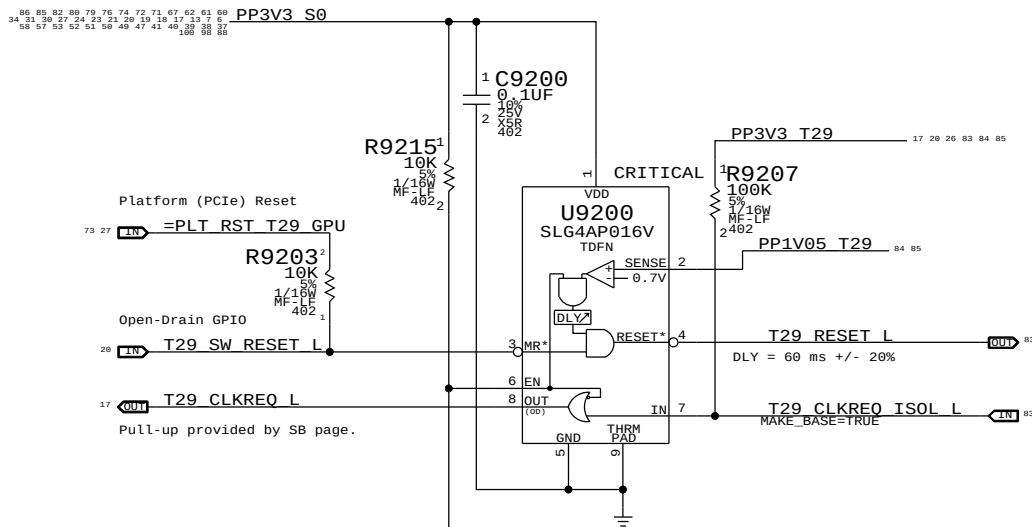
- T29\_CLKREQ\_L
- T29\_RESET\_L

BOM options provided by this page:

T29BST:Y - Stuffs 18V boost circuitry.

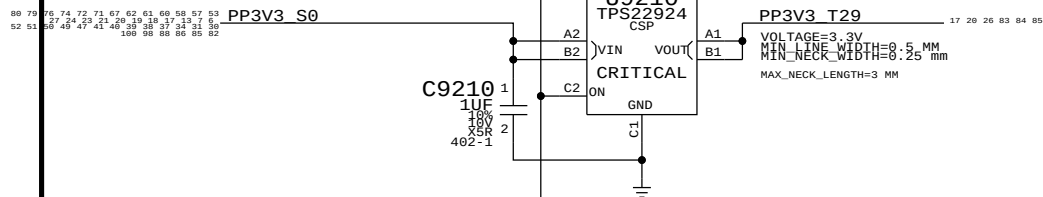


## Supervisor & CLKREQ# Isolation



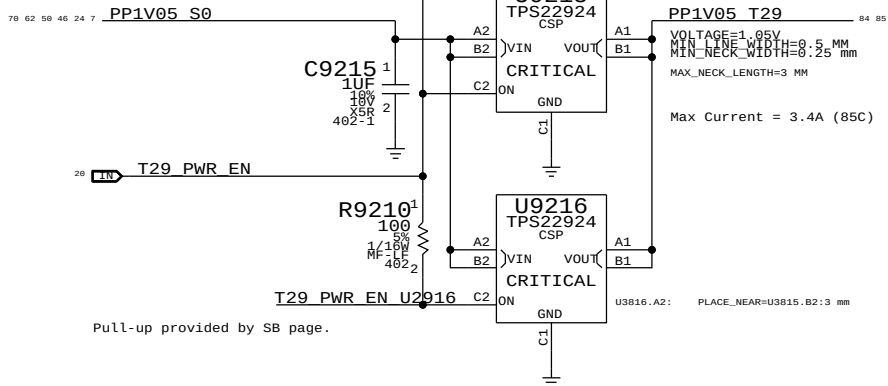
## 3.3V T29 Switch

Max Current = 1.7A (85C)



## 1.05V T29 Switch

Max Current = 3.4A (85C)



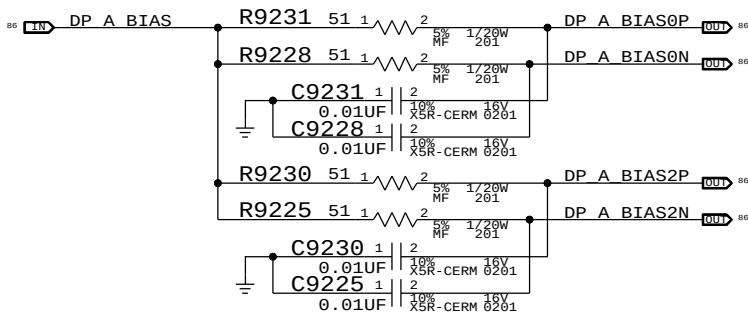
## U9210 & U9215/U9216

| Part  | TPS22924C                  |
|-------|----------------------------|
| Type  | Load Switch                |
| R(on) | 18 mOhm Typ<br>50 mOhm Max |

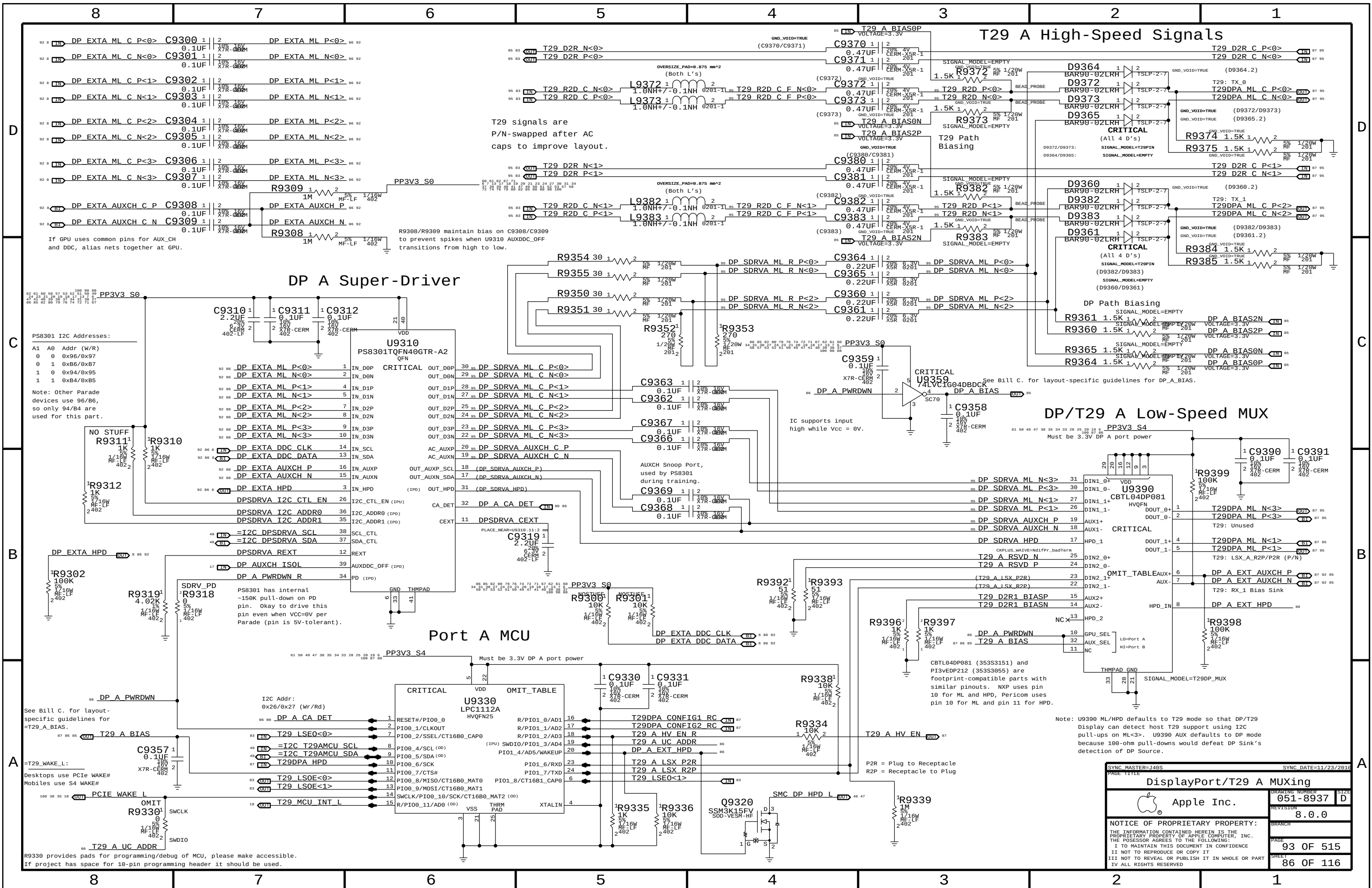
Max Output: 2A per IC

## DP/T29 Bias Filters

Prevents high-frequency coupling between 1.5K bias resistors.



|                                                                                                                             |  |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=146S                                                                                                            |  | SYNC DATE=11/23/2010 |           |
| PAGE TITLE                                                                                                                  |  |                      |           |
| T29 Power Support                                                                                                           |  |                      |           |
|  Apple Inc.                            |  | DRAWING NUMBER       | 051-8937  |
|                                                                                                                             |  | REVISION             | 8.0.0     |
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D

C

B

A

D

C

B

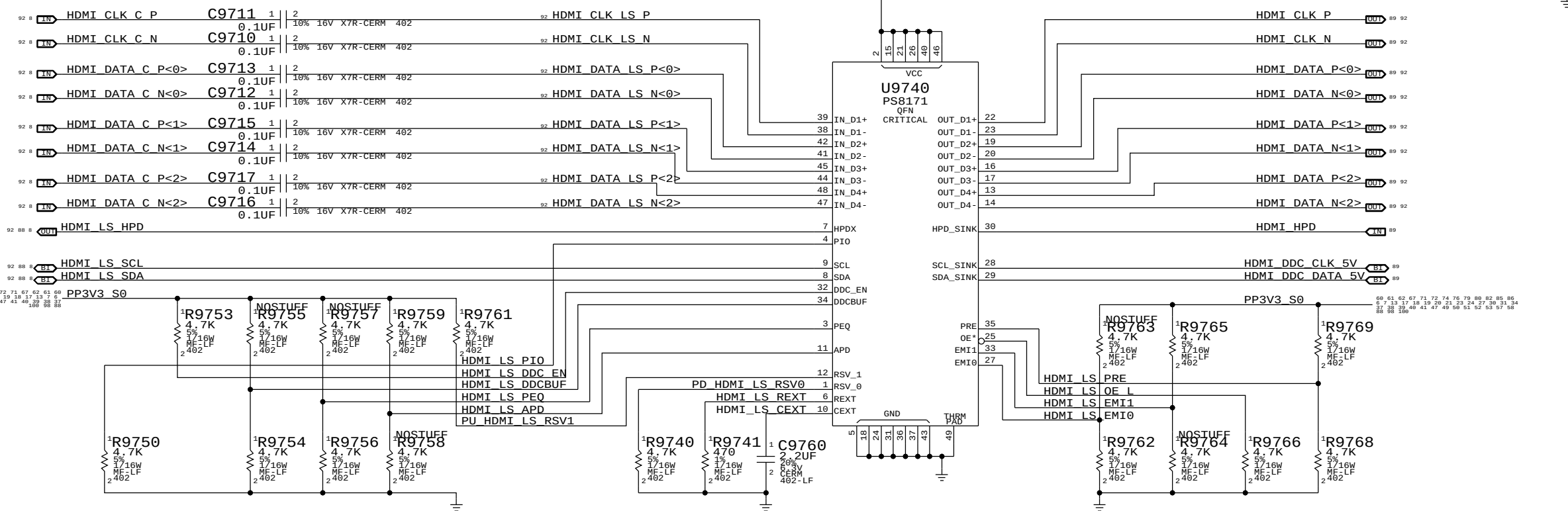
A

ONE 0.1UF AND ONE 0.01UF CAP PER POWER PIN.

06 01 02 03 04 05 06 07 08 09 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

PLACE DP CAPS NEXT TO U9740

ALL CHANNELS ARE EQUIVALENT.  
LANE SWAPPING AND POLARITY REVERSAL ARE ALLOWED.



PIO - HPD INPUT  
HIGH INVERT HPD\_SINK  
LOW NO INVERSION

PEQ - INPUT EQ  
HIGH HIGH EQ  
MID LOW EQ  
LOW MID EQ

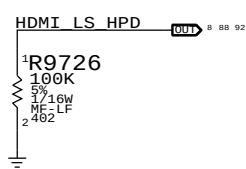
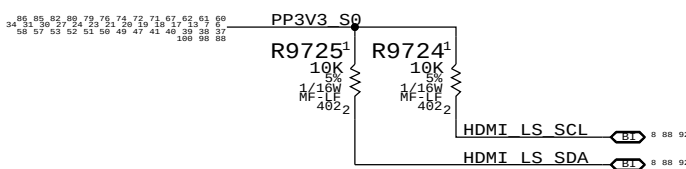
APD - AUTOMATIC POWER DOWN  
HIGH AUTO-POWER-DOWN WHEN HPD\_SINK IS LOW.  
MID (RESERVED)  
LOW DISABLE

DDCBUF - DDC LEVEL SHIFTER TYPE  
HIGH ACTIVE  
LOW PASSIVE

EMI0 - EMI FILTER ENABLE 0  
HIGH LARGE RISE/FALL TIME  
MID SMALL RISE/FALL TIME  
LOW NO EMI FILTER

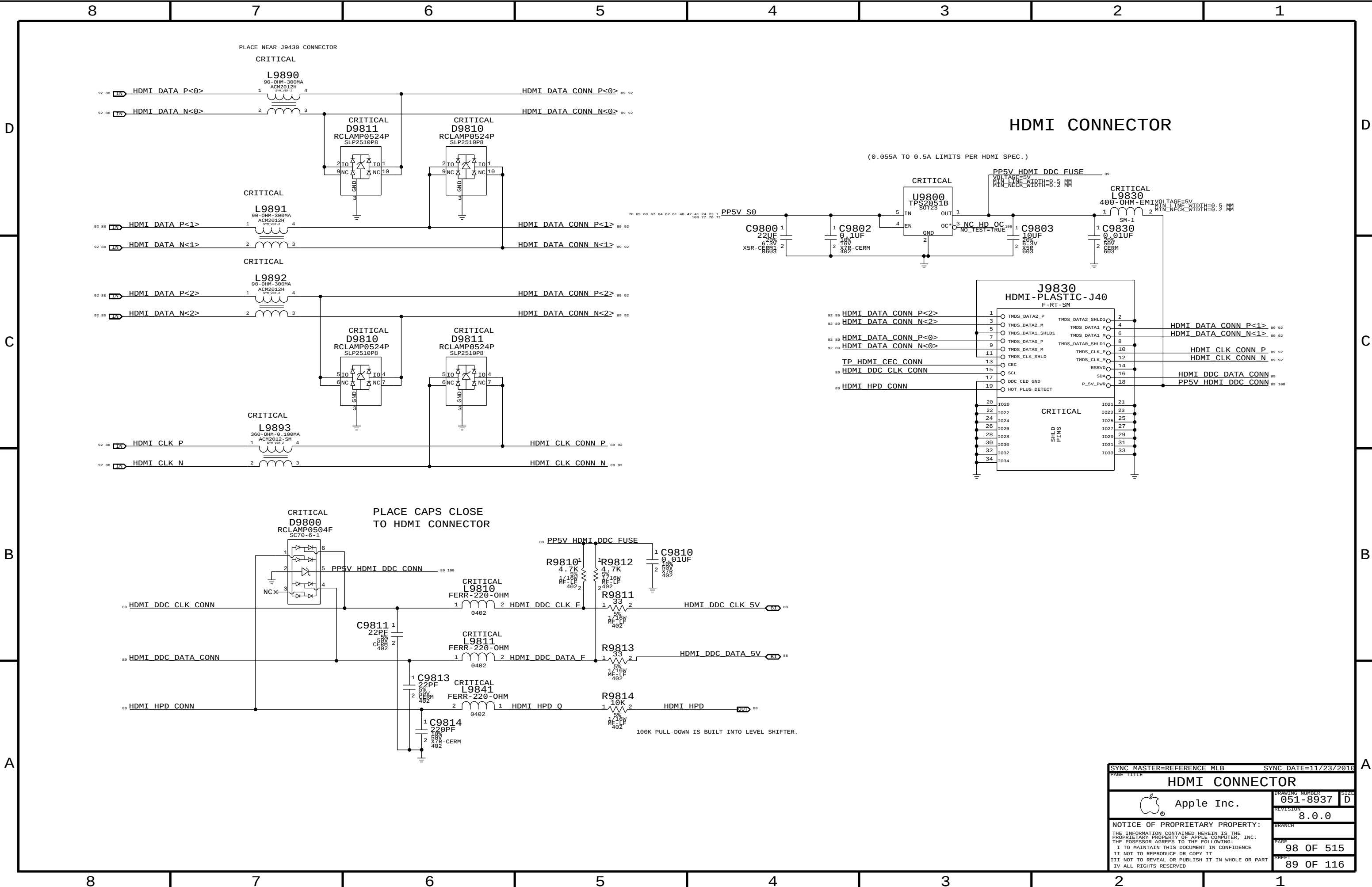
PRE - PRE-EMPHASIS  
HIGH LOW PRE-EMPHASIS  
MID HIGH PRE-EMPHASIS  
LOW NO PRE-EMPHASIS

OE\* - OUTPUT ENABLE.  
HIGH DISABLE TMDS DRIVERS  
LOW ENABLE TMDS DRIVERS

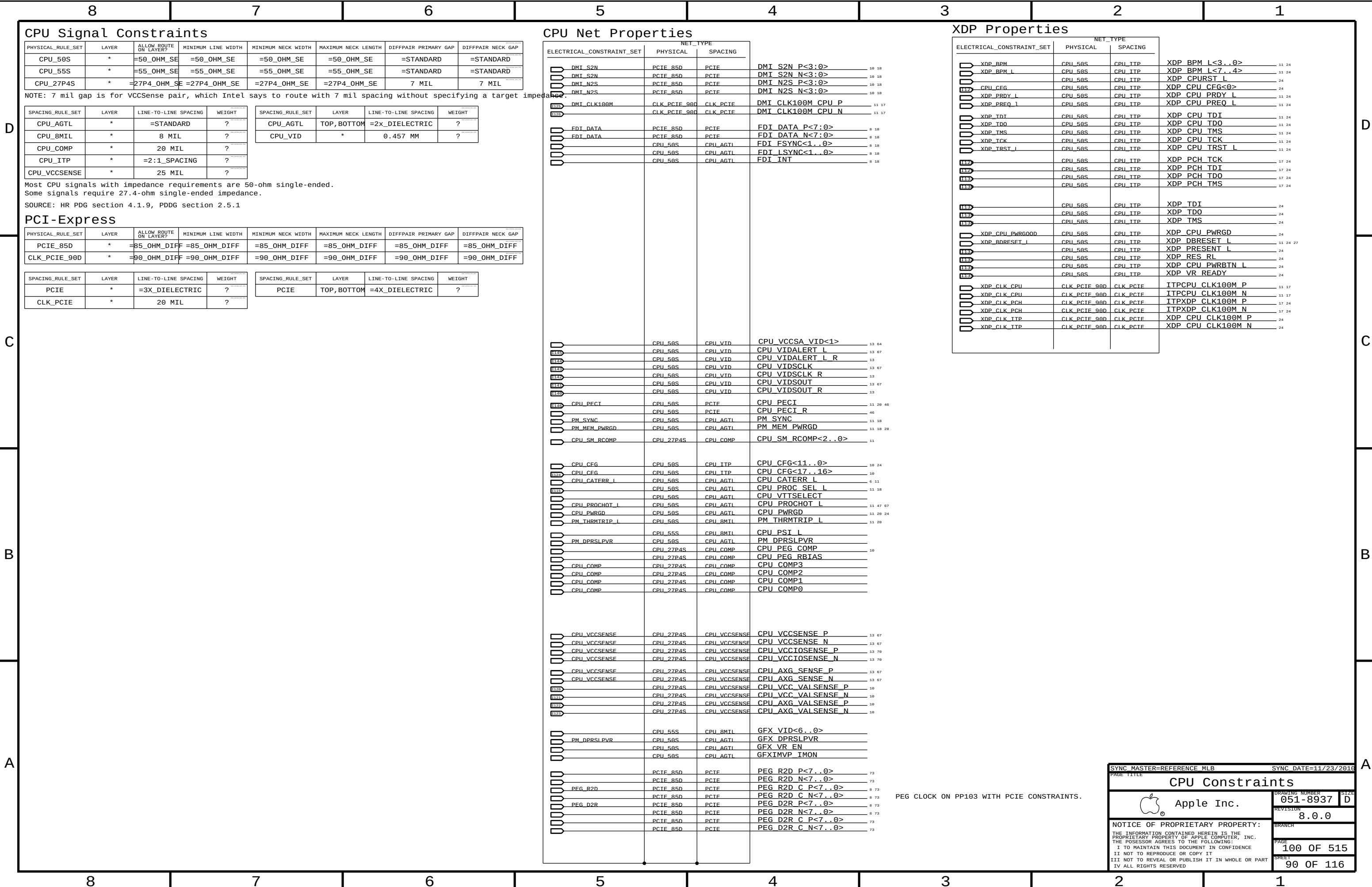


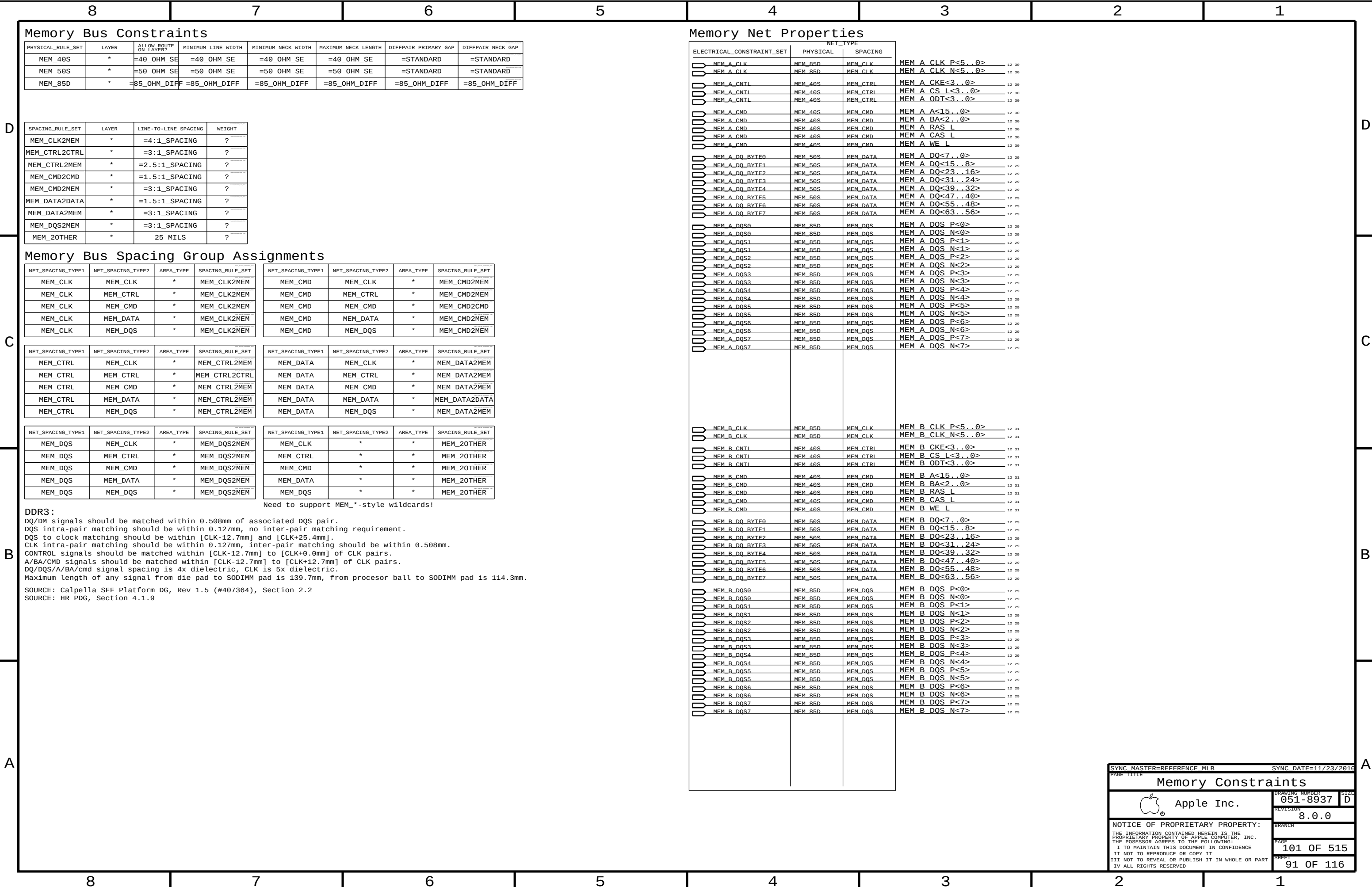
|                                                                                                                                                                                                                                                                                                                                   |  |                      |           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|
| SYNC MASTER=REFERENCE MLB                                                                                                                                                                                                                                                                                                         |  | SYNC DATE=11/23/2016 |           |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                        |  |                      |           |
| HDMI SHIFTER                                                                                                                                                                                                                                                                                                                      |  |                      |           |
|  Apple Inc.                                                                                                                                                                                                                                  |  | DRAWING NUMBER       | 051-8937  |
|                                                                                                                                                                                                                                                                                                                                   |  | SIZE                 | D         |
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|                                                                                                                                                                                                                                                                                                                                |  |                      |      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|------|
| SYNC MASTER=REFERENCE MLB                                                                                                                                                                                                                                                                                                      |  | SYNC DATE=11/23/2016 |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                                                     |  |                      |      |
| HDMI CONNECTOR                                                                                                                                                                                                                                                                                                                 |  |                      |      |
|  Apple Inc.                                                                                                                                                                                                                               |  | DRAWING NUMBER       | SIZE |
|                                                                                                                                                                                                                                                                                                                                |  | 051-8937             | D    |
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|                                                                                                                                                                                                                                                                                                                                |  |                      |      |
|                                                                                                                                                                                                                                                                                                                                |  | PAGE                 |      |
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## LPC Bus Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| LPC_50S           | *     | =50_OHM_SE           | =50_OHM_SE         | =50_OHM_SE         | =50_OHM_SE          | =STANDARD            | =STANDARD         |
| CLK_LPC_50S       | *     | =50_OHM_SE           | =50_OHM_SE         | =50_OHM_SE         | =50_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| LPC              | *     | 6 MIL                | ?      |
| CLK_LPC          | *     | 8 MIL                | ?      |

## SMBus Interface Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SMB_50S           | *     | =50_OHM_SE            | =50_OHM_SE         | =50_OHM_SE         | =50_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| SMB              | *     | =2x_DIELECTRIC       | ?      |

## HD Audio Interface Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| HDA_50S           | *     | =50_OHM_SE            | =50_OHM_SE         | =50_OHM_SE         | =50_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| HDA              | *     | =2x_DIELECTRIC       | ?      |

## SIO Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| CLK_SLOW_55S      | *     | =55_OHM_SE            | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| CLK_SLOW         | *     | 8 MIL                | ?      |

## SPI Interface Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW_ROUTE_ON_LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SPT_55S           | *     | =55_OHM_SE            | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| SPI              | *     | 8 MIL                | ?      |

## PCH Net Properties

| ELECTRICAL_CONSTRAINT_SET                                                             |                 | NET_TYPE    |         | PHYSICAL | SPACING              |           |
|---------------------------------------------------------------------------------------|-----------------|-------------|---------|----------|----------------------|-----------|
|    | LPC_AD          | LPC_56S     | LPC     |          | LPC_AD<3..0>         | 17 46 48  |
|    |                 | LPC_56S     | LPC     |          | LPC_R_AD<3..0>       | 17        |
|    | LPC_FRAME_L     | LPC_56S     | LPC     |          | LPC_FRAME_L          | 17 46 48  |
|    |                 | LPC_56S     | LPC     |          | LPC_FRAME_R_L        | 17        |
|    | LPC_RESET_L     | LPC_56S     | LPC     |          | LPCPLUS_RESET_L      | 27 48     |
|    | PCH_LPC_CLK0    | CLK_LPC_56S | CLK_LPC |          | LPC_CLK33M_SMC_R     | 19 27     |
|    |                 | CLK_LPC_56S | CLK_LPC |          | LPC_CLK33M_SMC       | 27 46     |
|    |                 | CLK_LPC_56S | CLK_LPC |          | LPC_CLK33M_LPCPLUS   | 27 48     |
|    |                 | CLK_LPC_56S | CLK_LPC |          | LPC_CLK33M_LPCPLUS_R | 19 27     |
|    |                 | SMB_56S     | SMB     |          | SMB_PCH_STR_SCL      | 49        |
|    |                 | SMB_56S     | SMB     |          | SMB_PCH_STR_SDA      | 49        |
|    |                 | SMB_56S     | SMB     |          | SMB_PCH_DIMM_SCL     | 49        |
|    |                 | SMB_56S     | SMB     |          | SMB_PCH_DIMM_SDA     | 49        |
|    |                 | SMB_56S     | SMB     |          | SMB_PCH_MKY_SCL      | 49        |
|    |                 | SMB_56S     | SMB     |          | SMB_PCH_MKY_SDA      | 49        |
|    |                 | SMB_56S     | SMB     |          | SMB_PCH_VRF_SCL      | 49        |
|    |                 | SMB_56S     | SMB     |          | SMB_PCH_VRF_SDA      | 49        |
|    | SMBUS_PCH_CLK   | SMB_56S     | SMB     |          | SMBUS_PCH_CLK        | 17 49     |
|    |                 | SMB_56S     | SMB     |          | SMBUS_PCH_DATA       | 17 49     |
|    | SMBUS_PCH_0_CLK | SMB_56S     | SMB     |          | SMB_PCH_XDP_SCL      | 49        |
|    |                 | SMB_56S     | SMB     |          | SMB_PCH_XDP_SDA      | 49        |
|    |                 | SMB_56S     | SMB     |          | SML_PCH_1_CLK        | 17 49     |
|    |                 | SMB_56S     | SMB     |          | SML_PCH_1_DATA       | 17 49     |
|    | SPT_CLK         | SPT_55S     | SPT     |          | SPI_CLK_R            | 17 48     |
|    |                 | SPT_55S     | SPT     |          | SPI_CLK              | 48 100    |
|    | SPT_MOST        | SPT_55S     | SPT     |          | SPI_MOSI_R           | 17 48     |
|    |                 | SPT_55S     | SPT     |          | SPI_MOSI             | 48 100    |
|    | SPT_MISO        | SPT_55S     | SPT     |          | SPI_MISO             | 17 48 100 |
|    | SPT_CS0         | SPT_55S     | SPT     |          | SPI_CS0_R_L          | 17 48     |
|    |                 | SPT_55S     | SPT     |          | SPI_CS0_L            | 48 100    |
|    | HDA_BIT_CLK     | HDA_56S     | HDA     |          | HDA_BIT_CLK          | 17 53     |
|    |                 | HDA_56S     | HDA     |          | HDA_BIT_CLK_R        | 17        |
|    | HDA_SYNC        | HDA_56S     | HDA     |          | HDA_SYNC             | 17 53     |
|    |                 | HDA_56S     | HDA     |          | HDA_SYNC_R           | 17        |
|    | HDA_RST_L       | HDA_56S     | HDA     |          | HDA_RST_R_L          | 17        |
|   |                 | HDA_56S     | HDA     |          | HDA_RST_L            | 17 53     |
|  | HDA_SDIN0       | HDA_56S     | HDA     |          | HDA_SDIN0            | 17 53     |
|  |                 | HDA_56S     | HDA     |          | AUD_SDI_R            | 53        |
|  | HDA_SDOUT       | HDA_56S     | HDA     |          | HDA_SDOUT            | 17 53     |
|  |                 | HDA_56S     | HDA     |          | HDA_SDOUT_R          | 17        |

|       |              |          |                     |       |
|-------|--------------|----------|---------------------|-------|
| 16550 | CLK_PCTF_90D | CLK_PCTF | PCH_CLK96M_DOT_P    | 17    |
| 16550 | CLK_PCTF_90D | CLK_PCTF | PCH_CLK96M_DOT_N    | 17    |
| 16550 | CLK_PCTF_90D | CLK_PCTF | PCH_CLK100M_SATA_P  | 17    |
| 16550 | CLK_PCTF_90D | CLK_PCTF | PCH_CLK100M_SATA_N  | 17    |
| 16550 | CPU1_56S     | CLK_PCTF | PCH_CLK14P3M_REECLK | 17    |
| 16550 | CPU1_56S     | CLK_PCTF | PCH_CLK33M_PCIIN    | 17 27 |

|  |               |          |      |                   |           |
|--|---------------|----------|------|-------------------|-----------|
|  |               | PCIE_85D | PCIE | PCIE ENET R2D P   | 37        |
|  |               | PCIE_85D | PCIE | PCIE ENET R2D N   | 37        |
|  | PCIE_ENET_R2D | PCIE_85D | PCIE | PCIE ENET R2D C P | 17 37     |
|  |               | PCIE_85D | PCIE | PCIE ENET R2D C N | 17 37     |
|  | PCIE_ENET_D2R | PCIE_85D | PCIE | PCIE ENET D2R P   | 17 37     |
|  |               | PCIE_85D | PCIE | PCIE ENET D2R N   | 17 37     |
|  |               | PCIE_85D | PCIE | PCIE ENET D2R C P | 37        |
|  |               | PCIE_85D | PCIE | PCIE ENET D2R C N | 37        |
|  |               | PCIE_85D | PCIE | PCIE AP R2D P     | 35 100    |
|  |               | PCIE_85D | PCIE | PCIE AP R2D N     | 35 100    |
|  | PCIE_AP_R2D   | PCIE_85D | PCIE | PCIE AP R2D C P   | 17 35     |
|  |               | PCIE_85D | PCIE | PCIE AP R2D C N   | 17 35     |
|  | PCIE_AP_D2R   | PCIE_85D | PCIE | PCIE AP D2R P     | 17 35 100 |
|  |               | PCIE_85D | PCIE | PCIE AP D2R N     | 17 35 100 |
|  |               | PCIE_85D | PCIE | PCIE FW R2D P     | 39        |
|  |               | PCIE_85D | PCIE | PCIE FW R2D N     | 39        |
|  | PCIE_FW_R2D   | PCIE_85D | PCIE | PCIE FW R2D C P   | 17 39     |
|  |               | PCIE_85D | PCIE | PCIE FW R2D C N   | 17 39     |
|  | PCIE_FW_D2R   | PCIE_85D | PCIE | PCIE FW D2R P     | 17 39     |
|  |               | PCIE_85D | PCIE | PCIE FW D2R N     | 17 39     |
|  |               | PCIE_85D | PCIE | PCIE FW D2R C P   | 39        |
|  |               | PCIE_85D | PCIE | PCIE FW D2R C N   | 39        |

|      |                  |              |                    |       |
|------|------------------|--------------|--------------------|-------|
| T209 | CLK_PCTF_90D     | CLK_PCTF     | PCIE_CLK100M_PCH_P | 17    |
| T254 | CLK_PCTF_90D     | CLK_PCTF     | PCIE_CLK100M_PCH_N | 17    |
| T265 | PCIE_CLK100M_T29 | CLK_PCTF_90D | PCIE_CLK100M_T29_P | 17 83 |
| T267 | CLK_PCTF_90D     | CLK_PCTF     | PCIE_CLK100M_T29_N | 17 83 |

|     |                   |              |          |                        |       |
|-----|-------------------|--------------|----------|------------------------|-------|
|     | PCIE_CLK100M      | CLK_PCIE_90D | CLK_PCIE | PEG_CLK100M_P          | 17_73 |
|     |                   |              |          | PEG_CLK100M_N          | 17_73 |
|     | PCIE_CLK100M_ENET | CLK_PCIE_90D | CLK_PCIE | PCIE_CLK100M_ENET_P    | 17_97 |
|     |                   |              |          | PCIE_CLK100M_ENET_N    | 17_97 |
| 250 | PCIE_CLK100M_AP   | CLK_PCIE_90D | CLK_PCIE | PCIE_CLK100M_AP_CONN_P | 35    |
| 250 |                   |              |          | PCIE_CLK100M_AP_CONN_N | 35    |
|     | PCIE_CLK100M_AP   | CLK_PCIE_90D | CLK_PCIE | PCIE_CLK100M_AP_P      | 17_95 |
|     |                   |              |          | PCIE_CLK100M_AP_N      | 17_95 |
|     | PCIE_CLK100M_FW   | CLK_PCIE_90D | CLK_PCIE | PCIE_CLK100M_FW_P      | 17_99 |
|     |                   |              |          | PCIE_CLK100M_FW_N      | 17_99 |

|              |          |      |                         |      |
|--------------|----------|------|-------------------------|------|
| PCIE T29_R2D | PCIE_85D | PCIE | PCIE T29_R2D C P<3...0> | 6 B3 |
| PCIE T29_R2D | PCIE_85D | PCIE | PCIE T29_R2D C N<3...0> | 6 B3 |
| PCIE T29_R2D | PCIE_85D | PCIE | PCIE T29_R2D P<3...0>   | 6 B3 |
| PCIE T29_R2D | PCIE_85D | PCIE | PCIE T29_R2D N<3...0>   | 6 B3 |
| PCIE T29_D2R | PCIE_85D | PCIE | PCIE T29_D2R P<3...0>   | 6 B3 |
| PCIE T29_D2R | PCIE_85D | PCIE | PCIE T29_D2R N<3...0>   | 6 B3 |
| PCIE T29_D2R | PCIE_85D | PCIE | PCIE T29_D2R C P<3...0> | 6 B3 |
| PCIE T29_D2R | PCIE_85D | PCIE | PCIE T29_D2R C N<3...0> | 6 B3 |

|                                                                                                                                                                                                                                                                                                |  |                       |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------|------|
| SYNCH MASTER=REFERENCE MLB                                                                                                                                                                                                                                                                     |  | SYNCH DATE=11/23/2016 |      |
| PAGE TITLE                                                                                                                                                                                                                                                                                     |  |                       |      |
| PCH Constraints 2                                                                                                                                                                                                                                                                              |  |                       |      |
|  Apple Inc.                                                                                                                                                                                               |  | DRAWING NUMBER        | SIZE |
|                                                                                                                                                                                                                                                                                                |  | 051-8937              | D    |
|                                                                                                                                                                                                                                                                                                |  | REVISION              |      |
|                                                                                                                                                                                                                                                                                                |  | 8.0.0                 |      |
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|                                                                                                                                                                                                                                                                                                |  | BRANCH                |      |
|                                                                                                                                                                                                                                                                                                |  | PAGE                  |      |
|                                                                                                                                                                                                                                                                                                |  | 103 OF 515            |      |
|                                                                                                                                                                                                                                                                                                |  | SHEET                 |      |
|                                                                                                                                                                                                                                                                                                |  | 93 OF 116             |      |

## ETHERNET CONSTRAINTS

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| ENET_50S          | *     | =50_OHM_SE            | =50_OHM_SE         | =50_OHM_SE         | =50_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| ENET_3X          | *     | =3:1_SPACING         | ?      |

SOURCE: Broadcom 5764-DS04-RDS Page 38

[illegible]

|                  |            |                      |        |                   |                   |           |                  |
|------------------|------------|----------------------|--------|-------------------|-------------------|-----------|------------------|
| SPACING_RULE_SET | LAYER      | LINE-TO-LINE SPACING | WEIGHT | NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
| ENET_MDI         | *          | 0.6 MM               | ?      | ENET_HV           | *                 | *         | 2KV_SPACING      |
| 2KV_SPACING      | *          | 0.50 MM              | 100    | ENET_HV           | ENET_HV           | *         | ENET_MDI         |
| 2KV_SPACING      | TOP,BOTTOM | 1.27 MM              | 100    |                   |                   |           |                  |

NOTE: 2.2kV isolation needed on all primary-side ethernet signals. Min 1.27mm separation from all other copper on the board!

## SD CARD INTERFACE CONSTRAINTS

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SD_55S            | *     | =55_OHM_SE            | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =STANDARD            | =STANDARD         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| SD_INTERFACE     | *     | =3X_DIELECTRIC       | ?      |

## FireWire Interface Constraints

[illegible]

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| FW_TP            | *     | =3:1_SPACING         | ?      |

## Ethernet Net Properties

| ELECTRICAL_CONSTRAINT_SET                                                           |          | NET_TYPE  |          |                       |       |
|-------------------------------------------------------------------------------------|----------|-----------|----------|-----------------------|-------|
|                                                                                     |          | PHYSICAL  | SPACING  |                       |       |
|  |          | ENET_50S  | ENET_3X  | BCM5764 CLK25M XTALI  | 26 37 |
|  |          | ENET_50S  | ENET_3X  | BCM5764 CLK25M XTALO  | 26 37 |
|  |          | ENET_50S  | ENET_3X  | ENET RESET L          | 27 34 |
|  | ENET_MDI | ENET_100D | ENET_MDI | ENET MDI P<3..0>      | 36 37 |
|  |          | ENET_100D | ENET_MDI | ENET MDI N<3..0>      | 36 37 |
|  |          |           |          |                       |       |
|  |          | ENET_100D | ENET_HV  | ENET MDI TRAN N<3..0> | 36    |
|  |          | ENET_100D | ENET_HV  | ENET MDI TRAN P<3..0> | 36    |
|  |          | ENET_50S  | ENET_HV  | ENET CENTER TAP<3..0> | 36    |
|  |          |           | ENET_HV  | ENET_CMODE_REF        | 36    |

High-voltage isolated ethernet signals.

## SD Interface Properties

| ELECTRICAL_CONSTRAINT_SET |  | NET_TYPE |              |                             |
|---------------------------|--|----------|--------------|-----------------------------|
|                           |  | PHYSICAL | SPACING      |                             |
| SD55S                     |  | SD 55S   | SD INTERFACE | SDCONN CLK 34 37 100        |
| SD55S                     |  | SD 55S   | SD INTERFACE | SDCONN CLK FF 34            |
| SD55S                     |  | SD 55S   | SD INTERFACE | SDCONN CMD 34 37 100        |
| SD55S                     |  | SD 55S   | SD INTERFACE | SDCONN DATA<7..0> 34 37 100 |
| SD55S                     |  | SD 55S   | SD INTERFACE | SDCONN CLK R 34             |
| SD55S                     |  | SD 55S   | SD INTERFACE | SDCONN CMD R 34             |
| SD55S                     |  | SD 55S   | SD INTERFACE | SDCONN DATA R<7..0> 34      |
| SD55S                     |  | SD 55S   | SD INTERFACE | ENET_CR CLK 37              |
| SD55S                     |  | SD 55S   | SD INTERFACE | ENET_CR CMD 37              |
| SD55S                     |  | SD 55S   | SD INTERFACE | ENET_CR DATA<7..0> 37       |
| SD55S                     |  | SD 55S   | SD INTERFACE | SDCONN_F CLK 34             |
| SD55S                     |  | SD 55S   | SD INTERFACE | SDCONN_F CLK_FF 34          |

## FireWire Net Properties

| ELECTRICAL_CONSTRAINT_SET |           | NET_TYPE  |          |                      |
|---------------------------|-----------|-----------|----------|----------------------|
|                           |           | PHYSICAL  | SPACING  |                      |
| LEADS                     | FW_P1_TPA | FW_118D   | FW_TP    | FW_P0_TPA_L_N 46     |
| LEADS                     |           | FW_118D   | FW_TP    | FW_P0_TPA_L_P 46     |
| LEADS                     | FW_P1_TPB | FW_118D   | FW_TP    | FW_P0_TPB_L_N 46     |
| LEADS                     |           | FW_118D   | FW_TP    | FW_P0_TPB_L_P 46     |
| PORTS                     |           | FW_118D   | FW_TP    | FW_PORT0_TPAL_N 41   |
| PORTS                     |           | FW_118D   | FW_TP    | FW_PORT0_TPAL_P 41   |
| LEADS                     |           | FW_118D   | FW_TP    | FW_PORT0_TPA_N 46 41 |
| PORTS                     |           | FW_118D   | FW_TP    | FW_PORT0_TPA_P 46 41 |
| LEADS                     |           | FW_118D   | FW_TP    | FW_PORT0_TPBL_N 41   |
| LEADS                     |           | FW_118D   | FW_TP    | FW_PORT0_TPBL_P 41   |
| PORTS                     |           | FW_118D   | FW_TP    | FW_PORT0_TPB_N 46 41 |
| PORTS                     |           | FW_118D   | FW_TP    | FW_PORT0_TPB_P 46 41 |
| LEADS                     |           | 50_OHM_SE | CLK_PCIE | CLK98M_FW_XI 39      |
| LEADS                     |           | 50_OHM_SE | CLK_PCIE | CLK98M_FW_XI_R 39    |

Port 0 and 2 Not Used



## DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

## T29 I2C Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| T29_I2C_55S       | *     | =55_OHM_SE            | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =STANDARD            | =STANDARD         |

|                  |       |                      |        |
|------------------|-------|----------------------|--------|
| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
| T29_I2C          | *     | =2x_DIELECTRIC       | ?      |

## T29 SPI Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| T29_SPI_55S       | *     | =55_OHM_SE            | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =STANDARD            | =STANDARD         |

|                  |       |                      |        |
|------------------|-------|----------------------|--------|
| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
| T29_SPI          | *     | =2x_DIELECTRIC       | ?      |

## T29/DP Connector Signal Constraints

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| T29DP_90D         | *     | =90_OHM_DIFF          | =90_OHM_DIFF       | =90_OHM_DIFF       | =90_OHM_DIFF        | =90_OHM_DIFF         | =90_OHM_DIFF      |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT | SPACING_RULE_SET | LAYER       | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|------------------|-------------|----------------------|--------|
| T29DP            | *     | =5x_DIELECTRIC       | ?      | T29DP            | TOP, BOTTOM | =7x_DIELECTRIC       | ?      |

SOURCE: Bill Cornelius's T29 Routing Notes

## T29 IC Net Properties

| ELECTRICAL_CONSTRAINT_SET | NET TYPE    |             |
|---------------------------|-------------|-------------|
|                           | PHYSICAL    | SPACING     |
|                           | T29_I2C_55S | T29_I2C     |
|                           | T29_I2C_55S | T29_I2C     |
|                           |             | I2C T29_SCL |
|                           |             | I2C T29_SDA |
|                           | T29_SPI_55S | T29_SPI     |
|                           | T29_SPI_55S | T29_SPI     |
|                           | T29_SPI_55S | T29_SPI     |
|                           | T29_SPI_55S | T29_SPI     |
|                           | T29DP_90D   | T29DP       |
|                           | T29DP_90D   | T29DP       |
|                           | T29DP_90D   | T29DP       |
|                           | T29DP_90D   | T29DP       |

## T29/DP Net Properties

| ELECTRICAL_CONSTRAINT_SET |           | NET_TYPE |                        |    |
|---------------------------|-----------|----------|------------------------|----|
|                           |           | PHYSICAL | SPACING                |    |
| T29_R2D0                  | T29DP_90D | T29DP    | T29_R2D P<0>           | 86 |
| T29_R2D0                  | T29DP_90D | T29DP    | T29_R2D N<0>           | 86 |
| T29_R2D1                  | T29DP_90D | T29DP    | T29_R2D P<1>           | 86 |
| T29_R2D1                  | T29DP_90D | T29DP    | T29_R2D N<1>           | 86 |
| T29_D2R0                  | T29DP_90D | T29DP    | T29_D2R C P<0>         | 87 |
| T29_D2R0                  | T29DP_90D | T29DP    | T29_D2R C N<0>         | 87 |
| T29_D2R1                  | T29DP_90D | T29DP    | T29_D2R C P<1>         | 87 |
| T29_D2R1                  | T29DP_90D | T29DP    | T29_D2R C N<1>         | 87 |
|                           | T29DP_90D | T29DP    | T29_R2D C F P<1..0>    | 86 |
|                           | T29DP_90D | T29DP    | T29_R2D C F N<1..0>    | 86 |
|                           | T29DP_90D | T29DP    | T29DPA_D2R1_AUXCH P    | 87 |
|                           | T29DP_90D | T29DP    | T29DPA_D2R1_AUXCH N    | 87 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_ML C P<3..0>  | 86 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_ML C N<3..0>  | 86 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_ML R P<3..0>  | 86 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_ML R N<3..0>  | 86 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_ML P<2>       | 86 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_ML N<2>       | 86 |
| DP_SDRVA_ML               | T29DP_90D | T29DP    | DP_SDRVA_ML P<3>       | 86 |
| DP_SDRVA_ML               | T29DP_90D | T29DP    | DP_SDRVA_ML N<3>       | 86 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_ML P<0>       | 86 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_ML N<0>       | 86 |
| DP_SDRVA_ML               | T29DP_90D | T29DP    | DP_SDRVA_ML P<1>       | 86 |
| DP_SDRVA_ML               | T29DP_90D | T29DP    | DP_SDRVA_ML N<1>       | 86 |
| DP_SDRVA_AUXCH            | T29DP_90D | T29DP    | DP_SDRVA_AUXCH P       | 86 |
| DP_SDRVA_AUXCH            | T29DP_90D | T29DP    | DP_SDRVA_AUXCH N       | 86 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_AUXCH C P     | 86 |
|                           | T29DP_90D | T29DP    | DP_SDRVA_AUXCH C N     | 86 |
|                           | T29DP_90D | T29DP    | T29DPA_ML P<3..0>      | 87 |
|                           | T29DP_90D | T29DP    | T29DPA_ML N<3..0>      | 87 |
|                           | T29DP_90D | T29DP    | T29DPA_ML C P<3..0>    | 87 |
|                           | T29DP_90D | T29DP    | T29DPA_ML C N<3..0>    | 87 |
|                           | T29DP_90D | T29DP    | DP_A_EXT_AUXCH P       | 87 |
|                           | T29DP_90D | T29DP    | DP_A_EXT_AUXCH N       | 87 |
|                           | T29DP_90D | T29DP    | T29DPA_D2R1_AUXCH_CC P |    |
|                           | T29DP_90D | T29DP    | T29DPA_D2R1_AUXCH_CC N |    |
|                           | T29DP_90D | T29DP    | T29_D2R_CC P<0>        |    |
|                           | T29DP_90D | T29DP    | T29_D2R_CC N<0>        |    |

## SMC SMBus Net Properties

A



|                                       |              |                       |                    |                    |                     |                      |                   |
|---------------------------------------|--------------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| 8                                     | 7            | 6                     | 5                  | 4                  | 3                   | 2                    | 1                 |
| GDDR5 Frame Buffer Signal Constraints |              |                       |                    |                    |                     |                      |                   |
| PHYSICAL_RULE_SET                     | LAYER        | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
| GDDR5_45R50SE                         | *            | =50_OHM_SE            | =50_OHM_SE         | =50_OHM_SE         | 12.7 MM             | =STANDARD            | =STANDARD         |
| GDDR5_45SE                            | *            | =45_OHM_SE            | =45_OHM_SE         | =45_OHM_SE         | =45_OHM_SE          | =STANDARD            | =STANDARD         |
| GDDR5_80D                             | *            | =80_OHM_DIFF          | =80_OHM_DIFF       | =80_OHM_DIFF       | =80_OHM_DIFF        | =80_OHM_DIFF         | =80_OHM_DIFF      |
| SPACING_RULE_SET                      | LAYER        | LINE-TO-LINE SPACING  | WEIGHT             |                    |                     |                      |                   |
| GDDR5_CLK                             | *            | =5x_DIELECTRIC        | ?                  |                    |                     |                      |                   |
| GDDR5_CMD                             | *            | =2x_DIELECTRIC        | ?                  |                    |                     |                      |                   |
| GDDR5_DATA                            | *            | =3x_DIELECTRIC        | ?                  |                    |                     |                      |                   |
| GDDR5_EDC                             | *            | =7x_DIELECTRIC        | ?                  |                    |                     |                      |                   |
| GDDR5 FB B Net Properties             |              |                       |                    |                    |                     |                      |                   |
|                                       |              |                       |                    | NET_TYPE           |                     |                      |                   |
| ELECTRICAL_CONSTRAINT_SET             |              | PHYSICAL              | SPACING            |                    |                     |                      |                   |
|                                       | FB_B0_CLK    | GDDR5_80D             | GDDR5_CLK          | FB_B0_CLK_P        | 75                  | 78                   |                   |
|                                       | FB_B0_CLK    | GDDR5_80D             | GDDR5_CLK          | FB_B0_CLK_N        | 75                  | 78                   |                   |
|                                       | FB_B1_CLK    | GDDR5_80D             | GDDR5_CLK          | FB_B1_CLK_P        | 75                  | 78                   |                   |
|                                       | FB_B1_CLK    | GDDR5_80D             | GDDR5_CLK          | FB_B1_CLK_N        | 75                  | 78                   |                   |
|                                       | FB_B0_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B0_A<8..0>      | 75                  | 78                   |                   |
|                                       | FB_B1_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B1_A<8..0>      | 75                  | 78                   |                   |
|                                       | FB_B0_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B0_ABI_L        | 75                  | 78                   |                   |
|                                       | FB_B1_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B1_ABI_L        | 75                  | 78                   |                   |
|                                       | FB_B0_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B0_RAS_L        | 75                  | 78                   |                   |
|                                       | FB_B1_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B1_RAS_L        | 75                  | 78                   |                   |
|                                       | FB_B0_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B0_CAS_L        | 75                  | 78                   |                   |
|                                       | FB_B1_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B1_CAS_L        | 75                  | 78                   |                   |
|                                       | FB_B0_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B0_WE_L         | 75                  | 78                   |                   |
|                                       | FB_B1_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B1_WE_L         | 75                  | 78                   |                   |
|                                       | FB_B0_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B0_CKE_L        | 75                  | 78                   |                   |
|                                       | FB_B1_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B1_CKE_L        | 75                  | 78                   |                   |
|                                       | FB_B0_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B0_CS_L         | 75                  | 78                   |                   |
|                                       | FB_B1_CMD    | GDDR5_45R50SE         | GDDR5_CMD          | FB_B1_CS_L         | 75                  | 78                   |                   |
|                                       | FB_B0_EDC0   | GDDR5_45SE            | GDDR5_EDC          | FB_B0_EDC<0>       | 75                  | 78                   |                   |
|                                       | FB_B0_EDC1   | GDDR5_45SE            | GDDR5_EDC          | FB_B0_EDC<1>       | 75                  | 78                   |                   |
|                                       | FB_B0_EDC2   | GDDR5_45SE            | GDDR5_EDC          | FB_B0_EDC<2>       | 75                  | 78                   |                   |
|                                       | FB_B0_EDC3   | GDDR5_45SE            | GDDR5_EDC          | FB_B0_EDC<3>       | 75                  | 78                   |                   |
|                                       | FB_B1_EDC0   | GDDR5_45SE            | GDDR5_EDC          | FB_B1_EDC<0>       | 75                  | 78                   |                   |
|                                       | FB_B1_EDC1   | GDDR5_45SE            | GDDR5_EDC          | FB_B1_EDC<1>       | 75                  | 78                   |                   |
|                                       | FB_B1_EDC2   | GDDR5_45SE            | GDDR5_EDC          | FB_B1_EDC<2>       | 75                  | 78                   |                   |
|                                       | FB_B1_EDC3   | GDDR5_45SE            | GDDR5_EDC          | FB_B1_EDC<3>       | 75                  | 78                   |                   |
|                                       | FB_B0_DBI_L0 | GDDR5_45SE            | GDDR5_DATA         | FB_B0_DBI_L<0>     | 75                  | 78                   |                   |
|                                       | FB_B0_DBI_L1 | GDDR5_45SE            | GDDR5_DATA         | FB_B0_DBI_L<1>     | 75                  | 78                   |                   |
|                                       | FB_B0_DBI_L2 | GDDR5_45SE            | GDDR5_DATA         | FB_B0_DBI_L<2>     | 75                  | 78                   |                   |
|                                       | FB_B0_DBI_L3 | GDDR5_45SE            | GDDR5_DATA         | FB_B0_DBI_L<3>     | 75                  | 78                   |                   |
|                                       | FB_B1_DBI_L0 | GDDR5_45SE            | GDDR5_DATA         | FB_B1_DBI_L<0>     | 75                  | 78                   |                   |
|                                       | FB_B1_DBI_L1 | GDDR5_45SE            | GDDR5_DATA         | FB_B1_DBI_L<1>     | 75                  | 78                   |                   |
|                                       | FB_B1_DBI_L2 | GDDR5_45SE            | GDDR5_DATA         | FB_B1_DBI_L<2>     | 75                  | 78                   |                   |
|                                       | FB_B1_DBI_L3 | GDDR5_45SE            | GDDR5_DATA         | FB_B1_DBI_L<3>     | 75                  | 78                   |                   |
|                                       | FB_B0_WCLK0  | GDDR5_80D             | GDDR5_CMD          | FB_B0_WCLK_P<0>    | 75                  | 78                   |                   |
|                                       | FB_B0_WCLK0  | GDDR5_80D             | GDDR5_CMD          | FB_B0_WCLK_N<0>    | 75                  | 78                   |                   |
|                                       | FB_B0_WCLK1  | GDDR5_80D             | GDDR5_CMD          | FB_B0_WCLK_P<1>    | 75                  | 78                   |                   |
|                                       | FB_B0_WCLK1  | GDDR5_80D             | GDDR5_CMD          | FB_B0_WCLK_N<1>    | 75                  | 78                   |                   |
|                                       | FB_B1_WCLK0  | GDDR5_80D             |                    |                    |                     |                      |                   |

| PHYSICAL_RULE_SET | LAYER | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-------------------|-------|-----------------------|--------------------|--------------------|---------------------|----------------------|-------------------|
| SENSE_1T01_55S    | *     | =1:1_DIFFPAIR         | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =1:1_DIFFPAIR        | =1:1_DIFFPAIR     |
| THERM_1T01_55S    | *     | =1:1_DIFFPAIR         | =55_OHM_SE         | =55_OHM_SE         | =55_OHM_SE          | =1:1_DIFFPAIR        | =1:1_DIFFPAIR     |
| DIFFPAIR          | *     | =1:1_DIFFPAIR         |                    |                    | =1:1_DIFFPAIR       | =1:1_DIFFPAIR        | =1:1_DIFFPAIR     |
| AUDIODIFF         | *     | =1:1_DIFFPAIR         | 0.1 MM             | 0.1 MM             | 10 MM               | 0.1 MM               | 0.1 MM            |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| SENSE            | *     | =2:1_SPACING         | ?      |
| THERM            | *     | =2:1_SPACING         | ?      |
| AUDIO            | *     | =2:1_SPACING         | ?      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CPU_COMP          | GND               | *         | GND_P2MM         |
| CPU_VCCSENSE      | GND               | *         | GND_P2MM         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| ENETCONN         | *     | 25 MILS              | ?      |

|                   |                   |           |                  |
|-------------------|-------------------|-----------|------------------|
| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
| ENET_MDI          | GND               | *         | GND_P2MM         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| GND              | *     | =STANDARD            | ?      |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CLK_PCIE          | GND               | *         | GND_P2MM         |

| SPACING_RULE_SET | LAYER | LINE-TO-LINE SPACING | WEIGHT |
|------------------|-------|----------------------|--------|
| GND_P2MM         | *     | 0.20 MM              | 1000   |
| PWR_P2MM         | *     | 0.20 MM              | 1000   |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| CLK_PCIE          | GND               | *         | GND_P2MM         |
| PCIE              | GND               | *         | GND_P2MM         |
| SATA              | GND               | *         | GND_P2MM         |
| USB               | GND               | *         | GND_P2MM         |
| CLK_PCIE          | SB_POWER          | *         | PwR_P2MM         |
| SATA              | SB_POWER          | *         | PwR_P2MM         |
| USB               | SB_POWER          | *         | PwR_P2MM         |

| NET_SPACING_TYPE1 | NET_SPACING_TYPE2 | AREA_TYPE | SPACING_RULE_SET |
|-------------------|-------------------|-----------|------------------|
| MEM_CLK           | GND               | *         | GND_P2MM         |
| MEM_CMD           | GND               | *         | GND_P2MM         |
| MEM_CTRL          | GND               | *         | GND_P2MM         |
| MEM_DATA          | GND               | *         | GND_P2MM         |
| MEM_DQS           | GND               | *         | GND_P2MM         |

| PHYSICAL_RULE_SET     | LAYER              | ALLOW ROUTE ON LAYER? | MINIMUM LINE WIDTH | MINIMUM NECK WIDTH  | MAXIMUM NECK LENGTH | DIFFPAIR PRIMARY GAP | DIFFPAIR NECK GAP |
|-----------------------|--------------------|-----------------------|--------------------|---------------------|---------------------|----------------------|-------------------|
| MEM_40S<br>OVERRIDE   | *                  | OVERRIDE              | OVERRIDE           | 0.09 MM<br>OVERRIDE | 2.54 MM<br>OVERRIDE | OVERRIDE             | OVERRIDE          |
| MEM_72D<br>OVERRIDE   | *                  | OVERRIDE              | OVERRIDE           | 0.09 MM<br>OVERRIDE | 2.54 MM<br>OVERRIDE | OVERRIDE             | OVERRIDE          |
| PCIE_85D<br>OVERRIDE  | *                  | OVERRIDE              | OVERRIDE           | 0.09 MM<br>OVERRIDE | 10 MM<br>OVERRIDE   | OVERRIDE             | OVERRIDE          |
| USB_85D<br>OVERRIDE   | TOP<br>OVERRIDE    | OVERRIDE              | OVERRIDE           | 0.1 MM<br>OVERRIDE  | 12.7 MM<br>OVERRIDE | OVERRIDE             | OVERRIDE          |
| CPU_27P4S<br>OVERRIDE | BOTTOM<br>OVERRIDE | OVERRIDE              | OVERRIDE           | 0.23 MM<br>OVERRIDE | 2.54 MM<br>OVERRIDE | OVERRIDE             | OVERRIDE          |

## Breakout Constraint Overrides

Alternate diffpair width/gap through BGA fanout areas

| NET_PHYSICAL_TYPE | AREA_TYPE | PHYSICAL_RULE_SET |
|-------------------|-----------|-------------------|
| CLK_PCIE_90D      | BGA       | 100_DIFF_BGA      |
| DP_90D            | BGA       | 100_DIFF_BGA      |
| TMD5_85D          | BGA       | TMD5_85D          |
| ENET_100D         | BGA       | 100_DIFF_BGA      |
| FW_110D           | BGA       | 100_DIFF_BGA      |
| SATA_90D          | BGA       | 100_DIFF_BGA      |
| PCIE_85D          | BGA       | PCIE_85D          |
| MEM_85D           | BGA       | MEM_85D           |
| USB_85D           | BGA       | USB_85D           |
| GDDR5_80D         | BGA       | GDDR5_80D         |

| NET_PHYSICAL_TYPE | AREA_TYPE | PHYSICAL_RULE_SET |
|-------------------|-----------|-------------------|
| MEM_40S           | BGA       | BGA_P075          |
| MEM_50S           | BGA       | BGA_P075          |

## J40\* SPECIFIC NET PROPERTIES

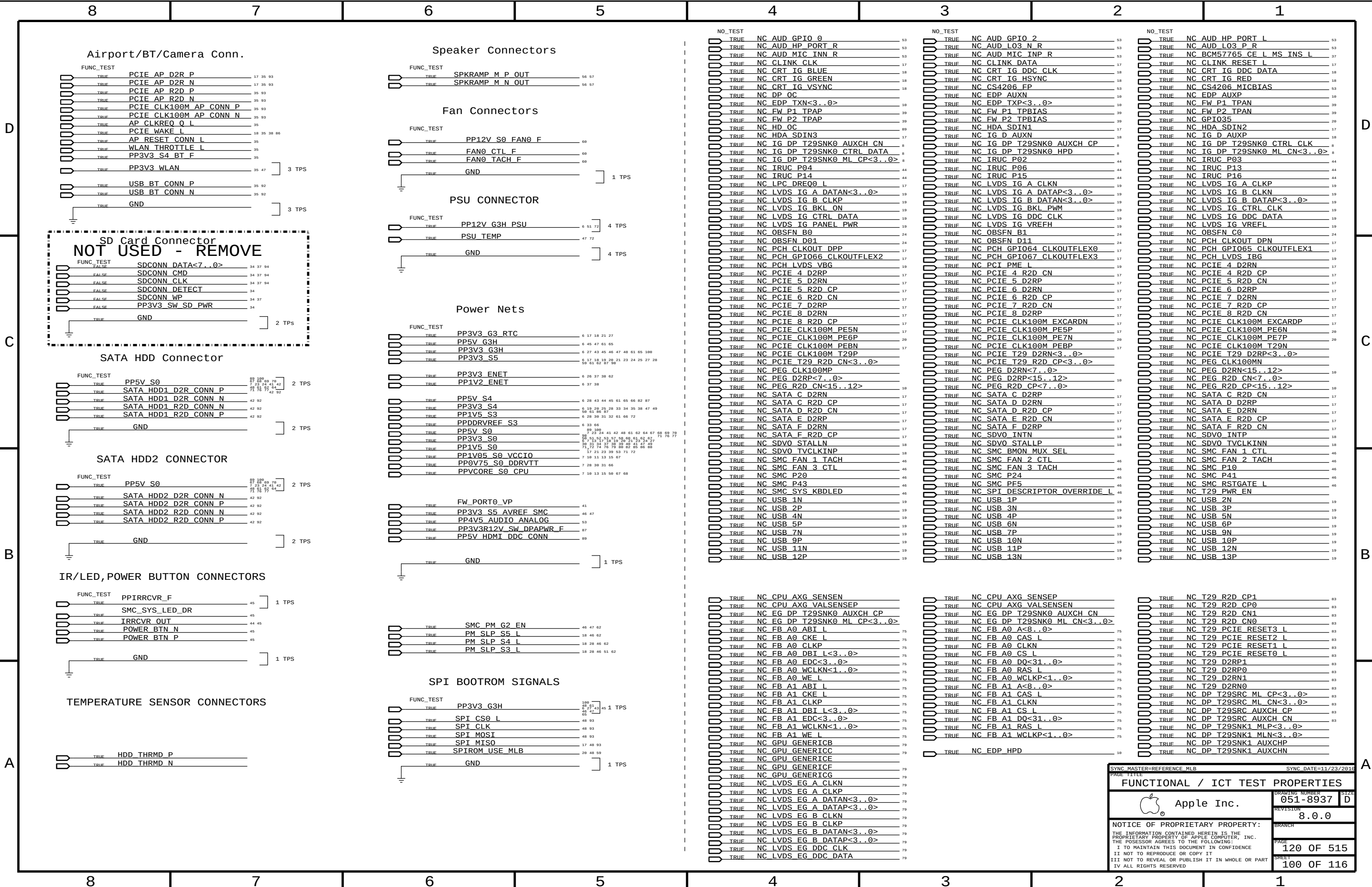
| ELECTRICAL_CONSTRAINT_SET                                                                          | NET_TYPE       |         |                 |       |
|----------------------------------------------------------------------------------------------------|----------------|---------|-----------------|-------|
|                                                                                                    | PHYSICAL       | SPACING |                 |       |
|  SENSE_DIFFPAIR | SENSE_1T01_55S | SENSE   | VCCSAS0_CS_P    | 50 64 |
|                 | SENSE_1T01_55S | SENSE   | VCCSAS0_CS_N    | 50 64 |
|  SENSE_DIFFPAIR | SENSE_1T01_55S | SENSE   | CPUVCCIOS0_CS_P | 50 70 |
|                 | SENSE_1T01_55S | SENSE   | CPUVCCIOS0_CS_N | 50 70 |
|  SENSE_DIFFPAIR | SENSE_1T01_55S | SENSE   | ISNS_HS_GPU_P   | 51    |
|                 | SENSE_1T01_55S | SENSE   | ISNS_HS_GPU_N   | 51    |

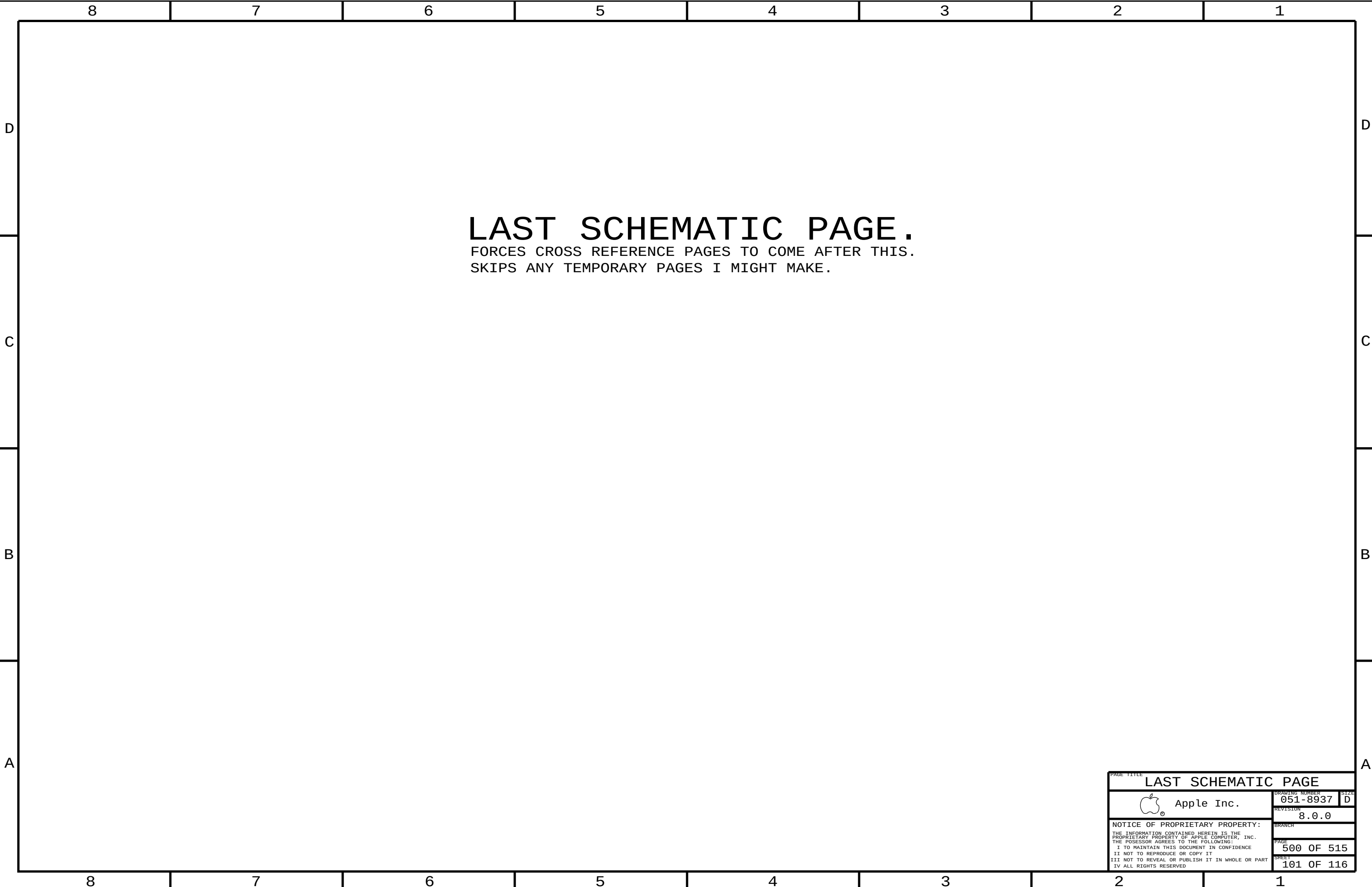
## J40\* SPECIFIC NET PROPERTIES

[illegible]

| ELECTRICAL_CONSTRAINT_SET                                                             |                | NET_TYPE       |         |                |
|---------------------------------------------------------------------------------------|----------------|----------------|---------|----------------|
|                                                                                       |                | PHYSICAL       | SPACING |                |
|  | SENSE_DIFFPAIR | THERM_1T01_55S | THERM   | CPUTHMSNS_D2_P |
|                                                                                       |                | THERM_1T01_55S | THERM   | CPUTHMSNS_D2_N |
|  | SENSE_DIFFPAIR | THERM_1T01_55S | THERM   | CPU_THERMD_P   |
|  |                | THERM_1T01_55S | THERM   | CPU_THERMD_N   |
|  | SENSE_DIFFPAIR | THERM_1T01_55S | THERM   | GPUTHMSNS_D_P  |
|                                                                                       |                | THERM_1T01_55S | THERM   | GPUTHMSNS_D_N  |
|  | SENSE_DIFFPAIR | THERM_1T01_55S | THERM   | GPU_TDIODE_P   |
|                                                                                       |                | THERM_1T01_55S | THERM   | GPU_TDIODE_N   |
|  | SENSE_DIFFPAIR | THERM_1T01_55S | THERM   | T29_THERMD_P   |
|  |                | THERM_1T01_55S | THERM   | T29_THERMD_N   |







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[illegible]

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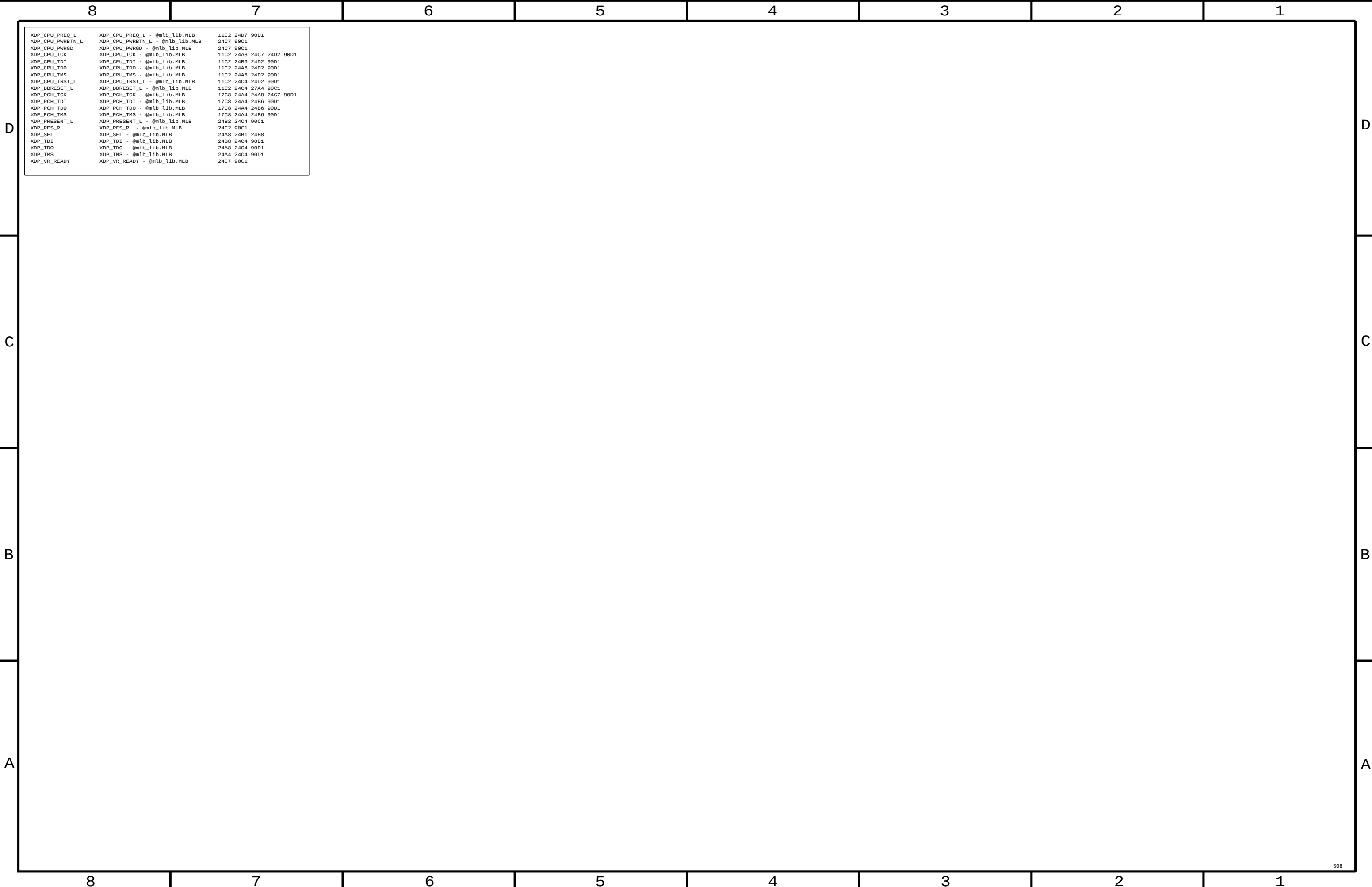


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|   |       |                         |           |       |                                     |       |          |           |       |         |           |
|---|-------|-------------------------|-----------|-------|-------------------------------------|-------|----------|-----------|-------|---------|-----------|
| D | L6053 | IND_0402                | m1b[5784] | Q5580 | SOT732-3                            | R1702 | RES_0603 | m1b[16A7] | R2491 | RES_402 | m1b[2385] |
|   | L6054 | IND_0402                | m1b[5784] | Q5590 | TRA_BC846BM3T5G_NPN_ m1b[52C2]      | R1800 | RES_201  | m1b[17A8] | R2500 | RES_402 | m1b[24C8] |
|   | L6092 | IND_0603                | m1b[57C2] | Q6100 | SOT732-3                            | R1801 | RES_201  | m1b[17A8] | R2501 | RES_402 | m1b[24C8] |
|   | L6100 | IND_0402                | m1b[58A3] | Q6101 | TRA_DUAL_SSM6N15FE_S m1b[58C5 58C6] | R1802 | RES_201  | m1b[17B7] | R2502 | RES_402 | m1b[24C7] |
|   | L6180 | IND_0402                | m1b[5803] | Q6102 | SOT732-3                            | R1803 | RES_201  | m1b[17B7] | R2504 | RES_402 | m1b[24C7] |
|   | L6500 | IND_0402                | m1b[60C3] | Q6103 | TRA_DUAL_SSM6N15FE_S m1b[58C7 58B6] | R1810 | RES_201  | m1b[17A5] | R2505 | RES_402 | m1b[24C4] |
|   | L6510 | FILTER_2P_0603          | m1b[6003] | Q6500 | OT563                               | R1811 | RES_201  | m1b[17A4] | R2510 | RES_402 | m1b[24D2] |
|   | L6520 | FILTER_2P_0603          | m1b[60C3] | Q6501 | TRA_DUAL_SSM6N15FE_S m1b[58C5 58C6] | R1812 | RES_201  | m1b[17A5] | R2511 | RES_402 | m1b[24D1] |
|   | L7100 | IND_FDV0630H-SM         | m1b[64C3] | Q6700 | OT563                               | R1813 | RES_201  | m1b[17A4] | R2512 | RES_402 | m1b[24D1] |
|   | L7201 | IND_MMD-06C2            | m1b[65C4] | Q6701 | TRA_SSM3K15FV_S0D-VE m1b[58A5]      | R1820 | RES_201  | m1b[17D5] | R2513 | RES_402 | m1b[24D2] |
| C | L7251 | IND_PIMB065T-SM         | m1b[6584] | Q6702 | SM-HF                               | R1830 | RES_201  | m1b[17C5] | R2514 | RES_402 | m1b[24D1] |
|   | L7302 | IND_PIMB042T-COMBO      | m1b[6605] | Q6703 | TRA_DUAL_SSM6N15FE_S m1b[58B3 58B2] | R1831 | RES_201  | m1b[17C5] | R2515 | RES_402 | m1b[24C4] |
|   | L7330 | IND_PIMB103E-SM-1       | m1b[66C3] | Q6710 | OT563                               | R1832 | RES_201  | m1b[17B5] | R2516 | RES_402 | m1b[24C4] |
|   | L7500 | IND_PCMB063T-MMD06C2-SM | m1b[6805] | Q6711 | TRA_FDC638P_SM m1b[60D4]            | R1833 | RES_201  | m1b[17A6] | R2520 | RES_402 | m1b[24B7] |
|   | L7501 | IND_PIMA104E-SM-1       | m1b[6803] | Q6712 | TRA_SSM3K15FV_S0D-VE m1b[60C6]      | R1840 | RES_201  | m1b[17A5] | R2521 | RES_402 | m1b[24B7] |
|   | L7521 | IND_PIMA104E-SM-1       | m1b[68C3] | Q6760 | SM-HF                               | R1841 | RES_201  | m1b[17A5] | R2522 | RES_402 | m1b[24A7] |
|   | L7601 | IND_PIMB065T-SM         | m1b[69C4] | Q6701 | TRA_MOSFET_PCHN_4P_S m1b[61D2]      | R1842 | RES_201  | m1b[17B0] | R2523 | RES_402 | m1b[24A7] |
|   | L7730 | IND_PIMB104T-SM         | m1b[70C3] | Q6702 | C70-6L                              | R1843 | RES_201  | m1b[17B7] | R2524 | RES_402 | m1b[24A5] |
|   | L7860 | IND_2520                | m1b[71D6] | Q6703 | TRA_MOSFET_NCHN_PWRP m1b[61B6]      | R1844 | RES_201  | m1b[17B0] | R2526 | RES_402 | m1b[24A6] |
|   | L8120 | IND_0402                | m1b[74C7] | Q6702 | K-1212-8-HF                         | R1845 | RES_201  | m1b[17B6] | R2530 | RES_402 | m1b[24C2] |
| B | L8130 | FILTER_2P_0603          | m1b[74C7] | Q6710 | TRA_DUAL_SSM6N15FE_S m1b[61B3 61D4] | R1846 | RES_201  | m1b[17B6] | R2531 | RES_402 | m1b[24B2] |
|   | L8131 | FILTER_2P_0603          | m1b[74B7] | Q6711 | OT563                               | R1847 | RES_201  | m1b[17A6] | R2532 | RES_402 | m1b[24B1] |
|   | L8132 | IND_0402                | m1b[74B7] | Q6710 | TRA_MOSFET_PCHN_4P_S m1b[61D6]      | R1848 | RES_201  | m1b[17B7] | R2540 | RES_402 | m1b[24D6] |
|   | L8140 | IND_0603                | m1b[74A7] | Q6711 | C70-6L                              | R1849 | RES_201  | m1b[17A3] | R2550 | RES_402 | m1b[24A3] |
|   | L8150 | IND_0402                | m1b[74D3] | Q6712 | TRA_DUAL_SSM6N15FE_S m1b[61D7]      | R1853 | RES_201  | m1b[17A3] | R2551 | RES_402 | m1b[24A3] |
|   | L8155 | IND_0603                | m1b[74A8] | Q6712 | OT563                               | R1854 | RES_201  | m1b[17A3] | R2552 | RES_402 | m1b[24A2] |
|   | L8300 | IND_PIMB042T-SM         | m1b[76D3] | Q6722 | TRA_DUAL_SSM6N15FE_S m1b[61C4]      | R1855 | RES_201  | m1b[17A2] | R2553 | RES_402 | m1b[24B3] |
|   | L8351 | IND_PIMB065T-SM         | m1b[76B3] | Q6730 | OT563                               | R1860 | RES_201  | m1b[17D5] | R2556 | RES_402 | m1b[24A2] |
|   | L8400 | IND_PIMB042T-SM         | m1b[77C4] | Q6740 | TRA_DUAL_SSM6N15FE_S m1b[61C2]      | R1861 | RES_201  | m1b[17D5] | R2557 | RES_402 | m1b[24B3] |
|   | L8600 | FILTER_2P_0603          | m1b[79B8] | Q6760 | OT563                               | R1862 | RES_201  | m1b[17D5] | R2600 | RES_402 | m1b[25C4] |
| A | L8601 | FILTER_2P_0603          | m1b[79A8] | Q6821 | TRA_MOSFET_PCHN_4P_S m1b[61C2]      | R1863 | RES_201  | m1b[17D5] | R2601 | RES_402 | m1b[25C8] |
|   | L8602 | IND_0402                | m1b[79A8] | Q6740 | C70-6L                              | R1864 | RES_201  | m1b[17D5] | R2602 | RES_402 | m1b[25C8] |
|   | L8702 | IND_0402                | m1b[80C3] | Q6760 | TRA_MOSFET_PCHN_4P_S m1b[61C2]      | R1866 | RES_201  | m1b[17A4] | R2603 | RES_402 | m1b[25C8] |
|   | L8703 | IND_0402                | m1b[80C3] | Q6760 | C70-6L                              | R1869 | RES_201  | m1b[17B0] | R2604 | RES_402 | m1b[25C8] |
|   | L8800 | FILTER_2P_0603          | m1b[81B7] | Q6760 | TRA_MOSFET_PCHN_8P4_ m1b[61B2]      | R1870 | RES_201  | m1b[17A2] | R2605 | RES_402 | m1b[25C7] |
|   | L8801 | FILTER_2P_0603          | m1b[81B5] | Q6821 | 23V1K-SM                            | R1871 | RES_201  | m1b[17A2] | R2606 | RES_402 | m1b[25C7] |
|   | L8802 | FILTER_2P_0603          | m1b[81A7] | Q6822 | TRA_SSM3K15FV_S0D-VE m1b[62A3]      | R1876 | RES_201  | m1b[17B5] | R2607 | RES_402 | m1b[25C8] |
|   | L8803 | FILTER_2P_0603          | m1b[81A5] | Q6822 | SM-HF                               | R1877 | RES_201  | m1b[17B5] | R2620 | RES_402 | m1b[25C4] |
|   | L8804 | FILTER_2P_0603          | m1b[81B7] | Q6822 | TRA_SINGLE_MOSFET_PC m1b[62B1]      | R1878 | RES_201  | m1b[17B0] | R2621 | RES_402 | m1b[25C3] |
|   | L8805 | FILTER_2P_0603          | m1b[81B5] | Q6825 | HN_SOT-23-HF                        | R1879 | RES_201  | m1b[17C5] | R2622 | RES_402 | m1b[25C3] |

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| 8 |       |         |           | 7 |       |         |           | 6 |       |         |           | 5 |       |         |           | 4 |       |         |           | 3 |       |         |           | 2 |       |         |           | 1 |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |          |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |                |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |                |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |                |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |         |           |   |       |                  |           |   |       |         |           |   |       |         |           |   |       |                |           |   |       |         |           |   |
|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|----------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|----------------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|----------------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|----------------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|------------------|-----------|---|-------|---------|-----------|---|-------|---------|-----------|---|-------|----------------|-----------|---|-------|---------|-----------|---|
| D | R3512 | RES_402 | m1b[3482] | C | R3514 | RES_402 | m1b[3483] | B | R3530 | RES_402 | m1b[3486] | A | R3531 | RES_402 | m1b[3483] | D | R3535 | RES_402 | m1b[34A3] | C | R3563 | RES_402 | m1b[34C4] | B | R3564 | RES_402 | m1b[34C4] | A | R3565 | RES_402 | m1b[34C4] | D | R3566 | RES_402 | m1b[34C4] | C | R3567 | RES_402 | m1b[34C4] | B | R3601 | RES_402 | m1b[35D7] | A | R3640 | RES_402 | m1b[35B4] | D | R3659 | RES_402 | m1b[35C2] | C | R3651 | RES_402 | m1b[35C2] | B | R3653 | RES_402 | m1b[35B4] | A | R3654 | RES_402 | m1b[35A4] | D | R3800 | RES_402 | m1b[36B4] | C | R3801 | RES_402 | m1b[36B4] | B | R3802 | RES_402 | m1b[36B4] | A | R3803 | RES_402 | m1b[36B3] | D | R3910 | RES_402 | m1b[37C6] | C | R3930 | RES_402 | m1b[37B2] | B | R3940 | RES_402 | m1b[37C6] | A | R3941 | RES_402 | m1b[37C6] | D | R3942 | RES_402 | m1b[37C6] | C | R3943 | RES_402 | m1b[37B7] | B | R3950 | RES_402 | m1b[37B2] | A | R3951 | RES_402 | m1b[37B2] | D | R3952 | RES_402 | m1b[37B2] | C | R3953 | RES_402 | m1b[37B2] | B | R3965 | RES_402 | m1b[37A6] | A | R3971 | RES_402 | m1b[37B2] | D | R3974 | RES_402 | m1b[37B2] | C | R3975 | RES_402 | m1b[37B2] | B | R3976 | RES_402 | m1b[37B2] | A | R3977 | RES_402 | m1b[37B2] | D | R3978 | RES_402 | m1b[37B2] | C | R3980 | RES_402 | m1b[37B2] | B | R3990 | RES_402 | m1b[37A7] | A | R3997 | RES_402 | m1b[37A6] | D | R4010 | RES_402 | m1b[38B4] | C | R4011 | RES_402 | m1b[38A4] | B | R4012 | RES_402 | m1b[38A4] | A | R4019 | RES_402 | m1b[38B4] | D | R4050 | RES_402 | m1b[38C2] | C | R4110 | RES_603 | m1b[39D7] | B | R4117 | RES_603 | m1b[39D7] | A | R4119 | RES_603 | m1b[39D8] | D | R4125 | RES_603 | m1b[39D2] | C | R4129 | RES_402 | m1b[39D2] | B | R4135 | RES_603 | m1b[39D1] | A | R4140 | RES_402 | m1b[39C6] | D | R4141 | RES_402 | m1b[39C7] | C | R4150 | RES_402 | m1b[39B7] | B | R4151 | RES_402 | m1b[39B7] | A | R4152 | RES_402 | m1b[39B7] | D | R4153 | RES_402 | m1b[39B7] | C | R4160 | RES_402 | m1b[39B7] | B | R4162 | RES_402 | m1b[39A7] | A | R4170 | RES_402 | m1b[39C2] | D | R4171 | RES_402 | m1b[39C2] | C | R4175 | RES_402 | m1b[39B2] | B | R4180 | RES_402 | m1b[39A7] | A | R4181 | RES_402 | m1b[39A8] | D | R4182 | RES_402 | m1b[39A8] | C | R4183 | RES_402 | m1b[39A7] | B | R4184 | RES_402 | m1b[39A7] | A | R4185 | RES_402 | m1b[39B2] | D | R4186 | RES_402 | m1b[39B2] | C | R4190 | RES_402 | m1b[39C1] | B | R4191 | RES_402 | m1b[39C2] | A | R4250 | RES_402 | m1b[40C3] | D | R4251 | RES_402 | m1b[40C3] | C | R4252 | RES_402 | m1b[40C3] | B | R4253 | RES_402 | m1b[40C3] | A | R4254 | RES_402 | m1b[40B3] | D | R4255 | RES_402 | m1b[40B7] | C | R4256 | RES_402 | m1b[40B6] | B | R4257 | RES_402 | m1b[40A7] | A | R4258 | RES_402 | m1b[40A7] | D | R4300 | RES_2512 | m1b[41D7] | C | R4310 | RES_402 | m1b[41C2] | B | R4311 | RES_402 | m1b[41B2] | A | R4312 | RES_402 | m1b[41C2] | D | R4313 | RES_402 | m1b[41B2] | C | R4320 | RES_402 | m1b[41D7] | B | R4321 | RES_402 | m1b[41D7] | A | R4322 | RES_402 | m1b[41C7] | D | R4323 | RES_402 | m1b[41D7] | C | R4324 | RES_402 | m1b[41C5] | B | R4325 | RES_402 | m1b[41C6] | A | R4326 | RES_402 | m1b[41C6] | D | R4327 | RES_402 | m1b[41B5] | C | R4328 | RES_402 | m1b[41B5] | B | R4329 | RES_402 | m1b[41B6] | A | R4335 | RES_402 | m1b[41B2] | D | R4340 | RES_402 | m1b[41A8] | C | R4341 | RES_402 | m1b[41A7] | B | R4342 | RES_402 | m1b[41A8] | A | R4343 | RES_402 | m1b[41A7] | D | R4350 | RES_402 | m1b[41A5] | C | R4351 | RES_402 | m1b[41A5] | B | R4352 | RES_402 | m1b[41A5] | A | R4353 | RES_402 | m1b[41A4] | D | R4354 | RES_402 | m1b[41A3] | C | R4390 | RES_402 | m1b[41B3] | B | R4450 | RES_201 | m1b[42C7] | A | R4451 | RES_201 | m1b[42C7] | D | R4452 | RES_201 | m1b[42B7] | C | R4453 | RES_201 | m1b[42B7] | B | R4460 | RES_201 | m1b[42C3] | A | R4461 | RES_201 | m1b[42C3] | D | R4462 | RES_201 | m1b[42B3] | C | R4463 | RES_201 | m1b[42B3] | B | R4600 | RES_402 | m1b[43B6] | A | R4601 | RES_402 | m1b[43D6] | D | R4650 | RES_402 | m1b[43D4] | C | R4651 | RES_402 | m1b[43C4] | B | R4652 | RES_402 | m1b[43C4] | A | R4700 | RES_402 | m1b[44D5] | D | R4790 | RES_402 | m1b[44C7] | C | R4794 | RES_402 | m1b[44B7] | B | R4798 | RES_402 | m1b[44C3] | A | R4799 | RES_402 | m1b[44C3] | D | R4800 | RES_402 | m1b[45C5] | C | R4801 | RES_402 | m1b[45D5] | B | R4802 | RES_402 | m1b[45D5] | A | R4803 | RES_402 | m1b[45C6] | D | R4805 | RES_402 | m1b[45D3] | C | R4808 | RES_402 | m1b[45D3] | B | R4820 | RES_402 | m1b[45B2] | A | R4821 | RES_805 | m1b[45A3] | D | R4822 | RES_805 | m1b[45A3] | C | R4823 | RES_402 | m1b[45A2] | B | R4901 | RES_201 | m1b[46D1] | A | R4902 | RES_201 | m1b[46C2] | D | R4909 | RES_201 | m1b[46D1] | C | R4910 | RES_402 | m1b[46A4] | B | R4911 | RES_402 | m1b[46A4] | A | R4912 | RES_402 | m1b[46A4] | D | R4998 | RES_201 | m1b[46C2] | C | R4999 | RES_402 | m1b[46D4] | B | R5000 | RES_402 | m1b[47D7] | A | R5001 | RES_603 | m1b[47D8] | D | R5010 | RES_402 | m1b[47B8] | C | R5012 | RES_402 | m1b[47C5] | B | R5020 | RES_201 | m1b[47C5] | A | R5051 | RES_402 | m1b[47A4] | D | R5060 | RES_402 | m1b[47D1] | C | R5061 | RES_402 | m1b[47D2] | B | R5062 | RES_402 | m1b[47D3] | A | R5071 | RES_201 | m1b[47C1] | D | R5072 | RES_201 | m1b[47C1] | C | R5073 | RES_201 | m1b[47B1] | B | R5074 | RES_201 | m1b[47B1] | A | R5075 | RES_201 | m1b[47B1] | D | R5076 | RES_402 | m1b[47B5] | C | R5077 | RES_201 | m1b[47B1] | B | R5078 | RES_201 | m1b[47B1] | A | R5079 | RES_201 | m1b[47B1] | D | R5080 | RES_201 | m1b[47B1] | C | R5081 | RES_201 | m1b[47B1] | B | R5084 | RES_201 | m1b[47B1] | A | R5085 | RES_201 | m1b[47B1] | D | R5086 | RES_201 | m1b[47B1] | C | R5087 | RES_201 | m1b[47B1] | B | R5088 | RES_201 | m1b[47B1] | A | R5089 | RES_201 | m1b[47A1] | D | R5090 | RES_201 | m1b[47B1] | C | R5091 | RES_201 | m1b[47B1] | B | R5092 | RES_201 | m1b[47B1] | A | R5093 | RES_201 | m1b[47B1] | D | R5110 | RES_402 | m1b[48B5] | C | R5111 | RES_402 | m1b[48B5] | B | R5112 | RES_402 | m1b[48A5] | A | R5120 | RES_402 | m1b[48B4] | D | R5121 | RES_402 | m1b[48B4] | C | R5122 | RES_402 | m1b[48A4] | B | R5123 | RES_402 | m1b[48A4] | A | R5125 | RES_402 | m1b[48B4] | D | R5126 | RES_402 | m1b[48B4] | C | R5127 | RES_402 | m1b[48B4] | B | R5128 | RES_402 | m1b[48B4] | A | R5200 | RES_402 | m1b[49D7] | D | R5201 | RES_402 | m1b[49D7] | C | R5202 | RES_402 | m1b[49D7] | B | R5203 | RES_402 | m1b[49D7] | A | R5204 | RES_402 | m1b[49D7] | D | R5205 | RES_402 | m1b[49D7] | C | R5206 | RES_402 | m1b[49D7] | B | R5207 | RES_402 | m1b[49D7] | A | R5208 | RES_402 | m1b[49C7] | D | R5209 | RES_402 | m1b[49C7] | C | R5210 | RES_402 | m1b[49B7] | B | R5211 | RES_402 | m1b[49B7] | A | R5212 | RES_402 | m1b[49C7] | D | R5213 | RES_402 | m1b[49C7] | C | R5220 | RES_402 | m1b[49A7] | B | R5221 | RES_402 | m1b[49A7] | A | R5222 | RES_402 | m1b[49A6] | D | R5223 | RES_402 | m1b[49A6] | C | R5230 | RES_402 | m1b[49B2] | B | R5231 | RES_402 | m1b[49B2] | A | R5234 | RES_402 | m1b[49B2] | D | R5235 | RES_402 | m1b[49B2] | C | R5236 | RES_402 | m1b[41C2] | B | R5237 | RES_402 | m1b[49B7] | A | R5250 | RES_402 | m1b[49D4] | D | R5251 | RES_402 | m1b[49D4] | C | R5260 | RES_402 | m1b[49A4] | B | R5261 | RES_402 | m1b[49A4] | A | R5262 | RES_402 | m1b[49A5] | D | R5263 | RES_402 | m1b[49A5] | C | R5264 | RES_402 | m1b[49A4] | B | R5265 | RES_402 | m1b[49A4] | A | R5270 | RES_402 | m1b[49C4] | D | R5271 | RES_402 | m1b[49C4] | C | R5290 | RES_402 | m1b[49C2] | B | R5291 | RES_402 | m1b[49C2] | A | R5300 | RES_SENSE_1206 | m1b[50D7] | D | R5301 | RES_402 | m1b[50D6] | C | R5302 | RES_402 | m1b[50D7] | B | R5304 | RES_402 | m1b[50D6] | A | R5305 | RES_SENSE_1206 | m1b[50C7] | D | R5306 | RES_402 | m1b[50C6] | C | R5320 | RES_402 | m1b[50C2] | B | R5322 | RES_402 | m1b[50A3] | A | R5330 | RES_402 | m1b[50B2] | D | R5333 | RES_402 | m1b[50A3] | C | R5341 | RES_402 | m1b[50D2] | B | R5343 | RES_402 | m1b[50D4] | A | R5345 | RES_402 | m1b[50D3] | D | R5351 | RES_402 | m1b[50C2] | C | R5353 | RES_402 | m1b[50B4] | B | R5355 | RES_402 | m1b[50B3] | A | R5361 | RES_402 | m1b[50B6] | D | R5366 | RES_402 | m1b[50A6] | C | R5371 | RES_402 | m1b[50A6] | B | R5376 | RES_402 | m1b[50C6] | A | R5400 | RES_SENSE_0612 | m1b[51C7] | D | R5403 | RES_402 | m1b[51C5] | C | R5411 | RES_402 | m1b[51C3] | B | R5412 | RES_402 | m1b[51C2] | A | R5413 | RES_402 | m1b[51D2] | D | R5414 | RES_402 | m1b[51C2] | C | R5420 | RES_SENSE_0612-3 | m1b[51B7] | B | R5423 | RES_402 | m1b[51A5] | A | R5433 | RES_402 | m1b[51D5] | D | R5493 | RES_SENSE_1206 | m1b[51D7] | C | R5500 | RES_402 | m1b[52D5] | B |

[illegible]

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